

PHYS 3114 Electrodynamics

Part B: Radiating Potentials and Relativistic Electrodynamics

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How to use these notes

These notes are organised as eighteen lectures. They are intended to be sufficiently explicit that the algebra can be followed line by line, but the main physical ideas are highlighted separately so that the derivations do not obscure the physics. Every lecture begins with two short orientation boxes. The **Aim of this lecture** box states what the lecture is intended to establish, while the **Motivation for this lecture** box identifies the physical or mathematical problem that makes the new material necessary and explains how it follows from what has already been established. The notes use SI units throughout.

The first thirteen lectures develop retarded potentials and radiation. Lectures 14–17 develop special relativity and the covariant formulation of electrodynamics. Lecture 18 completes Topic B5 by developing the action principle, the Euler–Lagrange equations, symmetry and conservation laws, Noether’s theorem, and the covariant action formulation of particles and electromagnetic fields.

Notation and convention

We use the Lorenz gauge, named after Ludvig Lorenz (not Lorentz):

$$\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0.$$

In Lectures 1–13 it is convenient to define the wave operator

$$\square_{\text{w}} \equiv \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}.$$

In the covariant lectures we use the metric $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ and therefore

$$\partial_\mu \partial^\mu = \frac{1}{c^2} \partial_t^2 - \nabla^2 = -\square_{\text{w}}.$$

This change of sign is purely conventional.

Lecture 1: Green's Functions for Static Problems

Aim of this lecture

Understand why Green's functions arise naturally in electrodynamics, interpret them as inverse differential operators or point-source responses, derive the free-space electrostatic Green's function, and obtain the general solution of Poisson's equation in a bounded region.

Motivation for this lecture

Green's functions enter because Maxwell's equations are differential equations, whereas the questions we usually want to answer are global: given sources and boundary conditions, what field is produced everywhere? In electrostatics this becomes Poisson's equation. Solving that equation afresh for every possible charge distribution would be hopelessly inefficient. The Green's function is the response to one elementary point source; linearity then lets us build the response to an arbitrary source by superposition. It is therefore both a calculational tool and the continuum analogue of an inverse matrix.

1.1 Where does the need for a Green's function arise?

Electrodynamics is, to a large extent, the problem of finding fields produced by specified sources. Maxwell's equations tell us how the fields vary locally, but in an actual calculation we usually want something more global: given a charge density, a current density, boundary conditions and perhaps initial data, what are the fields everywhere?

In electrostatics, Gauss's law and the fact that the electric field is conservative give

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \mathbf{E} = -\nabla\phi, \quad (1.1)$$

so that the scalar potential obeys Poisson's equation

$$\boxed{\nabla^2\phi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\epsilon_0}.} \quad (1.2)$$

This is already a major simplification: instead of solving directly for the three components of $\mathbf{E}$, we solve one scalar differential equation and then differentiate the answer.

But Poisson's equation still has to be solved for an *arbitrary* source $\rho(\mathbf{r})$ and with the correct boundary conditions. For a few highly symmetric charge distributions one can use Gauss's law directly. For a general distribution near conductors or other boundaries, symmetry is absent and direct solution of the differential equation can become difficult. Green's functions provide a systematic way of solving precisely this class of problems.

Physical meaning

The need for a Green's function appears whenever three ingredients occur together:

- (i) a *linear* differential equation;
- (ii) a specified source; and
- (iii) boundary or initial conditions that select the physical solution.

Electrodynamics contains this structure repeatedly: Poisson's equation in electrostatics, the vector Poisson equation in magnetostatics, and the inhomogeneous wave equation for time-dependent potentials.

1.2 The linear-algebra analogy: a Green's function is an inverse operator

It is useful to compare the differential equation with an ordinary system of linear equations. Suppose

$$A_{ij}x_j = b_i. \quad (1.3)$$

If the inverse matrix exists, then

$$x_i = (A^{-1})_{ij}b_j. \quad (1.4)$$

The matrix element $(A^{-1})_{ij}$ tells us how a unit source in component j contributes to the response in component i .

A differential equation is the continuum version of the same idea. The discrete label i becomes a position $\mathbf{r}$, the sum over j becomes an integral over $\mathbf{r}'$, and the inverse matrix becomes a two-point function $G(\mathbf{r}, \mathbf{r}')$. Schematically,

$$L\psi = f \quad \implies \quad \psi(\mathbf{r}) = \int d^3r' G(\mathbf{r}, \mathbf{r}') f(\mathbf{r}'), \quad (1.5)$$

up to the normalization convention used in defining G and possible homogeneous or boundary terms.

There is one important subtlety. A differential operator does not have a unique inverse until the boundary conditions have been specified. In the matrix analogy, the boundary conditions are part of the information that tells us which inverse we mean. Thus a Green's function contains not only the differential operator but also the physical boundary conditions.

1.3 Why the point source is the natural building block

Any source can be decomposed into point sources using the Dirac delta distribution:

$$\rho(\mathbf{r}) = \int d^3r' \rho(\mathbf{r}') \delta^{(3)}(\mathbf{r} - \mathbf{r}'). \quad (1.6)$$

Thus, if we know the response to a delta-function source located at every possible point $\mathbf{r}'$, linear superposition reconstructs the response to any continuous charge distribution.

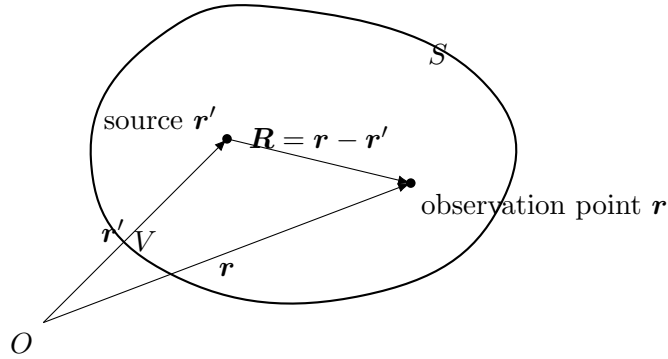
For a point charge q at $\mathbf{r}'$,

$$\rho(\mathbf{r}) = q \delta^{(3)}(\mathbf{r} - \mathbf{r}'), \quad (1.7)$$

and Poisson's equation becomes

$$\boxed{\nabla^2 \phi(\mathbf{r}) = -\frac{q}{\epsilon_0} \delta^{(3)}(\mathbf{r} - \mathbf{r}').} \quad (1.8)$$

The Green's function will be the solution of the same equation with a unit point source.



The notation deserves attention. The coordinate $\mathbf{r}'$ labels the *source point*; the coordinate $\mathbf{r}$ labels the *observation point*. In later integrals $\mathbf{r}'$ will be integrated over, while $\mathbf{r}$ is held fixed.

1.4 Deriving the free-space Green's function

In empty space, away from the point source, the potential satisfies Laplace's equation. The rotationally symmetric function $1/R$, where

$$R = |\mathbf{r} - \mathbf{r}'|, \quad (1.9)$$

has the property

$$\nabla^2 \frac{1}{R} = 0, \quad R \neq 0. \quad (1.10)$$

The only possible contribution is therefore concentrated at $R = 0$. To determine its strength, integrate over a small sphere of radius a surrounding the source point. By the divergence theorem,

$$\int_{R < a} d^3r \nabla^2 \frac{1}{R} = \oint_{R=a} d\mathbf{S} \cdot \nabla \frac{1}{R} \quad (1.11)$$

$$= \oint_{R=a} a^2 d\Omega \left(-\frac{1}{a^2} \right) \quad (1.12)$$

$$= -4\pi. \quad (1.13)$$

The only distribution with support at the origin and this integral is

$$\boxed{\nabla^2 \frac{1}{R} = -4\pi \delta^{(3)}(\mathbf{R}).} \quad (1.14)$$

Consequently the potential of a point charge is

$$\phi(\mathbf{r}) = \frac{q}{4\pi\epsilon_0 R} + \phi_s, \quad (1.15)$$

where ϕ_s is any solution of the homogeneous equation $\nabla^2 \phi_s = 0$.

This motivates the free-space electrostatic Green's function

$$\boxed{G_0(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|}, \quad \nabla^2 G_0(\mathbf{r}, \mathbf{r}') = -\frac{1}{\epsilon_0} \delta^{(3)}(\mathbf{r} - \mathbf{r}').} \quad (1.16)$$

Our convention includes the factor $1/\epsilon_0$ in G_0 . Some textbooks instead define the purely geometrical Green's function $g_0 = 1/(4\pi R)$ and keep $1/\epsilon_0$ outside the integral. The physics is of course identical.

Physical meaning

$G_0(\mathbf{r}, \mathbf{r}')$ answers the question: “what potential is produced at $\mathbf{r}$ by a unit point source placed at $\mathbf{r}'$ in free space?” The factor $1/R$ is not an arbitrary guess. It is the three-dimensional inverse of the Laplacian, in the same sense that A^{-1} is the inverse of a matrix.

1.5 From a point source to an arbitrary charge distribution

For point charges q_i at positions $\mathbf{r}_i$, linearity gives

$$\phi(\mathbf{r}) = \sum_i q_i G_0(\mathbf{r}, \mathbf{r}_i) + \phi_s. \quad (1.17)$$

Now divide a continuous charge distribution into small cells of volume ΔV_i , each carrying charge

$$q_i \simeq \rho(\mathbf{r}_i) \Delta V_i. \quad (1.18)$$

Taking the continuum limit gives

$$\phi(\mathbf{r}) = \int_V d^3 r' \rho(\mathbf{r}') G_0(\mathbf{r}, \mathbf{r}') + \phi_s. \quad (1.19)$$

For free space with the usual boundary condition $\phi \rightarrow 0$ at infinity, the homogeneous part is zero and this is simply Coulomb’s integral formula,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int d^3 r' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \quad (1.20)$$

Notice the conceptual gain: the differential equation has been solved *once and for all*. Every new charge distribution changes only the weighting factor $\rho(\mathbf{r}')$ in the integral.

1.6 Why boundaries require more work

The free-space result is not the whole story. Suppose the charge distribution lies near conducting surfaces. The potential must then satisfy conditions on those surfaces, and simply superposing Coulomb potentials of the known charges will generally violate those conditions.

The general solution has two logically different pieces:

$$\phi = \phi_{\text{source}} + \phi_{\text{hom}}. \quad (1.21)$$

The first is a particular solution produced by the charges. The second solves Laplace’s equation and is chosen so that the total potential satisfies the boundary conditions. A Green’s function adapted to the boundary lets us package these two tasks into a single construction.

Physical meaning

Boundary conditions are not a cosmetic correction added after solving the differential equation. They contain physical information about the environment: grounded conductors, fixed potentials, imposed normal fields, and so on. Two systems with the same charge density but different conducting boundaries generally have different electric fields.

1.7 Green's first and second identities

To incorporate boundaries systematically we begin with the divergence theorem. For scalar fields ϕ and ψ ,

$$\nabla \cdot (\phi \nabla \psi) = \nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi. \quad (1.22)$$

Integrating over a volume V bounded by S gives Green's first identity,

$$\boxed{\int_V d^3r [\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi] = \oint_S d\mathbf{S} \cdot (\phi \nabla \psi).} \quad (1.23)$$

Write the same equation with ϕ and ψ interchanged and subtract. The gradient-product terms cancel, leaving Green's second identity,

$$\boxed{\int_V d^3r [\phi \nabla^2 \psi - \psi \nabla^2 \phi] = \oint_S d\mathbf{S} \cdot (\phi \nabla \psi - \psi \nabla \phi).} \quad (1.24)$$

This identity is the engine behind the Green-function solution of Poisson's equation.

1.8 Deriving the general Green-function solution

Let $\phi(\mathbf{r}')$ be the physical potential and choose

$$\psi(\mathbf{r}') = G(\mathbf{r}', \mathbf{r}), \quad (1.25)$$

where $\mathbf{r}$ is the fixed observation point and $\mathbf{r}'$ is the integration variable. We therefore use primed derivatives when they act inside the integral. Green's second identity gives

$$\begin{aligned} \int_V d^3r' [\phi(\mathbf{r}') \nabla'^2 G(\mathbf{r}', \mathbf{r}) - G(\mathbf{r}', \mathbf{r}) \nabla'^2 \phi(\mathbf{r}')] \\ = \oint_S d\mathbf{S}' \cdot [\phi(\mathbf{r}') \nabla' G(\mathbf{r}', \mathbf{r}) - G(\mathbf{r}', \mathbf{r}) \nabla' \phi(\mathbf{r}')] . \end{aligned} \quad (1.26)$$

Now use the defining equations

$$\nabla'^2 G(\mathbf{r}', \mathbf{r}) = -\frac{1}{\epsilon_0} \delta^{(3)}(\mathbf{r}' - \mathbf{r}), \quad \nabla'^2 \phi(\mathbf{r}') = -\frac{\rho(\mathbf{r}')}{\epsilon_0}. \quad (1.27)$$

The first volume integral collapses by the delta function:

$$\int_V d^3r' \phi(\mathbf{r}') \nabla'^2 G = -\frac{\phi(\mathbf{r})}{\epsilon_0}. \quad (1.28)$$

The second becomes

$$-\int_V d^3r' G \nabla'^2 \phi = \frac{1}{\epsilon_0} \int_V d^3r' G(\mathbf{r}', \mathbf{r}) \rho(\mathbf{r}'). \quad (1.29)$$

Multiplying the resulting identity by ϵ_0 and solving for $\phi(\mathbf{r})$ gives

$$\boxed{\begin{aligned} \phi(\mathbf{r}) = & \int_V d^3r' \rho(\mathbf{r}') G(\mathbf{r}', \mathbf{r}) \\ & - \epsilon_0 \oint_S d\mathbf{S}' \cdot [\nabla' G(\mathbf{r}', \mathbf{r})] \phi(\mathbf{r}') \\ & + \epsilon_0 \oint_S d\mathbf{S}' \cdot [\nabla' \phi(\mathbf{r}')] G(\mathbf{r}', \mathbf{r}). \end{aligned}} \quad (1.30)$$

The three lines have a transparent interpretation:

- (i) the volume term is the response to the charge distribution;
- (ii) one surface term contains the value of the potential on the boundary;
- (iii) the other contains its normal derivative on the boundary.

At first sight this may seem to have made the problem worse because both ϕ and its normal derivative appear on the boundary. The key idea, developed in Lecture 2, is to choose the Green's function itself so that the term containing the *unknown* boundary data vanishes.

Checkpoint

A Green's function converts a differential equation into an integral representation. The source term tells us *what is driving the field*; the boundary terms tell us *which solution of the differential equation is physically realised*. The next lecture is about choosing the Green's function so that those boundary terms become useful rather than troublesome.

1.9 Worked example: the inverse of the one-dimensional Laplacian

Before doing more three-dimensional electrostatics, it is useful to see an exactly soluble version of the same idea in one dimension. Consider

$$-\frac{d^2u}{dx^2} = f(x), \quad 0 < x < L, \quad u(0) = u(L) = 0. \quad (1.31)$$

The differential operator is now

$$L_x = -\frac{d^2}{dx^2}. \quad (1.32)$$

The Green's function is defined by

$$-\frac{d^2G(x, x')}{dx^2} = \delta(x - x'), \quad G(0, x') = G(L, x') = 0. \quad (1.33)$$

Away from $x = x'$ the right hand side vanishes, so G is a straight line on each side of x' . Write

$$G(x, x') = \begin{cases} Ax, & 0 < x < x', \\ B(L - x), & x' < x < L. \end{cases} \quad (1.34)$$

The boundary conditions are already built in. Continuity at $x = x'$ gives

$$Ax' = B(L - x'). \quad (1.35)$$

The delta function produces a jump in the derivative. Integrate the defining equation from $x' - \eta$ to $x' + \eta$:

$$-\left. \frac{\partial G}{\partial x} \right|_{x=x'_+} + \left. \frac{\partial G}{\partial x} \right|_{x=x'_-} = 1. \quad (1.36)$$

Since the left derivative is A and the right derivative is $-B$, this gives

$$A + B = 1. \quad (1.37)$$

Solving the two algebraic equations,

$$A = \frac{L - x'}{L}, \quad B = \frac{x'}{L}. \quad (1.38)$$

Thus

$$G(x, x') = \begin{cases} \frac{x(L - x')}{L}, & x < x', \\ \frac{x'(L - x)}{L}, & x > x'. \end{cases} \quad (1.39)$$

The solution of the original problem is

$$u(x) = \int_0^L dx' G(x, x') f(x'). \quad (1.40)$$

This example is deliberately elementary, but it contains the whole logic of the method. The Green's function is the response to a point source; it is continuous at the source point; its derivative has a jump fixed by the delta function; and its form is fixed by the boundary conditions.

Checkpoint

In a discrete problem the inverse matrix is determined by algebra. In a differential problem the inverse operator is determined by a differential equation plus boundary conditions. The Green's function is that inverse operator written as a kernel.

Lecture 2: Boundary Conditions and the Inhomogeneous Wave Equation

Aim of this lecture

Understand how boundary conditions are built into Green's functions, distinguish Dirichlet and Neumann problems, apply the same logic to the magnetostatic vector potential, and derive step by step the inhomogeneous wave equations obeyed by the electromagnetic potentials in Lorenz gauge.

Motivation for this lecture

Lecture 1 showed how a Green's function inverts Poisson's equation, but a differential equation is not specified by its source alone: the apparatus surrounding the charges enters through boundary conditions. We therefore need to understand how Dirichlet and Neumann data are built into the Green's function itself. The second half of this lecture asks what changes when the sources begin to vary in time. The answer is fundamental: the static Poisson problem becomes a wave problem, information propagates at a finite speed, and the electromagnetic potentials become dynamical fields whose equations are most transparent after an appropriate gauge choice.

2.1 What the general solution from Lecture 1 is telling us

Lecture 1 ended with

$$\begin{aligned}\phi(\mathbf{r}) = & \int_V d^3r' \rho(\mathbf{r}') G(\mathbf{r}', \mathbf{r}) \\ & - \epsilon_0 \oint_S d\mathbf{S}' \cdot \nabla' G(\mathbf{r}', \mathbf{r}) \phi(\mathbf{r}') + \epsilon_0 \oint_S d\mathbf{S}' \cdot \nabla' \phi(\mathbf{r}') G(\mathbf{r}', \mathbf{r}).\end{aligned}\tag{2.1}$$

The volume term is the response to the physical charges. The two surface terms encode the boundary data. The practical problem is that in a physical boundary-value problem we normally know *either* the value of ϕ on the boundary *or* its normal derivative, but not both.

The Green's function is therefore chosen to make the unwanted surface term disappear. This is the crucial point: we do not choose G only to invert the Laplacian; we also choose it to respect the type of boundary condition in the problem.

2.2 Dirichlet boundary conditions

A Dirichlet problem specifies the potential itself on the boundary,

$$\phi|_S = \phi_S.\tag{2.2}$$

This is natural in electrostatics. A grounded conductor, for example, has $\phi_S = 0$; a conductor held at a fixed voltage has a prescribed nonzero value.

Because ϕ is known but $\partial\phi/\partial n$ is not, we choose a Dirichlet Green's function satisfying

$$G_D(\mathbf{r}_S, \mathbf{r}') = 0, \quad \mathbf{r}_S \in S. \quad (2.3)$$

Then the surface term proportional to $G_D \nabla' \phi$ vanishes, leaving

$$\phi(\mathbf{r}) = \int_V d^3r' \rho(\mathbf{r}') G_D(\mathbf{r}', \mathbf{r}) - \epsilon_0 \oint_S dS' \phi(\mathbf{r}') \frac{\partial G_D(\mathbf{r}', \mathbf{r})}{\partial n'}. \quad (2.4)$$

Everything on the right-hand side is now known once G_D has been constructed.

Physical meaning

The boundary value of the potential is not generated by the charges inside V alone. In a conductor problem it represents the effect of whatever external batteries or induced surface charges are required to maintain the prescribed potential. The surface term in the Green-function formula carries precisely this information.

2.3 Uniqueness: why the Dirichlet solution is the solution

Suppose two functions ϕ_1 and ϕ_2 solve the same Poisson equation and have the same value on the boundary. Their difference

$$u = \phi_1 - \phi_2 \quad (2.5)$$

satisfies

$$\nabla^2 u = 0, \quad u|_S = 0. \quad (2.6)$$

Green's first identity gives

$$\int_V d^3r |\nabla u|^2 = \oint_S dS u \frac{\partial u}{\partial n} - \int_V d^3r u \nabla^2 u = 0. \quad (2.7)$$

Since the integrand $|\nabla u|^2$ is non-negative, it must vanish everywhere. Hence u is constant, and because $u = 0$ on the boundary, $u = 0$ throughout V .

Thus the Dirichlet problem has a unique solution. Once we have found a function satisfying both Poisson's equation and the prescribed boundary potential, there is nothing else to check.

2.4 Reciprocity of the Dirichlet Green's function

The Dirichlet Green's function also satisfies

$$G_D(\mathbf{r}, \mathbf{r}') = G_D(\mathbf{r}', \mathbf{r}). \quad (2.8)$$

To see why, let $G_1(\mathbf{x}) = G_D(\mathbf{x}, \mathbf{r})$ and $G_2(\mathbf{x}) = G_D(\mathbf{x}, \mathbf{r}')$. Green's second identity gives

$$\int_V d^3x [G_1 \nabla^2 G_2 - G_2 \nabla^2 G_1] = \oint_S d\mathbf{S} \cdot (G_1 \nabla G_2 - G_2 \nabla G_1). \quad (2.9)$$

Both Green's functions vanish on S , so the surface integral is zero. The delta functions in the two Laplacians then give

$$-\frac{1}{\epsilon_0} G_D(\mathbf{r}', \mathbf{r}) + \frac{1}{\epsilon_0} G_D(\mathbf{r}, \mathbf{r}') = 0, \quad (2.10)$$

which proves reciprocity.

Physically, reciprocity says that the response measured at $\mathbf{r}$ from a unit source at $\mathbf{r}'$ equals the response at $\mathbf{r}'$ if source and observation point are interchanged, provided the same boundary conditions are kept.

2.5 Neumann boundary conditions

A Neumann problem specifies the normal derivative of the potential,

$$\left. \frac{\partial \phi}{\partial n} \right|_S = \hat{n}_S \cdot \nabla \phi|_S. \quad (2.11)$$

Since $\mathbf{E} = -\nabla \phi$,

$$\frac{\partial \phi}{\partial n} = -E_n. \quad (2.12)$$

Thus Neumann data correspond physically to specifying the normal electric field at the boundary. On a conductor this is closely related to specifying the surface charge density.

There are two important differences from the Dirichlet case.

First, integrating Poisson's equation over the volume gives the compatibility condition

$$\oint_S dS \frac{\partial \phi}{\partial n} = \int_V d^3r \nabla^2 \phi = -\frac{1}{\epsilon_0} \int_V d^3r \rho. \quad (2.13)$$

The prescribed normal derivative cannot be arbitrary; its total flux must agree with the total charge inside.

Second, if ϕ is a solution then $\phi + C$ is also a solution with exactly the same normal derivative. A pure Neumann problem therefore determines the potential only up to an additive constant. This is harmless because the electric field depends on $\nabla \phi$.

A corresponding Neumann Green's function can be chosen with vanishing normal derivative on the boundary, but its defining equation must be modified so that the same compatibility condition is respected. One convenient convention for a volume V is

$$\left. \frac{\partial G_N}{\partial n'} \right|_S = 0, \quad \nabla'^2 G_N(\mathbf{r}', \mathbf{r}) = -\frac{1}{\epsilon_0} \left[\delta^{(3)}(\mathbf{r}' - \mathbf{r}) - \frac{1}{V} \right]. \quad (2.14)$$

We will not need the explicit Neumann construction later, but the distinction is conceptually important.

Checkpoint

Dirichlet: specify ϕ on the boundary; the solution is unique. Neumann: specify $\partial \phi / \partial n$; the data must satisfy a flux constraint and the solution is determined only up to a constant.

2.6 The same method in magnetostatics

Green's functions are not special to the scalar potential. In magnetostatics,

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad (2.15)$$

and Ampère's law is

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (2.16)$$

Substituting $\mathbf{B} = \nabla \times \mathbf{A}$ gives

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J}. \quad (2.17)$$

Using

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (2.18)$$

and choosing the Coulomb gauge

$$\nabla \cdot \mathbf{A} = 0, \quad (2.19)$$

we obtain

$$\boxed{\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}.} \quad (2.20)$$

In Cartesian components this means

$$\nabla^2 A_x = -\mu_0 J_x, \quad \nabla^2 A_y = -\mu_0 J_y, \quad \nabla^2 A_z = -\mu_0 J_z. \quad (2.21)$$

Thus the vector problem is simply three Poisson equations. In free space,

$$\boxed{\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int d^3r' \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}.} \quad (2.22)$$

The same geometric kernel $1/(4\pi|\mathbf{r} - \mathbf{r}'|)$ has reappeared because the differential operator is again the Laplacian.

2.7 Why time dependence changes the problem qualitatively

For static sources, the mathematical operator is ∇^2 . When the sources vary in time, Maxwell's displacement current and Faraday induction couple space and time. The potentials no longer satisfy Poisson equations but wave equations.

The key physical consequence is finite propagation speed. If a source changes at some position $\mathbf{r}'$, an observer at a distant point $\mathbf{r}$ cannot respond instantly. Information must travel across the separation $R = |\mathbf{r} - \mathbf{r}'|$ at speed c . The Green's function of the wave equation will encode this delay automatically.

2.8 Introducing the time-dependent potentials

The equation $\nabla \cdot \mathbf{B} = 0$ permits us to write

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.} \quad (2.23)$$

Faraday's law then gives

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \frac{\partial \mathbf{A}}{\partial t}, \quad (2.24)$$

so

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0. \quad (2.25)$$

A curl-free vector can be written as a gradient, hence

$$\boxed{\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}.} \quad (2.26)$$

These two equations define the scalar and vector potentials for time-dependent electromagnetism.

2.9 The coupled equations before choosing a gauge

Insert the potentials into Gauss's law:

$$\nabla \cdot \mathbf{E} = -\nabla^2 \phi - \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = \frac{\rho}{\epsilon_0}. \quad (2.27)$$

Therefore

$$\boxed{\nabla^2 \phi + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{\rho}{\epsilon_0}.} \quad (2.28)$$

The scalar potential is coupled to the divergence of the vector potential.

Now use Ampère–Maxwell law,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (2.29)$$

The left-hand side is

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}, \quad (2.30)$$

while

$$\frac{\partial \mathbf{E}}{\partial t} = -\nabla \frac{\partial \phi}{\partial t} - \frac{\partial^2 \mathbf{A}}{\partial t^2}. \quad (2.31)$$

Using $c^{-2} = \mu_0 \epsilon_0$ and rearranging gives

$$\boxed{\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) \mathbf{A} - \nabla \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t}\right) = -\mu_0 \mathbf{J}.} \quad (2.32)$$

At this stage ϕ and $\mathbf{A}$ are still coupled.

2.10 Gauge freedom and the Lorenz gauge

The potentials are not unique. For any sufficiently smooth scalar function $\chi(\mathbf{r}, t)$,

$$\boxed{\mathbf{A} \rightarrow \mathbf{A}' = \mathbf{A} + \nabla \chi, \quad \phi \rightarrow \phi' = \phi - \frac{\partial \chi}{\partial t}.} \quad (2.33)$$

These transformations leave both $\mathbf{E}$ and $\mathbf{B}$ unchanged. Gauge freedom therefore allows us to impose one convenient condition on the potentials without changing the physical fields.

For time-dependent problems the most useful choice is the Lorenz gauge,

$$\boxed{\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0.} \quad (2.34)$$

Substituting this condition into the coupled equations immediately gives

$$\boxed{\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) \phi = -\frac{\rho}{\epsilon_0},} \quad (2.35)$$

$$\boxed{\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) \mathbf{A} = -\mu_0 \mathbf{J}.} \quad (2.36)$$

The scalar and vector potentials now obey the *same* wave operator, with different sources.

Physical meaning

The Lorenz gauge is especially powerful because it makes the propagation structure of electromagnetism explicit: both potentials satisfy wave equations with speed c . The gauge choice does not create causality; Maxwell's equations already contain it. It simply exposes that causal structure in a form that can be solved cleanly with a Green's function.

2.11 The generic inhomogeneous wave equation

Both potential equations have the form

$$\square_w \psi(\mathbf{r}, t) = -f(\mathbf{r}, t), \quad \square_w \equiv \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}. \quad (2.37)$$

The operator $\square_w$ is often called the wave operator or d'Alembertian, subject to sign convention.

The analogy with the static problem is now very close:

	Static problem	Time-dependent problem
Equation	$\nabla^2 \psi = -f(\mathbf{r})$	$\square_w \psi = -f(\mathbf{r}, t)$
Point source	$\delta^{(3)}(\mathbf{r} - \mathbf{r}')$	$\delta^{(3)}(\mathbf{r} - \mathbf{r}') \delta(t - t')$
Green function	response to a source at $\mathbf{r}'$	response to a source event ($\mathbf{r}', t'$)
Extra data	spatial boundary conditions	spatial boundary conditions plus initial data

We therefore define the wave Green's function by

$$\square_w G(\mathbf{r}, t; \mathbf{r}', t') = -\delta^{(3)}(\mathbf{r} - \mathbf{r}') \delta(t - t'). \quad (2.38)$$

2.12 Why initial conditions now appear

Poisson's equation contains spatial derivatives but no time derivatives. It is a boundary-value problem. The wave equation contains a second time derivative, so a complete dynamical problem also requires initial data such as

$$\psi(\mathbf{r}, t_1), \quad \left. \frac{\partial \psi}{\partial t} \right|_{t=t_1}. \quad (2.39)$$

This is the mathematical reason the time-dependent Green-function construction in Lecture 3 contains temporal boundary terms in addition to spatial ones.

Another useful way to phrase the distinction is that Poisson's equation is *elliptic*, whereas the wave equation is *hyperbolic*. Elliptic equations communicate boundary information throughout a region; hyperbolic equations possess characteristics and propagate disturbances at a finite speed.

Checkpoint

The static and dynamical problems are not two unrelated techniques. The whole strategy is the same: identify the linear differential operator, find its point-source response with the correct boundary conditions, and superpose. What changes in the time-dependent problem is the operator itself, and with it the physical possibility of waves, retardation and causality.

Lecture 3: The Green's Function of the Wave Equation

Aim of this lecture

Generalise Green's second identity to spacetime, solve the free-space wave Green's function in Fourier space, and understand why the same differential equation admits both retarded and advanced solutions.

Motivation for this lecture

Once the potentials obey an inhomogeneous wave equation, the static Green's function is no longer enough. A point source must now be localised in both space and time, so the relevant inverse operator is a spacetime Green's function. There is also a new conceptual issue: the wave equation by itself does not say whether disturbances propagate from the past to the future or from the future to the past. Deriving the retarded and advanced Green's functions will show precisely where this choice enters and why physical electrodynamics selects the causal, retarded solution.

3.1 From Poisson's equation to the wave equation

In the static problem the basic differential operator was the Laplacian ∇^2 . For time-dependent fields the corresponding operator is

$$\square_w = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}. \quad (3.1)$$

We wish to solve

$$\square_w \psi(\mathbf{r}, t) = -f(\mathbf{r}, t) \quad (3.2)$$

subject to spatial boundary conditions and initial data. The Green's function is defined by the response to a point source in both space and time,

$$\square_w G(\mathbf{r}, t; \mathbf{r}', t') = -\delta^{(3)}(\mathbf{r} - \mathbf{r}') \delta(t - t'). \quad (3.3)$$

The extra delta function is important: the source now acts at a particular *event* $(\mathbf{r}', t')$, not merely at a particular point in space.

3.2 The spacetime version of Green's second identity

The static derivation began from

$$\psi \nabla'^2 G - G \nabla'^2 \psi = \nabla' \cdot (\psi \nabla' G - G \nabla' \psi).$$

For the wave operator we use the same identity for the spatial derivatives and integrate the time derivatives by parts. Start from

$$\psi \square'_w G - G \square'_w \psi = \psi \nabla'^2 G - G \nabla'^2 \psi \quad (3.4)$$

$$- \frac{1}{c^2} \left(\psi \frac{\partial^2 G}{\partial t'^2} - G \frac{\partial^2 \psi}{\partial t'^2} \right). \quad (3.5)$$

The spatial part is a divergence,

$$\psi \nabla'^2 G - G \nabla'^2 \psi = \nabla' \cdot (\psi \nabla' G - G \nabla' \psi), \quad (3.6)$$

whereas the time part is a total time derivative,

$$\psi \frac{\partial^2 G}{\partial t'^2} - G \frac{\partial^2 \psi}{\partial t'^2} = \frac{\partial}{\partial t'} \left(\psi \frac{\partial G}{\partial t'} - G \frac{\partial \psi}{\partial t'} \right). \quad (3.7)$$

Integrating over a spatial volume V and over $t_1 < t' < t_2$ therefore gives

$$\begin{aligned} & \int_{t_1}^{t_2} dt' \int_V d^3 r' (\psi \square'_w G - G \square'_w \psi) \\ &= \int_{t_1}^{t_2} dt' \oint_S d\mathbf{S}' \cdot (\psi \nabla' G - G \nabla' \psi) \\ & \quad - \frac{1}{c^2} \int_V d^3 r' \left[\psi \frac{\partial G}{\partial t'} - G \frac{\partial \psi}{\partial t'} \right]_{t_1}^{t_2}. \end{aligned} \quad (3.8)$$

Now insert $\square'_w G = -\delta^{(3)}(\mathbf{r} - \mathbf{r}')\delta(t - t')$ and $\square'_w \psi = -f$. The point-source term immediately produces $-\psi(\mathbf{r}, t)$, and hence

$$\begin{aligned} \psi(\mathbf{r}, t) &= \int_{t_1}^{t_2} dt' \int_V d^3 r' G f \\ & \quad - \int_{t_1}^{t_2} dt' \oint_S d\mathbf{S}' \cdot (\psi \nabla' G - G \nabla' \psi) \\ & \quad + \frac{1}{c^2} \int_V d^3 r' \left[\psi \frac{\partial G}{\partial t'} - G \frac{\partial \psi}{\partial t'} \right]_{t_1}^{t_2}. \end{aligned} \quad (3.9)$$

This is the direct time-dependent analogue of the static Green-function formula. Its three lines have three distinct meanings: response to the source, spatial boundary data, and initial/final data.

Physical meaning

The wave equation requires more information than Poisson's equation. A static boundary-value problem asks what happens everywhere in space once the boundary values are specified. A dynamical problem must additionally say what the field and its time derivative were initially. This is why the Green-function formula acquires temporal boundary terms.

3.3 Free space and translation invariance

In unbounded vacuum there is no preferred position or time, so the free-space Green's function can depend only on the differences

$$\mathbf{R} = \mathbf{r} - \mathbf{r}', \quad \tau = t - t'. \quad (3.10)$$

We therefore write the Fourier representation

$$G(\mathbf{R}, \tau) = \int \frac{d^3 k}{(2\pi)^3} \frac{d\omega}{2\pi} \tilde{G}(\mathbf{k}, \omega) e^{i\mathbf{k} \cdot \mathbf{R} - i\omega\tau}. \quad (3.11)$$

Acting on a Fourier mode gives

$$\nabla^2 \rightarrow -k^2, \quad -\frac{1}{c^2} \partial_t^2 \rightarrow +\frac{\omega^2}{c^2}, \quad (3.12)$$

so the Green-function equation becomes

$$\left(-k^2 + \frac{\omega^2}{c^2}\right) \tilde{G} = -1. \quad (3.13)$$

Hence

$$\boxed{\tilde{G}(\mathbf{k}, \omega) = \frac{1}{k^2 - \omega^2/c^2}}. \quad (3.14)$$

3.4 Why the Fourier expression is not yet the answer

The denominator vanishes at

$$\omega = \pm ck. \quad (3.15)$$

The inverse Fourier integral therefore passes through poles on the real ω axis. We must specify how the contour is displaced around them. That prescription is not a small mathematical detail: it supplies the temporal boundary condition.

For a *retarded* response we displace the poles so that G vanishes for $t < t'$. For an *advanced* response we choose the opposite prescription so that G vanishes for $t > t'$. Symbolically one often writes the denominator with an infinitesimal $i0$ prescription to remember which contour has been chosen.

Physical meaning

The differential equation itself is time-reversal symmetric. It does not by itself say that causes precede effects. The retarded solution becomes the physical one because we impose the causal boundary condition that there is no response before the source acts.

3.5 Retarded and advanced Green's functions

Carrying out the Fourier integrals gives two particularly important free-space solutions,

$$\boxed{G_{\text{ret}}(\mathbf{R}, \tau) = \frac{\delta(\tau - R/c)}{4\pi R}}, \quad (3.16)$$

$$\boxed{G_{\text{adv}}(\mathbf{R}, \tau) = \frac{\delta(\tau + R/c)}{4\pi R}}, \quad (3.17)$$

where $R = |\mathbf{R}|$.

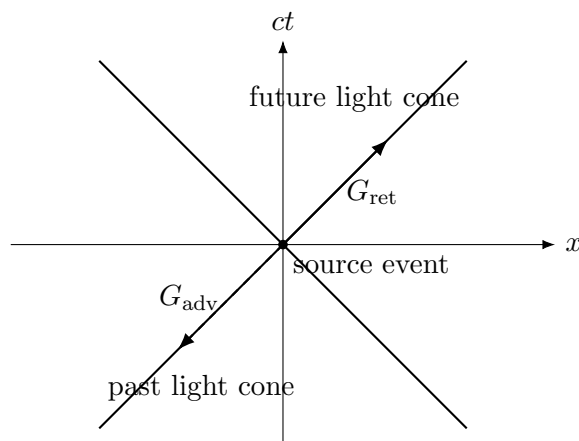
For the retarded Green's function the delta function has support only when

$$\tau = \frac{R}{c} > 0 \implies t = t' + \frac{R}{c} > t'. \quad (3.18)$$

A disturbance created at $(\mathbf{r}', t')$ reaches a point a distance R away only after the light-travel time R/c . The advanced Green's function instead has support at

$$t = t' - \frac{R}{c} < t', \quad (3.19)$$

which describes a response before the source event.



3.6 Support on the light cone

Notice something stronger than merely $t > t'$: in three spatial dimensions the free-space Green's function is proportional to

$$\delta\left(t - t' - \frac{R}{c}\right).$$

Thus a sharp disturbance propagates *on* the light cone at speed c . There is no contribution from points strictly inside the light cone for the ideal three-dimensional vacuum wave equation. This property is sometimes called Huygens' principle.

A useful dimensional check is

$$[G] \sim \frac{1}{R} \frac{1}{T},$$

which is exactly what is needed because $\delta^{(3)}(\mathbf{R})\delta(\tau)$ has dimensions $L^{-3}T^{-1}$ and $\square_w$ contributes L^{-2} .

3.7 The homogeneous solution and incoming radiation

The Green function constructs a particular solution generated by the source. We may always add a solution of the homogeneous equation,

$$\square_w \psi_{\text{hom}} = 0. \quad (3.20)$$

Physically this represents radiation that was already present independently of the source—for example an incoming electromagnetic wave. The usual statement “use the retarded solution with no incoming radiation” therefore contains two assumptions:

- (i) choose the retarded Green function; and
- (ii) set the independent incoming homogeneous solution to zero.

Checkpoint

Whenever a Green function is written down, ask two questions: what differential equation does it solve, and what boundary condition does it encode? The first determines the denominator in Fourier space; the second determines how the poles are treated and therefore whether the solution is retarded, advanced, outgoing, incoming, and so on.

Lecture 4: Retarded Waves, Potentials and Fields

Aim of this lecture

Use the retarded Green's function to obtain the retarded solution of the inhomogeneous wave equation, derive the electromagnetic retarded potentials, and differentiate them carefully to obtain the exact fields of arbitrary time-dependent sources.

Motivation for this lecture

The previous lecture produced the retarded Green's function formally. We now need to turn that mathematical object into the quantities that can actually be used in electrodynamics. Applying it to the scalar and vector potentials gives the retarded potentials, which express the finite propagation speed of electromagnetic influence. Differentiating those potentials then gives the electric and magnetic fields of arbitrary time-dependent charge and current distributions. These formulas are the starting point for essentially everything that follows about radiation.

4.1 The retarded solution of the wave equation

Consider

$$\square_w \psi(\mathbf{r}, t) = -f(\mathbf{r}, t) \quad (4.1)$$

in free space, with no incoming homogeneous wave. Substituting the retarded Green function gives

$$\psi_{\text{ret}}(\mathbf{r}, t) = \int d^3r' \int dt' f(\mathbf{r}', t') \frac{\delta(t - t' - R/c)}{4\pi R}, \quad R = |\mathbf{r} - \mathbf{r}'|. \quad (4.2)$$

The time integral is immediate. The argument of the delta function vanishes at

$$t' = t - \frac{R}{c}. \quad (4.3)$$

Because the derivative of $t - t' - R/c$ with respect to t' is simply -1 here—the source point $\mathbf{r}'$ is fixed during this integration—no additional Jacobian is generated. Thus

$$\psi_{\text{ret}}(\mathbf{r}, t) = \frac{1}{4\pi} \int d^3r' \frac{f(\mathbf{r}', t - R/c)}{R}. \quad (4.4)$$

We define

$$t_r = t - \frac{R}{c} \quad (4.5)$$

and call it the *retarded time* associated with the source point $\mathbf{r}'$ and observation point $\mathbf{r}$.

Physical meaning

The retarded time is not one universal time attached to the observation event. Different source points have different values of R , hence different retarded times. When an extended source is observed, we are literally seeing different parts of it at slightly different stages of their motion.

4.2 Wave equations for the electromagnetic potentials

Recall

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}. \quad (4.6)$$

In the Lorenz gauge,

$$\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0, \quad (4.7)$$

Maxwell's equations reduce to two inhomogeneous wave equations,

$$\boxed{\square_w \phi = -\frac{\rho}{\epsilon_0}}, \quad \boxed{\square_w \mathbf{A} = -\mu_0 \mathbf{J}}. \quad (4.8)$$

The scalar and vector potentials can therefore be obtained by applying the same Green-function solution component by component.

4.3 Retarded scalar and vector potentials

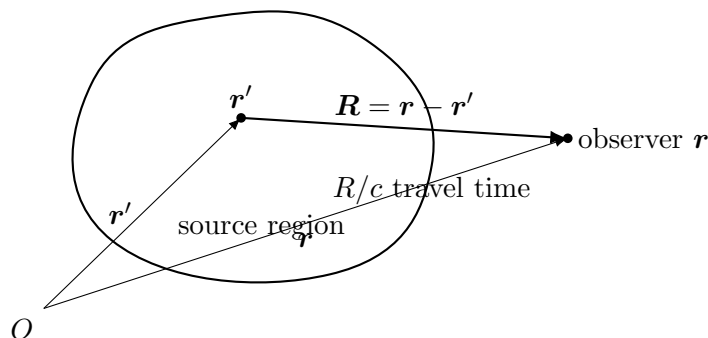
The exact free-space retarded potentials are

$$\boxed{\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{\rho(\mathbf{r}', t_r)}{R}}, \quad (4.9)$$

$$\boxed{\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int d^3r' \frac{\mathbf{J}(\mathbf{r}', t_r)}{R}}. \quad (4.10)$$

They look deceptively similar to the electrostatic and magnetostatic potentials. The decisive difference is that the sources are evaluated at t_r , not at the observation time t .

If the sources are truly static, then $\rho(\mathbf{r}', t_r) = \rho(\mathbf{r}')$ and $\mathbf{J}(\mathbf{r}', t_r) = \mathbf{J}(\mathbf{r}')$, so these expressions immediately reduce to the familiar Coulomb and magnetostatic potentials. Retardation therefore contains the static theory as a special case.



4.4 Differentiating a retarded quantity

The fields are obtained by differentiating the potentials, so we must be very precise about what depends on the observation point. For fixed source coordinate $\mathbf{r}'$,

$$R = |\mathbf{r} - \mathbf{r}'|, \quad \nabla R = \hat{\mathbf{n}}, \quad \nabla \frac{1}{R} = -\frac{\hat{\mathbf{n}}}{R^2}, \quad (4.11)$$

where

$$\hat{\mathbf{n}} = \frac{\mathbf{r} - \mathbf{r}'}{R}. \quad (4.12)$$

The retarded time obeys

$$\nabla t_r = -\frac{\hat{\mathbf{n}}}{c}, \quad \frac{\partial t_r}{\partial t} = 1. \quad (4.13)$$

Hence, for any source quantity $F(\mathbf{r}', t_r)$,

$$\boxed{\nabla F(\mathbf{r}', t_r) = -\frac{\hat{\mathbf{n}}}{c} \dot{F}(\mathbf{r}', t_r)}, \quad (4.14)$$

where the dot denotes differentiation with respect to the source-time argument.

As a concrete example,

$$\nabla \left(\frac{\rho(t_r)}{R} \right) = \rho(t_r) \nabla \frac{1}{R} + \frac{1}{R} \nabla \rho(t_r) \quad (4.15)$$

$$= -\frac{\rho(t_r)}{R^2} \hat{\mathbf{n}} - \frac{\dot{\rho}(t_r)}{cR} \hat{\mathbf{n}}. \quad (4.16)$$

This one line is the origin of two of the three terms in the exact electric field below.

4.5 Jefimenko's equations: exact fields of arbitrary sources

Using $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$ gives

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \left[\frac{\rho}{R^2} \hat{\mathbf{n}} + \frac{\dot{\rho}}{cR} \hat{\mathbf{n}} - \frac{\dot{\mathbf{J}}}{c^2 R} \right]_{\text{ret}}. \quad (4.17)$$

Thus

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \left[\frac{\rho \hat{\mathbf{n}}}{R^2} + \frac{\dot{\rho} \hat{\mathbf{n}}}{cR} - \frac{\dot{\mathbf{J}}}{c^2 R} \right]_{\text{ret}}}. \quad (4.18)$$

Likewise, since $\mathbf{B} = \nabla \times \mathbf{A}$,

$$\boxed{\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int d^3r' \left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR} \right]_{\text{ret}} \times \hat{\mathbf{n}}}. \quad (4.19)$$

These are Jefimenko's equations. They are exact consequences of Maxwell's equations for arbitrary localised sources in free space.

4.6 What each term means

The exact electric field contains three structures:

- (i) $[\rho]\hat{\mathbf{n}}/R^2$: the retarded Coulomb-like field;
- (ii) $[\dot{\rho}]\hat{\mathbf{n}}/(cR)$: a field generated by changing charge density;
- (iii) $-[\dot{\mathbf{J}}]/(c^2R)$: a field generated by changing current density.

Similarly, $\mathbf{B}$ has a retarded Biot–Savart-like term proportional to $1/R^2$ and a changing-current term proportional to $1/R$.

It would be wrong, however, to label every visible $1/R$ in the integrand as radiation without further thought. For an extended source the different contributions are linked by the continuity equation, and cancellations can occur. The cleanest way to identify radiation is to perform the far-zone expansion of the *complete* field and retain the net $1/r$ term. This is what we will do in the next lectures.

4.7 Near, induction and radiation zones

For a compact source of characteristic size a and angular frequency ω , define a characteristic wavelength

$$\lambda \sim \frac{c}{\omega}. \quad (4.20)$$

The spatial behaviour is then naturally divided into regimes:

Region	Typical condition	Physical character
Near zone	$r \ll \lambda$	quasistatic fields dominate; energy can be stored and returned to the source
Induction zone	$r \sim \lambda$	retardation is important and several powers of $1/r$ coexist
Radiation zone	$r \gg \lambda$	the transverse $1/r$ fields dominate and transport energy irreversibly outward

Because a sphere has area $4\pi r^2$, the asymptotic Poynting flux behaves as

$$P(r) \sim r^2 EB. \quad (4.21)$$

If $E, B \sim 1/r^2$, then $P(r) \sim 1/r^2 \rightarrow 0$. If $E, B \sim 1/r$, then $P(r)$ approaches a finite limit. This is the mathematical reason the $1/r$ fields are identified with radiation.

Physical meaning

The word *detachment* is useful. Near fields remain dynamically tied to the source: energy stored in them can flow back during another part of the cycle. A $1/r$ radiation field is different. Once formed, it propagates outward as an essentially free electromagnetic wave and carries finite energy through spheres of arbitrarily large radius.

4.8 A first consistency check

If the sources stop changing, $\dot{\rho} = 0$ and $\dot{\mathbf{J}} = 0$. Jefimenko's equations reduce to

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{\rho(\mathbf{r}')}{R^2} \hat{\mathbf{n}}, \quad (4.22)$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int d^3r' \frac{\mathbf{J}(\mathbf{r}')}{R^2} \times \hat{\mathbf{n}}, \quad (4.23)$$

which are exactly Coulomb's law and the Biot–Savart law. The time-dependent formalism therefore connects continuously to the static theory with which we began.

Checkpoint

There are now three equivalent levels of description available: Maxwell's differential equations, retarded potentials, and Jefimenko's integral equations. The potentials are usually the most efficient route for radiation calculations because they are easier to expand systematically in the far zone.

4.9 Worked example: a source that is switched on at one point

The retarded Green's function also gives a simple answer to the question: when does a distant observer first notice that a source has been switched on? Consider an idealised point source at the origin,

$$f(\mathbf{r}, t) = f_0 \Theta(t) \delta^{(3)}(\mathbf{r}), \quad (4.24)$$

where $\Theta(t)$ is the step function. In free space the retarded solution of

$$\square_w \psi = -f \quad (4.25)$$

is

$$\psi(\mathbf{r}, t) = \int d^3r' \frac{f(\mathbf{r}', t - |\mathbf{r} - \mathbf{r}'|/c)}{4\pi|\mathbf{r} - \mathbf{r}'|}. \quad (4.26)$$

The delta function fixes $\mathbf{r}' = 0$, hence

$$\psi(\mathbf{r}, t) = \frac{f_0}{4\pi r} \Theta\left(t - \frac{r}{c}\right). \quad (4.27)$$

The field is exactly zero until the light-travel time r/c has elapsed. This is the most elementary manifestation of retardation: the source changes at $t = 0$, but the observer at radius r cannot know about the change until $t = r/c$.

If the source is instead a short pulse,

$$f(\mathbf{r}, t) = f_0 \delta(t) \delta^{(3)}(\mathbf{r}), \quad (4.28)$$

then

$$\psi(\mathbf{r}, t) = \frac{f_0}{4\pi r} \delta\left(t - \frac{r}{c}\right). \quad (4.29)$$

The pulse remains sharp and propagates on the spherical light front $r = ct$, with amplitude proportional to $1/r$. The surface area of the sphere grows as r^2 , so a field amplitude falling as $1/r$ is precisely what is needed for a finite outward flux through a large sphere.

Lecture 5: Radiation from a Time-Dependent Electric Dipole

Aim of this lecture

Derive the fields of a time-dependent electric dipole, identify the three spatial zones, compute the angular distribution of radiated power, and obtain Larmor's formula as a special case.

Motivation for this lecture

Retarded fields exist for any time-dependent source, but not every part of those fields represents radiation that escapes to infinity. The simplest source for which the distinction can be seen clearly is a time-dependent electric dipole. Its exact fields contain terms falling as $1/r^3$, $1/r^2$ and $1/r$, allowing us to identify near, induction and radiation zones explicitly. The electric dipole is also universal: for a small neutral source it is usually the leading radiative multipole, so understanding it provides the prototype for antennas, atomic radiation and the general multipole expansion.

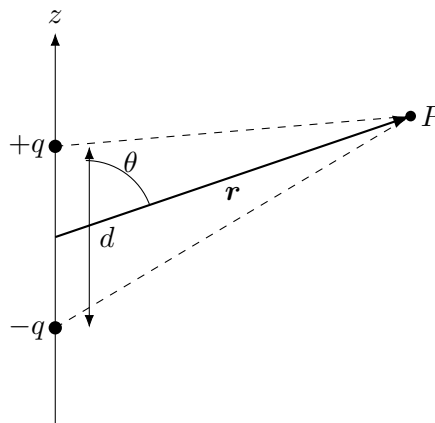
5.1 The electric dipole moment

Consider a localised neutral source with characteristic size d . Its electric dipole moment is

$$\mathbf{p}(t) = \int d^3r' \mathbf{r}' \rho(\mathbf{r}', t). \quad (5.1)$$

A concrete realisation is two opposite charges $\pm q$ separated by a small distance d along the z axis. If the separation oscillates harmonically,

$$\mathbf{p}(t) = p_0 \cos \omega t \hat{\mathbf{z}}, \quad p_0 = qd. \quad (5.2)$$



5.2 Exact dipole fields

For a point electric dipole at the origin with arbitrary time dependence, the exact fields away from the origin can be written compactly as

$$\boxed{\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon_0} \left\{ \frac{3\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{p}) - \mathbf{p}}{r^3} + \frac{3\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \dot{\mathbf{p}}) - \dot{\mathbf{p}}}{cr^2} + \frac{\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \ddot{\mathbf{p}})}{c^2 r} \right\}_{t_r}, \\ \mathbf{B}(\mathbf{r}, t) &= \frac{\mu_0}{4\pi} \left\{ \frac{\dot{\mathbf{p}} \times \hat{\mathbf{n}}}{r^2} + \frac{\ddot{\mathbf{p}} \times \hat{\mathbf{n}}}{cr} \right\}_{t_r}, \end{aligned}} \quad (5.3)$$

where now $t_r = t - r/c$ because the source is at the origin.

There are three distinct orders in $1/r$:

- $1/r^3$: the instantaneous-looking quasistatic dipole field, evaluated at retarded time;
- $1/r^2$: the induction field;
- $1/r$: the radiation field.

5.3 Radiation zone

At distances large compared with the wavelength scale,

$$r \gg \frac{c}{\omega},$$

we retain only the $1/r$ terms:

$$\boxed{\mathbf{E}_{\text{rad}} = \frac{1}{4\pi\epsilon_0 c^2} \frac{\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \ddot{\mathbf{p}}(t_r))}{r}}, \quad (5.4)$$

$$\boxed{\mathbf{B}_{\text{rad}} = \frac{1}{c} \hat{\mathbf{n}} \times \mathbf{E}_{\text{rad}}}. \quad (5.5)$$

Thus

$$\mathbf{E}_{\text{rad}} \perp \mathbf{B}_{\text{rad}} \perp \hat{\mathbf{n}}, \quad |\mathbf{E}_{\text{rad}}| = c|\mathbf{B}_{\text{rad}}|.$$

The radiation field locally looks like a plane electromagnetic wave even though the wavefronts are spherical.

For $\mathbf{p} = p_0 \cos \omega t \hat{\mathbf{z}}$,

$$\mathbf{E}_{\text{rad}} = \frac{p_0 \omega^2}{4\pi\epsilon_0 c^2} \frac{\sin \theta}{r} \cos \left[\omega \left(t - \frac{r}{c} \right) \right] \hat{\boldsymbol{\theta}}, \quad (5.6)$$

$$\mathbf{B}_{\text{rad}} = \frac{p_0 \omega^2}{4\pi\epsilon_0 c^3} \frac{\sin \theta}{r} \cos \left[\omega \left(t - \frac{r}{c} \right) \right] \hat{\boldsymbol{\phi}}. \quad (5.7)$$

5.4 Poynting vector and radiation pattern

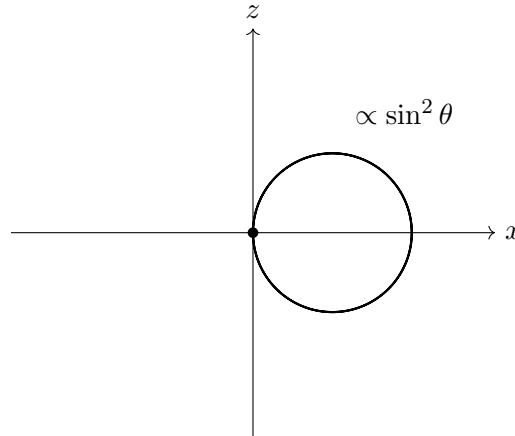
The far-zone Poynting vector is radial:

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E}_{\text{rad}} \times \mathbf{B}_{\text{rad}} = \epsilon_0 c E_{\text{rad}}^2 \hat{\mathbf{n}}.$$

The power per unit solid angle is therefore

$$\frac{dP}{d\Omega} = r^2 \mathbf{S} \cdot \hat{\mathbf{n}} = \frac{|\ddot{\mathbf{p}}(t_r)|^2}{16\pi^2\epsilon_0 c^3} \sin^2 \theta \quad (5.8)$$

when the polar axis is chosen along $\ddot{\mathbf{p}}$.



Radiation vanishes along the dipole axis and is maximal in the equatorial plane. Integrating over solid angle,

$$\int \sin^2 \theta d\Omega = \frac{8\pi}{3},$$

so

$$P(t_r) = \frac{|\ddot{\mathbf{p}}(t_r)|^2}{6\pi\epsilon_0 c^3}. \quad (5.9)$$

For harmonic motion, $\langle \ddot{p}^2 \rangle = p_0^2 \omega^4 / 2$, giving

$$\langle P \rangle = \frac{p_0^2 \omega^4}{12\pi\epsilon_0 c^3}. \quad (5.10)$$

5.5 Larmor's formula

For a nonrelativistic point charge, $\mathbf{p} = q\mathbf{r}_q$, hence $\ddot{\mathbf{p}} = q\mathbf{a}$. Therefore

$$P_{\text{Larmor}} = \frac{q^2 a^2}{6\pi\epsilon_0 c^3}. \quad (5.11)$$

Physical meaning

A charge does not radiate because it has a velocity; it radiates because its velocity changes. In the nonrelativistic regime the total radiated power depends only on a^2 . This is why both acceleration and deceleration generate radiation.

Lecture 6: Dipole Antennas and Magnetic Dipole Radiation

Aim of this lecture

Connect the ideal electric dipole to a short transmitting antenna, interpret the angular distribution and radiation resistance, derive the radiation of a small oscillating current loop, and compare electric and magnetic dipole radiation on equal footing.

Motivation for this lecture

The oscillating point dipole of Lecture 5 is an idealisation. To connect the theory with devices and experimentally realisable sources we should ask how a dipole is produced by an alternating current in a wire and how much power such an antenna emits. A small current loop provides the complementary magnetic-dipole source. Studying the two together reveals the close parallel between electric and magnetic dipole radiation, introduces quantities such as radiation resistance, and prepares us to recognise both contributions inside the radiation of an arbitrary localised source.

6.1 From an oscillating dipole to an antenna

The electric dipole of Lecture 5 was introduced as two opposite charges separated by a small distance. A real antenna provides a more physical way of creating the same time-dependent dipole moment: an alternating current moves charge back and forth along a conductor.

Consider a thin straight wire of length d oriented along z , with

$$d \ll \lambda. \quad (6.1)$$

Because the antenna is much shorter than a wavelength, the phase of the current does not vary appreciably over the wire. In the ideal *Hertzian dipole* model we therefore take the current to be approximately uniform along the short segment.

If charges $+q(t)$ and $-q(t)$ accumulate at the two ends, the electric dipole moment is

$$\mathbf{p}(t) = q(t)d \hat{\mathbf{z}}. \quad (6.2)$$

Charge conservation relates the endpoint charge to the current,

$$I(t) = \dot{q}(t), \quad (6.3)$$

so

$$\dot{\mathbf{p}}(t) = I(t)d \hat{\mathbf{z}}. \quad (6.4)$$

This identity is the bridge between the language of electric dipole moments and the language of antennas.

6.2 Radiation fields of the short dipole antenna

Take

$$I(t) = I_0 \cos \omega t. \quad (6.5)$$

Then one convenient phase choice is

$$p(t) = \frac{I_0 d}{\omega} \sin \omega t, \quad (6.6)$$

so

$$\ddot{p}(t) = -I_0 d \omega \sin \omega t. \quad (6.7)$$

The far-zone electric-dipole field from Lecture 5 is

$$\mathbf{E}_{\text{rad}} = \frac{1}{4\pi\epsilon_0 c^2 r} \hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times \ddot{\mathbf{p}}(t_r)]. \quad (6.8)$$

For a z -directed antenna this becomes

$$E_\theta(r, \theta, t) = \frac{I_0 d \omega}{4\pi\epsilon_0 c^2 r} \sin \theta \sin \left[\omega \left(t - \frac{r}{c} \right) \right], \quad (6.9)$$

up to the overall phase convention chosen for $I(t)$. The magnetic field is

$$B_\phi = \frac{1}{c} E_\theta. \quad (6.10)$$

Thus the wave is transverse: $\mathbf{E}$ points in the $\hat{\theta}$ direction, $\mathbf{B}$ in the $\hat{\phi}$ direction, and the energy propagates radially.

6.3 The antenna radiation pattern

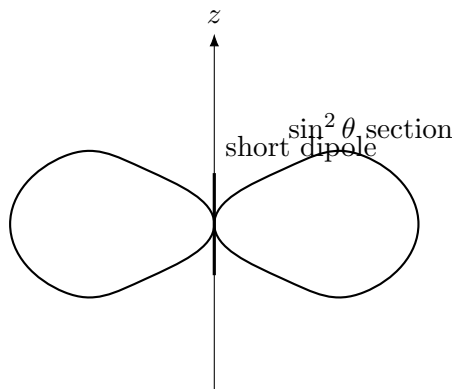
The time-averaged power per unit solid angle is

$$\left\langle \frac{dP}{d\Omega} \right\rangle = \frac{I_0^2 d^2 \omega^2}{32\pi^2 \epsilon_0 c^3} \sin^2 \theta. \quad (6.11)$$

The pattern has three immediate consequences:

- no power is radiated along the wire, $\theta = 0, \pi$;
- maximum power is radiated perpendicular to the wire, $\theta = \pi/2$;
- the pattern is independent of the azimuthal angle ϕ .

In three dimensions the familiar picture is a torus or “doughnut” surrounding the antenna axis.



Physical meaning

An antenna does not radiate most strongly in the direction in which it points. A short straight dipole radiates sideways. This follows directly from transversality: the radiation field depends on the component of $\ddot{\mathbf{p}}$ perpendicular to the line of sight.

6.4 Total power and radiation resistance

Using the dipole result

$$\langle P \rangle = \frac{p_0^2 \omega^4}{12\pi\epsilon_0 c^3} \quad (6.12)$$

and $p_0 = I_0 d / \omega$ gives

$$\langle P \rangle = \frac{I_0^2 d^2 \omega^2}{12\pi\epsilon_0 c^3}. \quad (6.13)$$

Antenna theory often rewrites this as though the radiated power were dissipated in an effective resistance,

$$\langle P \rangle = \frac{1}{2} I_0^2 R_{\text{rad}}, \quad (6.14)$$

where I_0 is the peak current. Thus for the ideal Hertzian dipole

$$R_{\text{rad}} = \frac{d^2 \omega^2}{6\pi\epsilon_0 c^3} \quad (6.15)$$

$$= \frac{2\pi}{3} Z_0 \left(\frac{d}{\lambda} \right)^2 \quad (6.16)$$

$$= 80\pi^2 \left(\frac{d}{\lambda} \right)^2 \Omega, \quad (6.17)$$

where $Z_0 = \sqrt{\mu_0/\epsilon_0} \simeq 377 \Omega$ is the impedance of free space.

This effective resistance represents real power carried away by electromagnetic waves. It is not Joule heating in the wire.

Notation and convention

The coefficient above is for the ideal Hertzian dipole with approximately uniform current over the length d . A practical short centre-fed wire has a current that falls toward its ends and therefore has a different numerical radiation resistance. The $\propto (d/\lambda)^2$ scaling is the robust point.

6.5 A small current loop as a magnetic dipole

Now consider an electrically neutral circular loop of radius b carrying

$$I(t) = I_0 \cos \omega t. \quad (6.18)$$

Its magnetic dipole moment is

$$\mathbf{m}(t) = I(t) \pi b^2 \hat{\mathbf{z}} = m_0 \cos \omega t \hat{\mathbf{z}}, \quad (6.19)$$

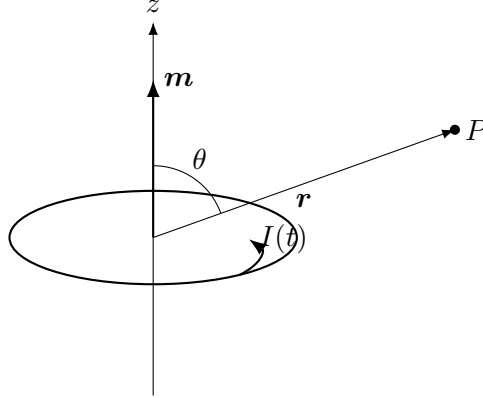
with

$$m_0 = \pi b^2 I_0. \quad (6.20)$$

The relevant approximations are

$$b \ll r, \quad b \ll \lambda, \quad r \gg \lambda \quad (6.21)$$

for a small loop observed in the radiation zone.



6.6 Retarded vector potential of the loop

Because the loop is neutral, its scalar potential has no leading monopole contribution. The vector potential is

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \oint \frac{I(t - R/c)}{R} d\ell'. \quad (6.22)$$

Expanding R and the retarded time across the small loop, the zeroth-order term vanishes because

$$\oint d\ell' = 0. \quad (6.23)$$

The first surviving term is proportional to the loop area and can be written entirely in terms of $\mathbf{m}$:

$$\boxed{\mathbf{A}_{M1}(\mathbf{r}, t) = \frac{\mu_0}{4\pi cr} \dot{\mathbf{m}}(t_r) \times \hat{\mathbf{n}}.} \quad (6.24)$$

For $\mathbf{m} = m_0 \cos \omega t \hat{\mathbf{z}}$,

$$\boxed{A_\phi = -\frac{\mu_0 m_0 \omega}{4\pi cr} \sin \theta \sin \left[\omega \left(t - \frac{r}{c} \right) \right].} \quad (6.25)$$

6.7 Magnetic-dipole radiation fields

Differentiating the potential gives

$$\boxed{\mathbf{B}_{\text{rad}} = \frac{\mu_0}{4\pi c^2 r} \hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times \ddot{\mathbf{m}}(t_r)],} \quad (6.26)$$

$$\boxed{\mathbf{E}_{\text{rad}} = -c \hat{\mathbf{n}} \times \mathbf{B}_{\text{rad}}.} \quad (6.27)$$

For a z -directed harmonic magnetic dipole,

$$E_\phi = \frac{\mu_0 m_0 \omega^2}{4\pi c} \frac{\sin \theta}{r} \cos \left[\omega \left(t - \frac{r}{c} \right) \right], \quad (6.28)$$

$$B_\theta = -\frac{1}{c} E_\phi. \quad (6.29)$$

Again the power pattern is proportional to $\sin^2 \theta$.

6.8 Radiated power of a magnetic dipole

The angular distribution is

$$\frac{dP_m}{d\Omega} = \frac{\mu_0 |\ddot{\mathbf{m}}|^2}{16\pi^2 c^3} \sin^2 \theta, \quad (6.30)$$

and the instantaneous total power is

$$P_m = \frac{\mu_0 |\ddot{\mathbf{m}}|^2}{6\pi c^3}. \quad (6.31)$$

For harmonic motion,

$$\langle P \rangle_m = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}. \quad (6.32)$$

6.9 Electric versus magnetic dipoles

The two dipoles have the same $\sin^2 \theta$ angular power pattern but different polarisation:

Source along z	electric field	magnetic field
electric dipole $\mathbf{p}$	$\hat{\theta}$	$\hat{\phi}$
magnetic dipole $\mathbf{m}$	$\hat{\phi}$	$-\hat{\theta}$

This is a manifestation of electric–magnetic duality in the source-free radiation zone.

Checkpoint

A short straight wire and a small current loop are the canonical electric- and magnetic-dipole antennas. Their far-zone patterns are identical in intensity, but the electric and magnetic polarisations are interchanged. The difference between them is therefore encoded in the source moment and polarisation, not in the gross doughnut-shaped radiation pattern.

Lecture 7: Radiation from an Arbitrary Localised Source

Aim of this lecture

Understand why a distant observer does not need the microscopic details of a compact source, expand the exact retarded potentials in the two small parameters a/r and $\omega a/c$, derive the electric-monopole and electric-dipole terms carefully, and set up the tensor structure that produces magnetic-dipole and electric-quadrupole radiation in Lecture 8.

Motivation for this lecture

So far we have analysed specially chosen sources for which the answer can be obtained almost by inspection. A real source may have an arbitrary charge and current distribution, and its lowest dipole moment may even vanish. We therefore need a systematic approximation scheme rather than a new calculation for every geometry. When the source is small compared with both the observation distance and the emitted wavelength, the retarded potentials can be expanded in powers of the source size. This is the origin of the multipole expansion and of the hierarchy of electric dipole, magnetic dipole, electric quadrupole, and higher radiation.

7.1 Why return to an arbitrary source?

Lectures 5 and 6 studied two special radiators: an oscillating electric dipole and an oscillating current loop. Those examples already revealed several universal features of radiation: the radiation fields fall as $1/r$, they are transverse, and their angular patterns reflect the symmetry of the source.

We now ask a more powerful question. Suppose the charge and current distributions inside some compact region are completely arbitrary. What can a distant observer learn about them?

The exact retarded potentials are

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{\rho(\mathbf{r}', t - R/c)}{R}, \quad (7.1)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int d^3r' \frac{\mathbf{J}(\mathbf{r}', t - R/c)}{R}, \quad (7.2)$$

where

$$\mathbf{R} = \mathbf{r} - \mathbf{r}', \quad R = |\mathbf{R}|. \quad (7.3)$$

These formulas are exact, but for a complicated source they are not yet very illuminating. The multipole expansion extracts the small amount of source information that survives at long distance.

Physical meaning

A distant observer cannot resolve arbitrarily fine spatial structure. If the source is much smaller than the emitted wavelength, all parts of the source are seen almost in phase. The observer first sees the total charge, then the electric dipole moment, then finer moments such as the magnetic dipole and electric quadrupole. The multipole expansion is therefore a hierarchy of spatial resolution.

7.2 Geometry of a localised source

Let the source be confined to a region of characteristic size a around the origin, so that

$$r' \lesssim a. \quad (7.4)$$

The observation point lies far away,

$$a \ll r. \quad (7.5)$$

Write

$$\hat{\mathbf{n}} = \frac{\mathbf{r}}{r}. \quad (7.6)$$

Then

$$R^2 = |\mathbf{r} - \mathbf{r}'|^2 = r^2 + r'^2 - 2\mathbf{r} \cdot \mathbf{r}' \quad (7.7)$$

$$= r^2 \left[1 + \frac{r'^2}{r^2} - 2\frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{r} \right]. \quad (7.8)$$

Hence

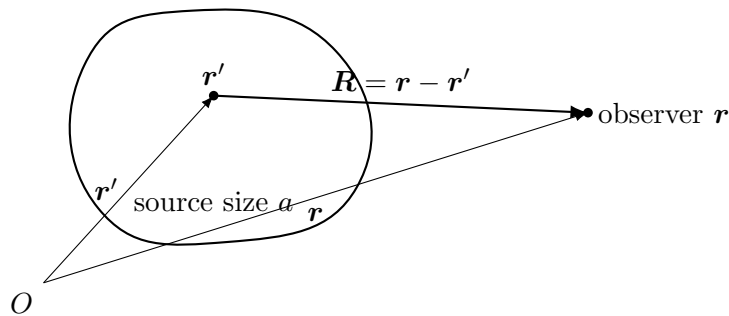
$$R = r \sqrt{1 + \frac{r'^2}{r^2} - 2\frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{r}}. \quad (7.9)$$

For $a/r \ll 1$, use $\sqrt{1+x} = 1 + x/2 + \mathcal{O}(x^2)$ to obtain

$$R = r - \hat{\mathbf{n}} \cdot \mathbf{r}' + \frac{r'^2 - (\hat{\mathbf{n}} \cdot \mathbf{r}')^2}{2r} + \mathcal{O}\left(\frac{a^3}{r^2}\right). \quad (7.10)$$

To first order,

$$R \simeq r - \hat{\mathbf{n}} \cdot \mathbf{r}'. \quad (7.11)$$



The same expansion gives

$$\frac{1}{R} = \frac{1}{r} \frac{1}{1 - (\hat{\mathbf{n}} \cdot \mathbf{r}')/r + \dots} \quad (7.12)$$

$$= \frac{1}{r} + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{r^2} + \mathcal{O}\left(\frac{a^2}{r^3}\right). \quad (7.13)$$

This is the first of the two expansions we shall need.

7.3 Retardation across the source

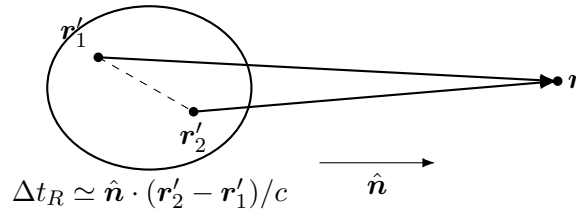
The retarded time associated with the source point $\mathbf{r}'$ is

$$t_R = t - \frac{R}{c}. \quad (7.14)$$

Using $R \simeq r - \hat{\mathbf{n}} \cdot \mathbf{r}'$,

$$t_R \simeq t_0 + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c}, \quad t_0 \equiv t - \frac{r}{c}. \quad (7.15)$$

The time t_0 is the retarded time from the origin. Different parts of the source are seen at slightly different source times, separated by an amount of order a/c .



Suppose the source varies on a characteristic timescale T or angular frequency $\omega \sim 1/T$. A Taylor expansion in the retardation difference is valid if

$$\frac{a/c}{T} \ll 1, \quad (7.16)$$

or equivalently

$$ka \equiv \frac{\omega a}{c} \ll 1. \quad (7.17)$$

For a monochromatic source with wavelength $\lambda = 2\pi c/\omega$,

$$ka = \frac{2\pi a}{\lambda}. \quad (7.18)$$

Thus the source must be small compared with a wavelength. More precisely, the expansion parameter is ka , not merely a/λ with the factor 2π forgotten.

7.4 Two different small parameters

It is essential not to confuse

$$\frac{a}{r} \ll 1 \quad (7.19)$$

with

$$ka = \frac{\omega a}{c} \ll 1. \quad (7.20)$$

They answer different questions.

- $a/r \ll 1$ says that the observer is far enough away that rays from different parts of the source are almost parallel and the geometric factor $1/R$ can be expanded.
- $ka \ll 1$ says that the light-travel time across the source is short compared with the timescale on which the source changes, so the retarded source can be Taylor expanded.

Later, when we want the *radiation zone*, we will impose a third condition,

$$r \gg \frac{c}{\omega}, \quad (7.21)$$

so that the $1/r$ fields dominate over the near and induction fields. A source can be spatially compact, $a \ll r$, without the observer yet being in the radiation zone.

Physical meaning

There are three length scales in the problem: source size a , wavelength λ , and observation distance r . The multipole expansion requires $a \ll r$ and $a \ll \lambda$; the radiation-zone simplification additionally requires $\lambda \ll r$ (up to factors of 2π depending on whether one uses λ or c/ω).

7.5 Systematic expansion of the scalar potential

Start from

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{\rho(\mathbf{r}', t_R)}{R}. \quad (7.22)$$

The source term is expanded about t_0 :

$$\rho(\mathbf{r}', t_R) = \rho\left(\mathbf{r}', t_0 + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c} + \dots\right) \quad (7.23)$$

$$= \rho(\mathbf{r}', t_0) + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c} \dot{\rho}(\mathbf{r}', t_0) + \frac{1}{2} \left(\frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c}\right)^2 \ddot{\rho}(\mathbf{r}', t_0) + \dots \quad (7.24)$$

To first order in a/r and ka , multiply this by

$$\frac{1}{R} \simeq \frac{1}{r} + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{r^2}. \quad (7.25)$$

Keeping only the required terms,

$$\frac{\rho(\mathbf{r}', t_R)}{R} \simeq \frac{\rho(\mathbf{r}', t_0)}{r} + \frac{(\hat{\mathbf{n}} \cdot \mathbf{r}')\rho(\mathbf{r}', t_0)}{r^2} + \frac{(\hat{\mathbf{n}} \cdot \mathbf{r}')\dot{\rho}(\mathbf{r}', t_0)}{cr} + \dots \quad (7.26)$$

Now define the total charge

$$Q(t) = \int d^3r' \rho(\mathbf{r}', t) \quad (7.27)$$

and the electric dipole moment

$$\mathbf{p}(t) = \int d^3r' \mathbf{r}' \rho(\mathbf{r}', t). \quad (7.28)$$

Then

$$\int d^3r' (\hat{\mathbf{n}} \cdot \mathbf{r}') \rho = \hat{\mathbf{n}} \cdot \mathbf{p}, \quad (7.29)$$

and

$$\int d^3r' (\hat{\mathbf{n}} \cdot \mathbf{r}') \dot{\rho} = \hat{\mathbf{n}} \cdot \dot{\mathbf{p}}. \quad (7.30)$$

Therefore

$$\phi(\mathbf{r}, t) \simeq \frac{1}{4\pi\epsilon_0} \left[\frac{Q(t_0)}{r} + \frac{\hat{\mathbf{n}} \cdot \mathbf{p}(t_0)}{r^2} + \frac{\hat{\mathbf{n}} \cdot \dot{\mathbf{p}}(t_0)}{cr} + \dots \right]. \quad (7.31)$$

Each term has a clear origin:

- (i) Q/r is the electric monopole term;
- (ii) $(\hat{\mathbf{n}} \cdot \mathbf{p})/r^2$ is the familiar quasistatic electric-dipole potential;
- (iii) $(\hat{\mathbf{n}} \cdot \dot{\mathbf{p}})/(cr)$ is a retardation correction that falls as $1/r$.

The appearance of a $1/r$ term in ϕ does not by itself prove that this term is radiation. The potential is gauge dependent, and the physical electric field combines derivatives of both ϕ and $\mathbf{A}$. Radiation must be identified from the complete fields or the Poynting flux.

7.6 Why there is no electric-monopole radiation

For a localised source, charge conservation gives

$$\frac{\partial \rho}{\partial t} + \nabla' \cdot \mathbf{J} = 0. \quad (7.32)$$

Integrating over all space,

$$\dot{Q} = \int d^3r' \dot{\rho} = - \int d^3r' \nabla' \cdot \mathbf{J} \quad (7.33)$$

$$= - \oint_{S_\infty} d\mathbf{S}' \cdot \mathbf{J}. \quad (7.34)$$

For a localised source, $\mathbf{J}$ vanishes at infinity, so

$$\boxed{\dot{Q} = 0.} \quad (7.35)$$

The total charge may be nonzero, but it is constant. Hence the monopole term is Coulombic and cannot carry time-dependent radiation.

Physical meaning

This is the first example of a general rule: conservation laws eliminate possible radiation channels. Electric monopole radiation would require the total charge of an isolated system to oscillate in time, which charge conservation forbids. The first possible electric radiation is therefore dipolar.

7.7 Expansion of the vector potential

Now start from

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int d^3r' \frac{\mathbf{J}(\mathbf{r}', t_R)}{R}. \quad (7.36)$$

Expanding both the geometric factor and the retarded current gives, in Cartesian components,

$$\begin{aligned} A_j(\mathbf{r}, t) \simeq \frac{\mu_0}{4\pi} & \left[\frac{1}{r} \int d^3r' J_j(\mathbf{r}', t_0) + \frac{n_i}{r^2} \int d^3r' r'_i J_j(\mathbf{r}', t_0) \right. \\ & \left. + \frac{n_i}{cr} \frac{d}{dt_0} \int d^3r' r'_i J_j(\mathbf{r}', t_0) + \cdots \right]. \end{aligned} \quad (7.37)$$

The first integral can be related directly to the electric dipole moment.

Write the continuity equation in components,

$$\dot{\rho} = -\partial_j J_j. \quad (7.38)$$

Then

$$\dot{p}_i = \int d^3r' r'_i \dot{\rho} \quad (7.39)$$

$$= - \int d^3r' r'_i \partial'_j J_j. \quad (7.40)$$

Use the product rule,

$$\partial'_j(r'_i J_j) = \delta_{ij} J_j + r'_i \partial'_j J_j = J_i + r'_i \partial'_j J_j, \quad (7.41)$$

so

$$r'_i \partial'_j J_j = \partial'_j(r'_i J_j) - J_i. \quad (7.42)$$

Therefore

$$\dot{p}_i = - \int d^3r' \partial'_j(r'_i J_j) + \int d^3r' J_i \quad (7.43)$$

$$= - \oint_{S_\infty} dS'_j r'_i J_j + \int d^3r' J_i. \quad (7.44)$$

The surface term vanishes for a localised source, giving the important identity

$$\int d^3r' \mathbf{J}(\mathbf{r}', t) = \dot{\mathbf{p}}(t). \quad (7.45)$$

Hence the leading vector potential is

$$\mathbf{A}_{E1}(\mathbf{r}, t) = \frac{\mu_0}{4\pi r} \dot{\mathbf{p}}(t_0). \quad (7.46)$$

This is exactly the electric-dipole vector potential encountered earlier, now derived for an arbitrary localised source rather than for a particular pair of charges.

7.8 What is hiding in the next term?

Define

$$I_{ij}(t) = \int d^3r' r'_i r'_j J_j(\mathbf{r}', t). \quad (7.47)$$

Then the vector-potential expansion can be written schematically as

$$A_j \simeq \frac{\mu_0}{4\pi} \left[\frac{\dot{p}_j}{r} + \frac{n_i I_{ij}}{r^2} + \frac{n_i \dot{I}_{ij}}{cr} + \dots \right]_{t_0}. \quad (7.48)$$

This equation is worth reading slowly.

The first correction, $n_i I_{ij}/r^2$, falls faster than $1/r$ and therefore does not contribute to the asymptotic radiated power. The second correction, $n_i \dot{I}_{ij}/(cr)$, still falls as $1/r$ and therefore contains the next radiation channels. The object I_{ij} has two Cartesian indices, so it is a tensor rather than a vector. In Lecture 8 we will split it into symmetric and antisymmetric parts and discover that they are precisely the electric quadrupole and magnetic dipole moments.

7.9 Order counting and the multipole hierarchy

For a source oscillating at frequency ω , the leading electric-dipole vector potential scales as

$$A_{E1} \sim \frac{\mu_0}{4\pi r} \omega p. \quad (7.49)$$

Since I_{ij} contains one extra factor of source coordinate, we expect roughly

$$I \sim a \int J d^3 r' \sim a \omega p. \quad (7.50)$$

Therefore the next radiative term scales as

$$A_{\text{next}} \sim \frac{\mu_0}{4\pi c r} \omega I \sim A_{E1} \frac{\omega a}{c}. \quad (7.51)$$

Thus

$$\boxed{\frac{A_{\text{next}}}{A_{E1}} \sim ka \ll 1.} \quad (7.52)$$

This is why electric-dipole radiation is normally dominant, while magnetic-dipole and electric-quadrupole radiation are smaller by one power of ka in field amplitude, unless symmetry forces the electric dipole moment to vanish.

Checkpoint

At this point the logic of the multipole expansion is complete even though the algebra is not. The exact retarded potentials depend on the entire functions $\rho(\mathbf{r}', t)$ and $\mathbf{J}(\mathbf{r}', t)$. Far away and at long wavelength, that information collapses into a hierarchy of moments. Charge conservation removes electric-monopole radiation; the leading $1/r$ term is controlled by $\dot{\mathbf{p}}$; the next $1/r$ term contains the tensor I_{ij} , which Lecture 8 will decompose into magnetic-dipole and electric-quadrupole pieces.

Lecture 8: Cartesian Multipole Radiation

Aim of this lecture

Complete the long-wavelength expansion of an arbitrary localised source, show explicitly how electric-dipole, magnetic-dipole and electric-quadrupole moments emerge, and understand the hierarchy and symmetry of multipole radiation.

Motivation for this lecture

The leading electric-dipole term does not always tell the whole story. Symmetry may force the total dipole moment to vanish, or the next corrections may be needed for quantitative accuracy. The arbitrary-source expansion of Lecture 7 has therefore left us with a tensorial current moment whose physical content is not yet obvious. Decomposing that tensor into symmetric and antisymmetric pieces will uncover two distinct next-order sources: the electric quadrupole and the magnetic dipole. This shows how the familiar multipoles emerge from the general retarded-potential formalism rather than being introduced as unrelated constructions.

8.1 Where we left the expansion

For a source confined to a region of size a with $a \ll r$ and $a \ll \lambda$, Lecture 7 gave

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi r} \int d^3r' \mathbf{J}\left(\mathbf{r}', t_0 + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c}\right) + \dots, \quad t_0 = t - \frac{r}{c}. \quad (8.1)$$

Taylor expanding the current in the small retardation across the source gives

$$\mathbf{J}\left(\mathbf{r}', t_0 + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c}\right) = \mathbf{J}(\mathbf{r}', t_0) + \frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c} \dot{\mathbf{J}}(\mathbf{r}', t_0) + \frac{1}{2} \left(\frac{\hat{\mathbf{n}} \cdot \mathbf{r}'}{c}\right)^2 \ddot{\mathbf{J}}(\mathbf{r}', t_0) + \dots \quad (8.2)$$

The first term produced the electric-dipole vector potential,

$$\mathbf{A}_{E1}(\mathbf{r}, t) = \frac{\mu_0}{4\pi r} \dot{\mathbf{p}}(t_0), \quad (8.3)$$

because $\int \mathbf{J} d^3r' = \dot{\mathbf{p}}$.

The next term contains the tensor integral

$$I_{ij}(t) = \int d^3r' r'_i J_j(\mathbf{r}', t). \quad (8.4)$$

Understanding the second multipole order therefore reduces to understanding the symmetric and antisymmetric parts of I_{ij} .

8.2 Why a tensor appears at the next order

The first-order retardation correction to the vector potential is

$$A_j^{(1)}(\mathbf{r}, t) = \frac{\mu_0}{4\pi cr} \int d^3r' (\hat{\mathbf{n}} \cdot \mathbf{r}') \dot{J}_j(\mathbf{r}', t_0) \quad (8.5)$$

$$= \frac{\mu_0}{4\pi cr} n_i \dot{I}_{ij}(t_0). \quad (8.6)$$

Repeated Cartesian indices are summed. Now decompose

$$I_{ij} = \frac{1}{2}(I_{ij} + I_{ji}) + \frac{1}{2}(I_{ij} - I_{ji}). \quad (8.7)$$

The two pieces have different physical meanings.

8.3 The symmetric part and the electric quadrupole

Define the second moment of the charge distribution

$$M_{ij} = \int d^3r' \rho(\mathbf{r}', t) r'_i r'_j. \quad (8.8)$$

Using the continuity equation $\dot{\rho} = -\nabla' \cdot \mathbf{J}$,

$$\dot{M}_{ij} = - \int d^3r' r'_i r'_j \partial'_k J_k \quad (8.9)$$

$$= \int d^3r' J_k \partial'_k (r'_i r'_j), \quad (8.10)$$

where the surface term vanishes for a localised source. Since

$$\partial'_k (r'_i r'_j) = \delta_{ki} r'_j + \delta_{kj} r'_i, \quad (8.11)$$

we obtain

$$\dot{M}_{ij} = I_{ij} + I_{ji}. \quad (8.12)$$

Thus the symmetric part of I_{ij} is one half of $\dot{M}_{ij}$.

It is conventional to remove the trace and define the traceless electric quadrupole tensor

$$Q_{ij} = \int d^3r' \rho(\mathbf{r}', t) (3r'_i r'_j - r'^2 \delta_{ij}). \quad (8.13)$$

The trace does not contribute to the transverse far-zone radiation, so Q_{ij} contains precisely the physically useful quadrupolar information.

8.4 The antisymmetric part and the magnetic dipole

Define

$$\mathbf{m}(t) = \frac{1}{2} \int d^3r' \mathbf{r}' \times \mathbf{J}(\mathbf{r}', t). \quad (8.14)$$

In components,

$$m_k = \frac{1}{2} \epsilon_{k\ell m} I_{\ell m}. \quad (8.15)$$

Multiplying by ϵ_{ijk} and using

$$\epsilon_{ijk} \epsilon_{k\ell m} = \delta_{i\ell} \delta_{jm} - \delta_{im} \delta_{j\ell} \quad (8.16)$$

gives

$$\frac{1}{2} (I_{ij} - I_{ji}) = \epsilon_{ijk} m_k. \quad (8.17)$$

Therefore the very same next-order term in the current expansion automatically splits into an electric-quadrupole contribution and a magnetic-dipole contribution.

Physical meaning

The multipoles are not arbitrary labels attached after the calculation. They arise from irreducible pieces of the spatial moments of the source: scalar total charge, vector electric dipole, antisymmetric current moment (magnetic dipole), symmetric traceless charge moment (electric quadrupole), and so on.

8.5 Far-zone electric-dipole fields

From

$$\mathbf{A}_{E1} = \frac{\mu_0}{4\pi r} \dot{\mathbf{p}}(t_0), \quad (8.18)$$

the leading curl comes from differentiating the retarded time. Since

$$\nabla t_0 = -\frac{\hat{\mathbf{n}}}{c}, \quad (8.19)$$

we have

$$\mathbf{B}_{E1} = \nabla \times \mathbf{A}_{E1} \quad (8.20)$$

$$\simeq -\frac{\mu_0}{4\pi r} (\nabla t_0) \times \dot{\mathbf{p}}(t_0) \quad (8.21)$$

$$= \boxed{\frac{\mu_0}{4\pi cr} \ddot{\mathbf{p}}(t_0) \times \hat{\mathbf{n}}}. \quad (8.22)$$

The radiation fields must form a transverse electromagnetic wave, hence

$$\boxed{\mathbf{E}_{E1} = c\mathbf{B}_{E1} \times \hat{\mathbf{n}} = \frac{1}{4\pi\epsilon_0 c^2 r} \hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times \ddot{\mathbf{p}}(t_0)]}. \quad (8.23)$$

This recovers the exact far-zone result of Lecture 5 without assuming a particular two-charge model.

The angular power distribution is

$$\frac{dP_{E1}}{d\Omega} = \frac{1}{16\pi^2\epsilon_0 c^3} |\hat{\mathbf{n}} \times \ddot{\mathbf{p}}(t_0)|^2, \quad (8.24)$$

and integration over solid angle gives

$$\boxed{P_{E1} = \frac{|\ddot{\mathbf{p}}|^2}{6\pi\epsilon_0 c^3}}. \quad (8.25)$$

For a sinusoidal dipole $\mathbf{p} = \mathbf{p}_0 \cos \omega t$, the time-averaged power is therefore proportional to $\omega^4 p_0^2$.

8.6 Far-zone magnetic-dipole fields

The antisymmetric part of I_{ij} gives the leading magnetic-dipole vector potential

$$\boxed{\mathbf{A}_{M1}(\mathbf{r}, t) = \frac{\mu_0}{4\pi cr} \dot{\mathbf{m}}(t_0) \times \hat{\mathbf{n}}}. \quad (8.26)$$

Differentiating once more through t_0 gives the radiation fields. Their electric and magnetic roles are interchanged relative to the electric dipole. A convenient form is

$$\boxed{\mathbf{B}_{M1} = \frac{\mu_0}{4\pi c^2 r} \hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times \ddot{\mathbf{m}}(t_0)]}, \quad (8.27)$$

$$\boxed{\mathbf{E}_{M1} = -c \hat{\mathbf{n}} \times \mathbf{B}_{M1}.} \quad (8.28)$$

The total instantaneous power is

$$\boxed{P_{M1} = \frac{\mu_0}{6\pi c^3} |\ddot{\mathbf{m}}|^2.} \quad (8.29)$$

For the oscillating current loop of Lecture 6 this reproduces the earlier result.

8.7 Electric quadrupole radiation

The symmetric traceless part produces electric-quadrupole radiation. Its far-zone electric field is proportional to the third time derivative of the quadrupole tensor,

$$\mathbf{E}_{E2} \propto \frac{1}{c^3 r} \hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times (\ddot{\mathbf{Q}}(t_0) \cdot \hat{\mathbf{n}})]. \quad (8.30)$$

With the convention for Q_{ij} used above, the full coefficient is

$$\boxed{\mathbf{E}_{E2} = \frac{1}{24\pi\epsilon_0 c^3 r} \hat{\mathbf{n}} \times [\hat{\mathbf{n}} \times (\ddot{\mathbf{Q}}(t_0) \cdot \hat{\mathbf{n}})].} \quad (8.31)$$

The important feature for the present course is the extra factor of source size divided by wavelength compared with electric-dipole radiation.

8.8 Why M1 and E2 are both one order smaller than E1

Suppose the source oscillates at frequency ω and has characteristic size a . Very roughly,

$$\dot{\mathbf{p}} \sim \omega \mathbf{p}, \quad \mathbf{m} \sim a \dot{\mathbf{p}} \sim \omega a \mathbf{p}, \quad Q \sim a p. \quad (8.32)$$

Therefore

$$\frac{E_{M1}}{E_{E1}} \sim \frac{\omega a}{c} = ka, \quad \frac{E_{E2}}{E_{E1}} \sim \frac{\omega a}{c} = ka, \quad (8.33)$$

where

$$\boxed{ka = \frac{\omega a}{c} = \frac{2\pi a}{\lambda} \ll 1.} \quad (8.34)$$

Since power is quadratic in the fields,

$$\frac{P_{M1}}{P_{E1}} \sim (ka)^2, \quad \frac{P_{E2}}{P_{E1}} \sim (ka)^2 \quad (8.35)$$

for a generic source whose moments have their natural sizes.

8.9 The multipole hierarchy as a symmetry hierarchy

The usual ordering is

$$E1 \longrightarrow M1 \text{ and } E2 \longrightarrow M2, E3, \dots \quad (8.36)$$

but “leading” means leading among the moments that are actually allowed to vary.

- A localised isolated source cannot radiate as an electric monopole because total charge is conserved.
- A neutral system can radiate strongly through a time-dependent electric dipole moment.

- A symmetric source may have $\mathbf{p} = 0$ identically. Then the nominally smaller $M1$ or $E2$ term becomes the leading radiation.
- A small current loop is the canonical example with a magnetic dipole moment but no electric dipole moment.

Checkpoint

Before doing a long multipole calculation, inspect the symmetry. Ask whether Q , $\mathbf{p}$, $\mathbf{m}$ and Q_{ij} are allowed to be nonzero and time dependent. Symmetry can eliminate entire orders of the radiation field before any algebra is done.

8.10 Why only the $1/r$ terms matter for radiated power

For any multipole, the far-zone fields have the wave relation

$$\mathbf{B}_{\text{rad}} = \frac{1}{c} \hat{\mathbf{n}} \times \mathbf{E}_{\text{rad}} \quad (8.37)$$

and scale as $1/r$. Thus

$$\mathbf{S}_{\text{rad}} = \frac{1}{\mu_0} \mathbf{E}_{\text{rad}} \times \mathbf{B}_{\text{rad}} \propto \frac{1}{r^2} \hat{\mathbf{n}}. \quad (8.38)$$

The total power through a sphere is

$$P = \lim_{r \rightarrow \infty} \oint r^2 d\Omega \mathbf{S} \cdot \hat{\mathbf{n}}, \quad (8.39)$$

which is finite. Any field falling faster than $1/r$ gives vanishing power at infinity.

Physical meaning

A multipole expansion is simultaneously an expansion in spatial structure and a diagnostic of what a distant observer can learn about the source. At very long wavelength the observer cannot resolve fine details of the charge distribution, so only its lowest nonvanishing moments survive.

Worked example: when the electric dipole vanishes

The multipole hierarchy is most useful when the leading allowed moment is not the electric dipole. A small circular wire loop carrying an alternating current is the cleanest example. The source is electrically neutral and no charge is separated from one side of the loop to the other, so

$$Q = 0, \quad \mathbf{p} = \int \rho \mathbf{r}' d^3 r' = 0. \quad (8.40)$$

The electric-dipole channel is therefore absent. The first nonzero moment is the magnetic dipole

$$\mathbf{m}(t) = I(t) A \hat{\mathbf{z}}, \quad (8.41)$$

where A is the area of the loop. The leading radiated power is consequently

$$\langle P \rangle_{M1} = \frac{\mu_0 \omega^4 m_0^2}{12\pi c^3}. \quad (8.42)$$

This is not a small correction to electric-dipole radiation; it is the leading radiation because the electric-dipole moment is forbidden by the geometry and charge distribution.

For a source of size a , the ratio between magnetic-dipole and electric-dipole amplitudes is typically of order

$$\frac{E_{M1}}{E_{E1}} \sim ka. \quad (8.43)$$

Thus, when an electric dipole is allowed, it usually dominates. But when symmetry or conservation laws set $\mathbf{p} = 0$, the $M1$ or $E2$ terms become physically visible rather than merely formal corrections.

Lecture 9: Liénard–Wiechert Potentials

Aim of this lecture

Derive the retarded potentials of a moving point charge carefully, understand why the retarded time is implicit, and give a geometric interpretation of the Jacobian factor $1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}$.

Motivation for this lecture

The multipole expansion is ideal for a compact distribution whose internal structure is more important than the motion of any one constituent. A single point charge following an arbitrary trajectory presents the opposite problem: its source density is singular and its position changes during the light-travel time to the observer. We therefore need the exact retarded potentials of a moving point charge. Their derivation introduces the crucial factor $1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}$, which encodes the changing separation between source and observer and ultimately controls relativistic beaming, radiation from accelerated charges and Cherenkov-like geometrical effects.

9.1 A moving point charge

Let a charge q move along the trajectory $\mathbf{r}_q(t')$. Its charge and current densities are

$$\rho(\mathbf{r}', t') = q \delta^{(3)}[\mathbf{r}' - \mathbf{r}_q(t')] , \quad (9.1)$$

$$\mathbf{J}(\mathbf{r}', t') = q \mathbf{v}(t') \delta^{(3)}[\mathbf{r}' - \mathbf{r}_q(t')] . \quad (9.2)$$

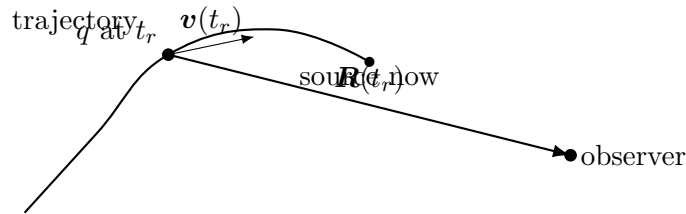
For an observation event $(\mathbf{r}, t)$ define

$$\mathbf{R}(t') = \mathbf{r} - \mathbf{r}_q(t'), \quad R(t') = |\mathbf{R}(t')|, \quad \hat{\mathbf{n}}(t') = \frac{\mathbf{R}(t')}{R(t')}. \quad (9.3)$$

The retarded time satisfies

$$t - t_r = \frac{R(t_r)}{c}. \quad (9.4)$$

Unlike the retarded time for a fixed source element in Lecture 4, this equation is implicit: the distance R depends on where the moving charge was at the unknown time t_r .



9.2 Why the retarded root is unique for a subluminal charge

Define

$$g(t') = t - t' - \frac{R(t')}{c}. \quad (9.5)$$

The retarded time is a zero of g . Since

$$\frac{dR}{dt'} = -\hat{\mathbf{n}} \cdot \mathbf{v}, \quad (9.6)$$

we have

$$\frac{dg}{dt'} = -1 + \frac{\hat{\mathbf{n}} \cdot \mathbf{v}}{c} = -(1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}). \quad (9.7)$$

For a massive particle $|\boldsymbol{\beta}| < 1$, so

$$1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta} > 0. \quad (9.8)$$

Therefore $g(t')$ decreases monotonically and can cross zero only once. A subluminal point charge has a unique retarded point on its trajectory for a given observation event.

This fact will cease to be true in the same simple way when the relevant propagation speed is reduced by a medium, which is precisely why Cherenkov geometry can involve multiple retarded intersections.

9.3 Using the spacetime delta function

Start from the Green-function representation before the source-time integral is performed:

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int d^3r' dt' \frac{\rho(\mathbf{r}', t')}{|\mathbf{r} - \mathbf{r}'|} \delta\left(t - t' - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right). \quad (9.9)$$

Insert the point-charge density. The spatial delta function first fixes

$$\mathbf{r}' = \mathbf{r}_q(t'), \quad (9.10)$$

so

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \int dt' \frac{1}{R(t')} \delta[g(t')]. \quad (9.11)$$

Now use the standard identity

$$\delta[g(t')] = \sum_i \frac{\delta(t' - t_i)}{|g'(t_i)|}, \quad (9.12)$$

where t_i are the roots of g .

For the unique retarded root,

$$|g'(t_r)| = 1 - \frac{\hat{\mathbf{n}} \cdot \mathbf{v}}{c}. \quad (9.13)$$

Define

$$\kappa = 1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}. \quad (9.14)$$

The scalar potential is therefore

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\kappa R} \right]_{\text{ret}}. \quad (9.15)$$

The same calculation for the current gives

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 q}{4\pi} \left[\frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}}. \quad (9.16)$$

These are the Liénard–Wiechert potentials.

Notation and convention

It is tempting to obtain the moving-charge potential by taking the extended-source retarded potential and simply replacing ρ by a delta function evaluated at the naive retarded time. That misses the factor $1/\kappa$. The factor is generated because the delta function itself contains the moving source position and therefore requires a Jacobian.

9.4 Physical meaning of the Jacobian

Consider two nearby emission events separated by dt_r . Their signals arrive separated by

$$dt = dt_r + \frac{1}{c}dR. \quad (9.17)$$

Since

$$dR = -\hat{\mathbf{n}} \cdot \mathbf{v} dt_r, \quad (9.18)$$

we obtain

$$\boxed{dt = (1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}) dt_r = \kappa dt_r.} \quad (9.19)$$

Thus

$$\boxed{\frac{dt_r}{dt} = \frac{1}{\kappa}.} \quad (9.20)$$

If the source moves toward the observer, $\hat{\mathbf{n}} \cdot \boldsymbol{\beta} > 0$ and $\kappa < 1$: a longer interval of source history is compressed into a shorter interval of observer time. If it moves away, $\kappa > 1$ and the received history is stretched.

Physical meaning

The same geometry produces several effects that at first look unrelated: the $1/\kappa$ factor in the Liénard–Wiechert potentials, the relativistic angular enhancement of moving-charge fields, the difference between emitted and detected power, and Doppler-like time compression. All originate in the map between source time and observation time.

9.5 A one-dimensional moving-segment picture

The handwritten derivation can be visualised with a short segment moving directly toward a distant observer. Let its physical endpoint separation in the observation frame be $L = x_2 - x_1$. The observer at x receives at one instant signals emitted when the endpoints were at the retarded positions $\tilde{x}_1$ and $\tilde{x}_2$.

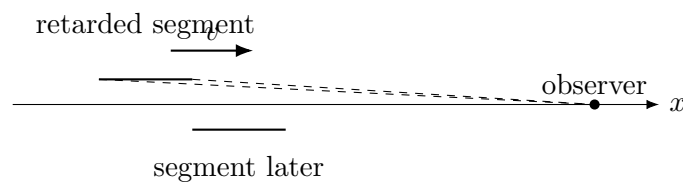
For motion toward the observer with speed v ,

$$x_i - \tilde{x}_i = v \frac{x - \tilde{x}_i}{c}. \quad (9.21)$$

Solving for each retarded endpoint and subtracting gives

$$\boxed{\tilde{L} \equiv \tilde{x}_2 - \tilde{x}_1 = \frac{L}{1 - v/c}.} \quad (9.22)$$

This is an *apparent retarded extent*, not a Lorentz-contracted proper length. The factor is again $1/\kappa$ for $\hat{\mathbf{n}} \parallel \mathbf{v}$.



9.6 Limiting cases

If the charge is instantaneously at rest,

$$\beta = 0, \quad \kappa = 1, \quad (9.23)$$

and the scalar potential reduces to the ordinary Coulomb form evaluated at the retarded position,

$$\phi = \frac{q}{4\pi\epsilon_0 R}. \quad (9.24)$$

For motion almost directly toward the observer at relativistic speed,

$$\kappa = 1 - \beta \ll 1, \quad (9.25)$$

so the potentials can be strongly enhanced. This is the precursor of the strong forward beaming that appears once the fields and radiated power are calculated.

Checkpoint

For a moving point charge, retardation does two things at once: it tells us *where* on the trajectory the source must be evaluated, and it changes the weight of that source through the Jacobian $1/\kappa$. Both are essential parts of the Liénard–Wiechert potentials.

Lecture 10: Fields of a Moving Charge

Aim of this lecture

Differentiate the Liénard–Wiechert potentials carefully, obtain the exact electric and magnetic fields of a moving point charge, and separate the bound velocity field from the radiative acceleration field.

Motivation for this lecture

Potentials are extremely convenient for solving Maxwell’s equations, but the observable quantities are the electric and magnetic fields. For a moving point charge, obtaining those fields is not a routine differentiation because the source position, velocity and retarded time all depend implicitly on the observation event. Working through these derivatives carefully produces the Liénard–Wiechert fields and, more importantly, separates them into a velocity field that falls as $1/R^2$ and an acceleration field that falls as $1/R$. That separation tells us where radiation from a moving charge actually resides.

10.1 Recap of the Liénard–Wiechert potentials

For a point charge following the trajectory $\mathbf{r}_0(t)$, define at the retarded time

$$\mathbf{R} = \mathbf{r} - \mathbf{r}_0(t_r), \quad R = |\mathbf{R}|, \quad \hat{\mathbf{n}} = \frac{\mathbf{R}}{R}, \quad \boldsymbol{\beta} = \frac{\mathbf{v}}{c}, \quad (10.1)$$

and

$$\kappa = 1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}. \quad (10.2)$$

The potentials derived in Lecture 9 are

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\kappa R} \right]_{\text{ret}}, \quad (10.3)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0 q}{4\pi} \left[\frac{\mathbf{v}}{\kappa R} \right]_{\text{ret}} = \frac{1}{c^2} [\mathbf{v}\phi]_{\text{ret}}. \quad (10.4)$$

The difficulty is that every quantity inside the brackets depends on the implicitly defined retarded time.

10.2 Retarded-time derivatives

The defining equation is

$$t = t_r + \frac{R(t_r)}{c}. \quad (10.5)$$

Differentiating with respect to t_r while holding the observation point fixed gives

$$\frac{dR}{dt_r} = \hat{\mathbf{n}} \cdot \frac{d\mathbf{R}}{dt_r} = -\hat{\mathbf{n}} \cdot \mathbf{v} = -c\hat{\mathbf{n}} \cdot \boldsymbol{\beta}. \quad (10.6)$$

Therefore

$$\frac{dt}{dt_r} = 1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta} = \kappa, \quad \frac{dt_r}{dt} = \frac{1}{\kappa}. \quad (10.7)$$

Similarly,

$$\frac{d\hat{\mathbf{n}}}{dt_r} = \frac{1}{R} \frac{d\mathbf{R}}{dt_r} - \frac{\mathbf{R}}{R^2} \frac{dR}{dt_r} \quad (10.8)$$

$$= -\frac{\mathbf{v}}{R} + \frac{c(\hat{\mathbf{n}} \cdot \boldsymbol{\beta})}{R} \hat{\mathbf{n}} \quad (10.9)$$

$$= \left[\frac{c}{R} [(\hat{\mathbf{n}} \cdot \boldsymbol{\beta}) \hat{\mathbf{n}} - \boldsymbol{\beta}] \right]. \quad (10.10)$$

For spatial differentiation at fixed observation time, differentiate $t = t_r + R/c$ with respect to $\mathbf{r}$. Because the source position changes when t_r changes,

$$\boldsymbol{\nabla} R = \hat{\mathbf{n}} - c(\hat{\mathbf{n}} \cdot \boldsymbol{\beta}) \boldsymbol{\nabla} t_r. \quad (10.11)$$

Using $0 = \boldsymbol{\nabla} t_r + (1/c) \boldsymbol{\nabla} R$ gives

$$\left[\boldsymbol{\nabla} t_r = -\frac{\hat{\mathbf{n}}}{c\kappa} \right], \quad \left[\boldsymbol{\nabla} R = \frac{\hat{\mathbf{n}}}{\kappa} \right]. \quad (10.12)$$

These extra factors of $1/\kappa$ are the spatial analogue of the Jacobian found in Lecture 9.

Physical meaning

The factor $\kappa = 1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}$ appears repeatedly because observer time and source time are not the same clock parameter. When the charge moves toward the observer, successive wavefronts arrive closer together and κ becomes small. This is the same kinematic compression that underlies Doppler shifts and relativistic beaming.

10.3 A useful intermediate denominator

It is convenient to define

$$D = \kappa R. \quad (10.13)$$

After a short calculation using the derivative identities above, one finds

$$\boldsymbol{\nabla} D = -\boldsymbol{\beta} + \frac{\hat{\mathbf{n}}}{\kappa} \left(1 - \beta^2 + \frac{R}{c} \hat{\mathbf{n}} \cdot \dot{\boldsymbol{\beta}} \right), \quad (10.14)$$

where the dot means d/dt_r . Therefore

$$\boldsymbol{\nabla} \frac{1}{D} = \frac{\boldsymbol{\beta}}{\kappa^2 R^2} - \frac{\hat{\mathbf{n}}}{\kappa^3 R^2} \left(1 - \beta^2 + \frac{R}{c} \hat{\mathbf{n}} \cdot \dot{\boldsymbol{\beta}} \right). \quad (10.15)$$

Likewise,

$$\frac{\partial}{\partial t} \frac{1}{D} = \frac{c(\hat{\mathbf{n}} \cdot \boldsymbol{\beta} - \beta^2) + R \hat{\mathbf{n}} \cdot \dot{\boldsymbol{\beta}}}{\kappa^3 R^2}. \quad (10.16)$$

These two relations contain most of the algebra required for the fields.

10.4 Constructing the electric field

Write the potentials as

$$\phi = \frac{q}{4\pi\epsilon_0} \frac{1}{D}, \quad \mathbf{A} = \frac{q}{4\pi\epsilon_0 c} \frac{\boldsymbol{\beta}}{D}. \quad (10.17)$$

Then

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}. \quad (10.18)$$

Because

$$\frac{\partial\boldsymbol{\beta}}{\partial t} = \frac{\dot{\boldsymbol{\beta}}}{\kappa}, \quad (10.19)$$

we obtain the useful intermediate expression

$$\mathbf{E} = -\frac{q}{4\pi\epsilon_0} \left[\nabla \frac{1}{D} + \frac{1}{c} \left(\frac{\dot{\boldsymbol{\beta}}}{\kappa D} + \boldsymbol{\beta} \frac{\partial}{\partial t} \frac{1}{D} \right) \right]_{\text{ret}}. \quad (10.20)$$

Substituting the derivatives of D and collecting separately the terms proportional to $1/R^2$ and $1/R$ gives the exact Liénard–Wiechert field

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \left[\frac{(1 - \beta^2)(\hat{\mathbf{n}} - \boldsymbol{\beta})}{\kappa^3 R^2} + \frac{\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}]}{c\kappa^3 R} \right]_{\text{ret}}. \quad (10.21)$$

The magnetic field simplifies beautifully to

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} [\hat{\mathbf{n}} \times \mathbf{E}]_{\text{ret}}. \quad (10.22)$$

10.5 Velocity field and acceleration field

It is useful to write

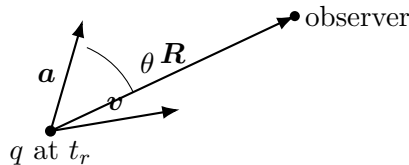
$$\mathbf{E} = \mathbf{E}_v + \mathbf{E}_a, \quad (10.23)$$

with

$$\mathbf{E}_v = \frac{q}{4\pi\epsilon_0} \left[\frac{(1 - \beta^2)(\hat{\mathbf{n}} - \boldsymbol{\beta})}{\kappa^3 R^2} \right]_{\text{ret}}, \quad (10.24)$$

$$\mathbf{E}_a = \frac{q}{4\pi\epsilon_0 c} \left[\frac{\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}]}{\kappa^3 R} \right]_{\text{ret}}. \quad (10.25)$$

The first scales as $1/R^2$ and survives even when the charge moves uniformly. The second scales as $1/R$ and vanishes when $\dot{\boldsymbol{\beta}} = 0$.



Physical meaning

A uniformly moving charge does not radiate, even though its electric field is not spherically symmetric in the laboratory frame. Radiation is associated with the $1/R$ acceleration field: it is the outward-propagating information that the charge has changed its state of motion.

10.6 Why the acceleration field is transverse

Take the dot product with $\hat{\mathbf{n}}$:

$$\hat{\mathbf{n}} \cdot \mathbf{E}_a \propto \hat{\mathbf{n}} \cdot \left\{ \hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}] \right\} = 0. \quad (10.26)$$

Hence

$$\boxed{\hat{\mathbf{n}} \cdot \mathbf{E}_a = 0.} \quad (10.27)$$

Since $\mathbf{B}_a = (1/c)\hat{\mathbf{n}} \times \mathbf{E}_a$, the three vectors $\mathbf{E}_a$, $\mathbf{B}_a$ and $\hat{\mathbf{n}}$ are mutually perpendicular and satisfy $E_a = cB_a$. The far field is therefore locally a plane electromagnetic wave.

10.7 Uniform motion in terms of the instantaneous position

For constant velocity the acceleration term vanishes. The retarded expression can be rewritten in terms of the vector $\mathcal{R}$ from the *present* position of the charge to the observation point. If θ is the angle between $\mathcal{R}$ and $\mathbf{v}$,

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{1 - \beta^2}{(1 - \beta^2 \sin^2 \theta)^{3/2}} \frac{\hat{\mathcal{R}}}{\mathcal{R}^2}.} \quad (10.28)$$

Thus the field is radial with respect to the instantaneous position, but its magnitude is anisotropic. Along the direction of motion,

$$E_{\parallel} = \frac{1}{\gamma^2} E_{\text{Coulomb}}, \quad (10.29)$$

whereas in the transverse direction,

$$E_{\perp} = \gamma E_{\text{Coulomb}}. \quad (10.30)$$

The field lines are therefore compressed into a transverse “pancake” at relativistic speed.

This same result will be recovered in Lecture 16 by Lorentz-transforming a Coulomb field. The agreement is an important consistency check between retardation and relativity.

Checkpoint

The exact moving-charge field contains two physically distinct pieces. The $1/R^2$ velocity field is the relativistically distorted Coulomb field. The $1/R$ acceleration field is transverse and radiative. Keeping those two scalings visible is much more useful than memorising the full formula as one undifferentiated expression.

Worked example: uniformly moving charge from the Liénard–Wiechert fields

For a charge moving with constant velocity, $\dot{\boldsymbol{\beta}} = 0$. The acceleration field vanishes, and the Liénard–Wiechert electric field reduces to

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \left[\frac{(1 - \beta^2)(\hat{\mathbf{n}} - \boldsymbol{\beta})}{(1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta})^3 R^2} \right]_{\text{ret}}. \quad (10.31)$$

This expression is written in terms of retarded quantities. After eliminating the retarded time one obtains the field at the observation time in the familiar form

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{1 - \beta^2}{(1 - \beta^2 \sin^2 \theta)^{3/2}} \frac{\mathbf{R}_{\text{inst}}}{R_{\text{inst}}^3}.} \quad (10.32)$$

Here $\mathbf{R}_{\text{inst}}$ points from the instantaneous position of the charge to the observation point, and θ is the angle between $\mathbf{R}_{\text{inst}}$ and the velocity.

This field is radial with respect to the instantaneous position, but its magnitude is anisotropic. Along the direction of motion, $\theta = 0$, the field is reduced by a factor $1 - \beta^2 = 1/\gamma^2$. Transverse to the motion, $\theta = \pi/2$, it is enhanced by a factor γ :

$$E_{\perp} = \gamma \frac{q}{4\pi\epsilon_0 R^2}. \quad (10.33)$$

Thus the Coulomb field of a fast charge is compressed into a transverse pancake. This is the same result obtained in Lecture 16 by Lorentz-transforming the Coulomb field of a charge at rest.

Lecture 11: Radiation from an Accelerating Charge

Aim of this lecture

Use the acceleration field to derive the relativistic angular distribution of emitted power, understand the difference between emission time and observation time, recover the Larmor formula, and interpret relativistic beaming.

Motivation for this lecture

Lecture 10 revealed a $1/R$ field proportional to the charge's acceleration. A field with this radial dependence can carry a finite energy flux through an arbitrarily large sphere, so it is the part responsible for genuine radiation. We now turn that observation into quantitative statements: the angular distribution of emitted power, the difference between emission time and detection time, the nonrelativistic Larmor formula, and its relativistic Liénard generalisation. This lecture provides the master radiation formula from which the important special cases in the next lecture follow.

11.1 The part of the field that carries energy to infinity

From Lecture 10 the radiative electric field is

$$\mathbf{E}_a = \frac{q}{4\pi\epsilon_0 c} \left[\frac{\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}]}{\kappa^3 R} \right]_{\text{ret}}, \quad (11.1)$$

with

$$\kappa = 1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}. \quad (11.2)$$

It is transverse, so

$$\mathbf{B}_a = \frac{1}{c} \hat{\mathbf{n}} \times \mathbf{E}_a. \quad (11.3)$$

Consequently the Poynting vector is purely radial,

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E}_a \times \mathbf{B}_a \quad (11.4)$$

$$= \frac{1}{\mu_0 c} \mathbf{E}_a \times (\hat{\mathbf{n}} \times \mathbf{E}_a) \quad (11.5)$$

$$= \epsilon_0 c E_a^2 \hat{\mathbf{n}}, \quad (11.6)$$

where we used $\mathbf{E}_a \cdot \hat{\mathbf{n}} = 0$.

11.2 Detected power versus emitted power

At first sight one might write

$$\frac{dP}{d\Omega} = R^2 \mathbf{S} \cdot \hat{\mathbf{n}}. \quad (11.7)$$

This is the power *received* per unit solid angle per unit observer time. For a moving source, however, an interval dt_r of emission does not correspond to the same interval dt at the detector.

The retarded-time relation gives

$$\boxed{dt = \kappa dt_r.} \quad (11.8)$$

Therefore

$$\frac{dU}{dt_r d\Omega} = \frac{dt}{dt_r} \frac{dU}{dt d\Omega} = \kappa R^2 \mathbf{S} \cdot \hat{\mathbf{n}}. \quad (11.9)$$

The extra factor of κ reduces the sixth power in E_a^2 to a fifth power in the emitted-power formula.

Substituting E_a gives

$$\boxed{\frac{dP}{d\Omega} = \frac{q^2}{16\pi^2 \epsilon_0 c} \left[\frac{|\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \dot{\boldsymbol{\beta}}]|^2}{(1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta})^5} \right]_{\text{ret}}}. \quad (11.10)$$

Since $\mathbf{a} = c\dot{\boldsymbol{\beta}}$,

$$\boxed{\frac{dP}{d\Omega} = \frac{q^2}{16\pi^2 \epsilon_0 c^3} \left[\frac{|\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \mathbf{a}]|^2}{(1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta})^5} \right]_{\text{ret}}}. \quad (11.11)$$

Physical meaning

The factor $(1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta})^{-5}$ is partly field enhancement and partly clock conversion. A charge moving toward the observer emits successive pieces of radiation from successively closer positions, so the arrival times are compressed. Relativistic beaming is therefore both an angular and a timing effect.

11.3 Nonrelativistic limit

When $\beta \ll 1$,

$$\kappa \simeq 1, \quad \hat{\mathbf{n}} - \boldsymbol{\beta} \simeq \hat{\mathbf{n}}. \quad (11.12)$$

Hence

$$\frac{dP}{d\Omega} \simeq \frac{q^2}{16\pi^2 \epsilon_0 c^3} |\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \mathbf{a})|^2. \quad (11.13)$$

Using

$$\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \mathbf{a}) = \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{a}) - \mathbf{a}, \quad (11.14)$$

its magnitude is the component of $\mathbf{a}$ perpendicular to the line of sight. If the z axis is chosen along $\mathbf{a}$,

$$|\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \mathbf{a})|^2 = a^2 \sin^2 \theta, \quad (11.15)$$

so

$$\boxed{\frac{dP}{d\Omega} = \frac{q^2 a^2}{16\pi^2 \epsilon_0 c^3} \sin^2 \theta.} \quad (11.16)$$

This is exactly the dipole pattern: there is no radiation along the acceleration axis and maximum radiation perpendicular to it.

11.4 Integration over angle: Larmor's formula

The total power is

$$P = \int \frac{dP}{d\Omega} d\Omega \quad (11.17)$$

$$= \frac{q^2 a^2}{16\pi^2 \epsilon_0 c^3} \int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta d\theta. \quad (11.18)$$

The angular integrals are

$$\int_0^{2\pi} d\phi = 2\pi, \quad \int_0^\pi \sin^3 \theta d\theta = \frac{4}{3}. \quad (11.19)$$

Therefore

$$P = \frac{q^2 a^2}{6\pi \epsilon_0 c^3}. \quad (11.20)$$

This is Larmor's formula for the instantaneous power radiated by a nonrelativistic point charge.

11.5 The relativistic Liénard formula

Integrating the exact angular distribution gives

$$P = \frac{q^2 \gamma^6}{6\pi \epsilon_0 c^3} [a^2 - |\boldsymbol{\beta} \times \mathbf{a}|^2]. \quad (11.21)$$

Decompose the acceleration into components parallel and perpendicular to the velocity,

$$\mathbf{a} = \mathbf{a}_\parallel + \mathbf{a}_\perp. \quad (11.22)$$

Since $|\boldsymbol{\beta} \times \mathbf{a}|^2 = \beta^2 a_\perp^2$,

$$a^2 - |\boldsymbol{\beta} \times \mathbf{a}|^2 = a_\parallel^2 + (1 - \beta^2) a_\perp^2 \quad (11.23)$$

$$= a_\parallel^2 + \frac{a_\perp^2}{\gamma^2}. \quad (11.24)$$

Hence the Liénard formula can be written in the very useful form

$$P = \frac{q^2}{6\pi \epsilon_0 c^3} \left(\gamma^6 a_\parallel^2 + \gamma^4 a_\perp^2 \right). \quad (11.25)$$

Thus longitudinal acceleration is enhanced by γ^6 , whereas transverse acceleration is enhanced by γ^4 when the acceleration is specified in the laboratory frame.

11.6 Relativistic beaming from the factor kappa

Let θ be the angle between the observation direction and the velocity. Then

$$\kappa = 1 - \beta \cos \theta. \quad (11.26)$$

For $\gamma \gg 1$ and small θ ,

$$\beta \simeq 1 - \frac{1}{2\gamma^2}, \quad \cos \theta \simeq 1 - \frac{\theta^2}{2}. \quad (11.27)$$

Therefore

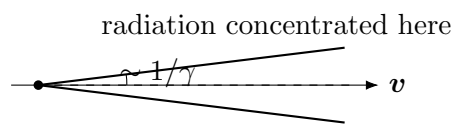
$$\kappa \simeq 1 - \left(1 - \frac{1}{2\gamma^2}\right) \left(1 - \frac{\theta^2}{2}\right) \quad (11.28)$$

$$\simeq \frac{1}{2\gamma^2} + \frac{\theta^2}{2} \quad (11.29)$$

$$= \boxed{\frac{1 + \gamma^2 \theta^2}{2\gamma^2}}. \quad (11.30)$$

The denominator therefore changes significantly once $\gamma\theta \sim 1$. This gives the characteristic beaming angle

$$\boxed{\Delta\theta \sim \frac{1}{\gamma}}. \quad (11.31)$$



Checkpoint

The nonrelativistic limit is controlled by the acceleration direction and gives the broad $\sin^2\theta$ dipole pattern. In the relativistic limit the velocity direction becomes equally important: the factor $1 - \hat{\mathbf{n}} \cdot \boldsymbol{\beta}$ squeezes the radiation into a narrow forward cone.

Lecture 12: Bremsstrahlung and Synchrotron Radiation

Aim of this lecture

Apply the general accelerated-charge formula to longitudinal and transverse acceleration, derive their angular patterns, and understand physically why relativistic circular motion produces intense, strongly beamed, broadband synchrotron radiation.

Motivation for this lecture

The general accelerated-charge formula is powerful but abstract. Its physical content becomes much clearer when it is applied to two opposite geometries: acceleration parallel to the velocity and acceleration perpendicular to it. The first describes braking and gives the characteristic pattern of bremsstrahlung; the second describes circular motion and leads from cyclotron radiation to highly relativistic synchrotron radiation. These examples show how relativistic motion reshapes the angular pattern, concentrates radiation into a narrow forward cone, and produces very high characteristic frequencies.

12.1 Two geometries worth separating

The Liénard formula distinguishes acceleration parallel and perpendicular to the velocity,

$$P = \frac{q^2}{6\pi\epsilon_0 c^3} \left(\gamma^6 a_{\parallel}^2 + \gamma^4 a_{\perp}^2 \right). \quad (12.1)$$

These cases correspond to different physical situations:

- $\mathbf{a} \parallel \mathbf{v}$: the particle changes its speed;
- $\mathbf{a} \perp \mathbf{v}$: the particle changes its direction of motion.

The first is the natural model for linear acceleration and braking. The second is the natural model for magnetic bending and circular motion.

12.2 Longitudinal acceleration and bremsstrahlung

Take

$$\mathbf{v} = v\hat{\mathbf{z}}, \quad \mathbf{a} = a\hat{\mathbf{z}}. \quad (12.2)$$

Then $\boldsymbol{\beta} \times \mathbf{a} = 0$. In the numerator of the general angular formula,

$$\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \mathbf{a}] = \hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \mathbf{a}) - \hat{\mathbf{n}} \times (\boldsymbol{\beta} \times \mathbf{a}) \quad (12.3)$$

$$= \hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \mathbf{a}). \quad (12.4)$$

Therefore

$$|\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \mathbf{a}]|^2 = a^2 \sin^2 \theta, \quad (12.5)$$

and

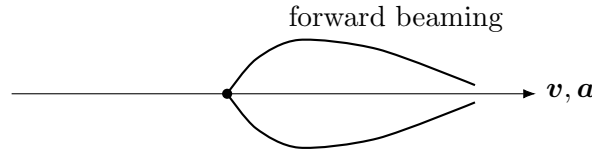
$$\left(\frac{dP}{d\Omega}\right)_{\parallel} = \frac{q^2 a^2}{16\pi^2 \epsilon_0 c^3} \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^5}. \quad (12.6)$$

The formula depends on a^2 , so reversing the acceleration does not change the instantaneous radiated power pattern. Radiation associated with deceleration is traditionally called *bremsstrahlung*, literally “braking radiation.”

Physical meaning

Bremsstrahlung does not require the particle to stop. Any deflection or deceleration caused, for example, by the electric field of another charged particle can radiate. In matter, energetic electrons repeatedly accelerated by nuclei produce a broad bremsstrahlung spectrum.

For $\beta \ll 1$ the denominator is nearly unity and the familiar $\sin^2 \theta$ dipole pattern is recovered. For $\gamma \gg 1$, the factor $(1 - \beta \cos \theta)^{-5}$ moves most of the radiation into the forward direction.



12.3 Transverse acceleration: setting up the geometry

Now take

$$\mathbf{v} = v\hat{\mathbf{z}}, \quad \mathbf{a} = a\hat{\mathbf{x}}. \quad (12.7)$$

The observation direction is

$$\hat{\mathbf{n}} = \hat{\mathbf{x}} \sin \theta \cos \phi + \hat{\mathbf{y}} \sin \theta \sin \phi + \hat{\mathbf{z}} \cos \theta. \quad (12.8)$$

Since $\boldsymbol{\beta} = \beta\hat{\mathbf{z}}$,

$$\kappa = 1 - \beta \cos \theta. \quad (12.9)$$

We need

$$\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \mathbf{a}]. \quad (12.10)$$

Using $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$,

$$\hat{\mathbf{n}} \times [(\hat{\mathbf{n}} - \boldsymbol{\beta}) \times \mathbf{a}] = (\hat{\mathbf{n}} - \boldsymbol{\beta})(\hat{\mathbf{n}} \cdot \mathbf{a}) - \mathbf{a}[\hat{\mathbf{n}} \cdot (\hat{\mathbf{n}} - \boldsymbol{\beta})] \quad (12.11)$$

$$= a \sin \theta \cos \phi (\hat{\mathbf{n}} - \beta\hat{\mathbf{z}}) - a\kappa\hat{\mathbf{x}}. \quad (12.12)$$

Squaring and simplifying gives

$$\left(\frac{dP}{d\Omega}\right)_{\perp} = \frac{q^2 a^2}{16\pi^2 \epsilon_0 c^3} \frac{1}{(1 - \beta \cos \theta)^3} \left[1 - \frac{(1 - \beta^2) \sin^2 \theta \cos^2 \phi}{(1 - \beta \cos \theta)^2} \right]. \quad (12.13)$$

This is the angular distribution appropriate to instantaneous circular motion.

In the nonrelativistic limit,

$$\left(\frac{dP}{d\Omega}\right)_{\perp} \longrightarrow \frac{q^2 a^2}{16\pi^2 \epsilon_0 c^3} (1 - \sin^2 \theta \cos^2 \phi). \quad (12.14)$$

This is simply the ordinary dipole pattern measured relative to the acceleration axis $\hat{\mathbf{x}}$.

12.4 Cyclotron versus synchrotron radiation

A charged particle in circular motion always has transverse acceleration. The terminology depends mainly on the speed:

- *cyclotron radiation*: nonrelativistic circular motion;
- *synchrotron radiation*: relativistic circular motion.

There is no sharp physical boundary between them. Synchrotron radiation is the relativistic limit of the same acceleration-radiation mechanism.

For transverse acceleration the total power is

$$P_{\perp} = \frac{q^2 \gamma^4 a^2}{6\pi \epsilon_0 c^3}. \quad (12.15)$$

The angular distribution is concentrated into a cone of characteristic width

$$\Delta\theta \sim \frac{1}{\gamma} \quad (12.16)$$

around the instantaneous direction of motion.

12.5 Why a distant observer sees a short pulse

Consider a particle moving around a circle of curvature radius ρ_c . Its velocity direction rotates with angular speed approximately

$$\omega_{\text{orb}} \simeq \frac{v}{\rho_c}. \quad (12.17)$$

The observer receives substantial radiation only while the $1/\gamma$ cone sweeps across the line of sight. The corresponding emission-time interval is roughly

$$\Delta t_{\text{em}} \sim \frac{1/\gamma}{\omega_{\text{orb}}} \sim \frac{\rho_c}{\gamma c} \quad (12.18)$$

for $v \simeq c$.

But during this interval the particle is moving almost toward the observer, so arrival times are further compressed. For forward emission

$$\frac{dt}{dt_r} = 1 - \beta \simeq \frac{1}{2\gamma^2}. \quad (12.19)$$

Hence

$$\Delta t_{\text{obs}} \sim (1 - \beta) \Delta t_{\text{em}} \sim \frac{\rho_c}{2\gamma^3 c}. \quad (12.20)$$

A short pulse contains Fourier components up to frequencies of order

$$\omega \sim \frac{1}{\Delta t_{\text{obs}}} \sim \gamma^3 \frac{c}{\rho_c}. \quad (12.21)$$

A more precise calculation defines the characteristic synchrotron frequency

$$\omega_c = \frac{3}{2} \gamma^3 \frac{c}{\rho_c}. \quad (12.22)$$

Physical meaning

The broad synchrotron spectrum is a time-domain effect. The source motion may be smooth and periodic, but a relativistic observer receives narrow flashes. Narrow flashes require many Fourier harmonics, which is why synchrotron radiation can reach frequencies far above the orbital frequency.

12.6 Magnetic bending and the orbit radius

For a particle moving perpendicular to a magnetic field,

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}. \quad (12.23)$$

Because the force is perpendicular to the velocity it changes the direction, not the instantaneous energy. The relativistic transverse equation of motion is

$$|q|vB = \frac{\gamma m v^2}{\rho_c}, \quad (12.24)$$

so

$$\rho_c = \frac{\gamma m v}{|q|B}. \quad (12.25)$$

The acceleration is therefore

$$a = \frac{v^2}{\rho_c} = \frac{|q|Bv}{\gamma m}. \quad (12.26)$$

Substituting into the transverse Liénard power gives

$$P = \frac{q^4 B^2 \gamma^2 v^2}{6\pi\epsilon_0 m^2 c^3}. \quad (12.27)$$

For an ultrarelativistic particle $v \simeq c$, the radiated power grows approximately as γ^2 at fixed magnetic field.

12.7 Why synchrotron losses matter especially for light particles

At fixed charge, field and energy scale, the curvature and radiated power depend strongly on the particle mass. Electrons therefore radiate much more readily than heavy particles in magnetic storage rings. This is why synchrotron-radiation losses are a central design issue for high-energy electron rings, while they are far less severe for proton rings at comparable energy.

Checkpoint

Bremsstrahlung and synchrotron radiation are not separate fundamental laws. Both are applications of the same Liénard–Wiechert acceleration field. Their apparently different phenomenology comes from the geometry of the acceleration and, at relativistic speed, from beaming and arrival-time compression.

Lecture 13: Cherenkov Radiation, Radiation Reaction and Thomas Precession

Aim of this lecture

Understand radiation from motion faster than the phase speed of light in a medium, see how energy conservation forces an accelerated charge to react to its own radiation, and understand Thomas precession as the rotational residue of non-collinear Lorentz boosts.

Motivation for this lecture

The standard picture developed so far assumes radiation in vacuum and treats the particle trajectory as given. Several important phenomena force us beyond those assumptions. In a dielectric, a particle can outrun the phase velocity of light and generate Cherenkov radiation. An accelerating charge loses energy, so consistency requires its own radiation to feed back on its motion through radiation reaction. Finally, successive non-collinear Lorentz boosts do not combine to a pure boost, producing Thomas precession. These three effects complete the discussion of moving-charge radiation while also pointing directly toward the relativistic formulation that follows.

13.1 Cherenkov radiation: faster than light in a medium, not in vacuum

In a dielectric medium with refractive index $n_r(\omega)$, the phase speed of light is

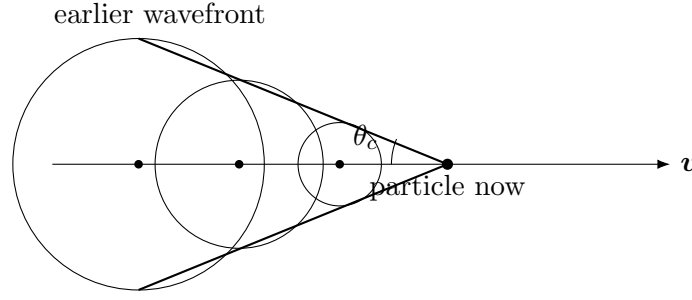
$$c_n(\omega) = \frac{c}{n_r(\omega)}. \quad (13.1)$$

A massive particle must always satisfy $v < c$, but it can perfectly well satisfy

$$v > \frac{c}{n_r}. \quad (13.2)$$

This does not violate special relativity: the particle is faster than the phase velocity of light *in the material*, not faster than c in vacuum.

The passing charge polarises the medium. Each point along the track is disturbed as the particle passes, and the medium subsequently emits an electromagnetic response. When the particle is slower than c_n , the wavefronts do not build up coherently in a fixed direction. When $v > c_n$, they possess a common tangent and interfere constructively on a cone.



13.2 Geometry of the Cherenkov angle

Consider two emission events separated by a time Δt . During that time the particle travels

$$v\Delta t, \quad (13.3)$$

whereas the wavefront from the earlier event expands by

$$c_n \Delta t. \quad (13.4)$$

The right triangle formed by the particle track and the tangent to the wavefront gives

$$\cos \theta_c = \frac{c_n \Delta t}{v \Delta t} = \frac{c}{n_r v}. \quad (13.5)$$

Thus

$$\boxed{\cos \theta_c = \frac{1}{n_r \beta}}. \quad (13.6)$$

The threshold is therefore

$$\boxed{n_r \beta > 1}. \quad (13.7)$$

Some drawings define the complementary cone angle, which converts the cosine to a sine. The physics is of course unchanged provided the angle is defined explicitly.

Physical meaning

Cherenkov radiation is the electromagnetic analogue of a sonic boom. The cone is a phase-coherence effect: the particle continuously excites the medium, and the emitted disturbances add constructively only along the Cherenkov direction.

13.3 Dispersion and the spectrum

Real materials are dispersive, so $n_r = n_r(\omega)$. The Cherenkov condition must therefore be satisfied frequency by frequency,

$$\beta n_r(\omega) > 1. \quad (13.8)$$

For a transparent, nonmagnetic medium the classical Frank–Tamm result for the energy radiated per unit path length per unit angular frequency is

$$\boxed{\frac{d^2 U}{dx d\omega} = \frac{q^2 \omega}{4\pi \epsilon_0 c^2} \left(1 - \frac{1}{\beta^2 n_r^2(\omega)} \right)}, \quad (13.9)$$

for frequencies satisfying the threshold, and zero otherwise. The explicit factor of ω helps explain why visible Cherenkov light is characteristically weighted toward the blue, although the actual spectrum is also limited by material dispersion and absorption.

13.4 Radiation reaction: why the trajectory cannot be prescribed forever

Up to now the calculation followed a convenient two-step logic:

- (1) determine the trajectory from the external forces while ignoring radiation;
- (2) insert that trajectory into the radiation formula.

This cannot be exact because the radiation carries away energy and momentum. If a particle radiates power P , the source of that energy must ultimately be the mechanical work done on the particle. The motion must therefore experience a recoil or *radiation-reaction* force.

In the nonrelativistic point-particle theory the Abraham–Lorentz force is

$$\mathbf{F}_{\text{rr}} = \frac{q^2}{6\pi\epsilon_0 c^3} \dot{\mathbf{a}}. \quad (13.10)$$

The equation of motion becomes

$$m\mathbf{a} = \mathbf{F}_{\text{ext}} + m\tau_0 \dot{\mathbf{a}}, \quad (13.11)$$

where

$$\tau_0 = \frac{q^2}{6\pi\epsilon_0 mc^3}. \quad (13.12)$$

For an electron, $\tau_0 \simeq 6.3 \times 10^{-24}$ s, an extraordinarily short timescale.

13.5 Energy balance and the Schott term

Dot the radiation-reaction force with the velocity:

$$\mathbf{F}_{\text{rr}} \cdot \mathbf{v} = m\tau_0 \dot{\mathbf{a}} \cdot \mathbf{v} \quad (13.13)$$

$$= m\tau_0 \frac{d}{dt}(\mathbf{a} \cdot \mathbf{v}) - m\tau_0 a^2. \quad (13.14)$$

Using

$$m\tau_0 = \frac{q^2}{6\pi\epsilon_0 c^3}, \quad (13.15)$$

we obtain

$$\mathbf{F}_{\text{rr}} \cdot \mathbf{v} = \frac{d}{dt} \left(\frac{q^2}{6\pi\epsilon_0 c^3} \mathbf{a} \cdot \mathbf{v} \right) - P_{\text{Larmor}}. \quad (13.16)$$

The second term is precisely minus the power radiated to infinity. The total derivative is associated with energy stored in the near field and is called the *Schott energy*. Thus instantaneous mechanical-energy loss need not equal the radiated power at every instant, even though the balance works after the near-field contribution is included.

Notation and convention

The Abraham–Lorentz equation is third order in position because it contains $\dot{\mathbf{a}}$. Treated as an exact fundamental equation it admits unphysical runaway and pre-accelerating solutions. In ordinary applications radiation reaction is instead treated as a small perturbation. To leading order one substitutes the acceleration generated by the external force into the radiation-reaction term (“order reduction”).

For a simple external force that is not strongly velocity dependent,

$$\mathbf{a} \simeq \frac{\mathbf{F}_{\text{ext}}}{m} \quad (13.17)$$

inside the small correction, giving schematically

$$\mathbf{F}_{\text{rr}} \simeq \tau_0 \frac{d\mathbf{F}_{\text{ext}}}{dt}. \quad (13.18)$$

The tiny factor τ_0 explains why radiation reaction is negligible in most nonrelativistic laboratory mechanics but important in sufficiently intense fields.

13.6 Thomas precession: non-collinear boosts do not commute

A Lorentz boost changes the time axis of an inertial frame. If a particle accelerates, its instantaneous rest frame changes from one moment to the next. When successive boosts are along the same line, their composition is another boost. When their directions differ, their composition contains not only a boost but also a spatial rotation.

That residual rotation is the origin of Thomas precession. The instantaneous angular velocity of the particle's momentarily comoving axes relative to the laboratory frame is

$$\boxed{\boldsymbol{\omega}_T = \frac{\gamma^2}{\gamma + 1} \frac{\mathbf{a} \times \mathbf{v}}{c^2}}. \quad (13.19)$$

Using

$$\frac{\gamma^2 \beta^2}{\gamma + 1} = \gamma - 1, \quad (13.20)$$

this can also be written as

$$\boxed{\boldsymbol{\omega}_T = \frac{\gamma - 1}{v^2} \mathbf{a} \times \mathbf{v}}. \quad (13.21)$$

In the nonrelativistic limit $\gamma - 1 \simeq v^2/(2c^2)$,

$$\boxed{\boldsymbol{\omega}_T \simeq \frac{1}{2c^2} \mathbf{a} \times \mathbf{v}}. \quad (13.22)$$

13.7 Circular motion as the clearest example

For uniform circular motion of radius R , the acceleration is inward and perpendicular to the velocity,

$$\mathbf{a} = -\frac{v^2}{R} \hat{\mathbf{r}}. \quad (13.23)$$

The orbital angular speed is

$$\omega_{\text{orb}} = \frac{v}{R}. \quad (13.24)$$

Hence

$$|\boldsymbol{\omega}_T| = \frac{\gamma - 1}{v^2} a v = (\gamma - 1) \frac{v}{R} = \boxed{(\gamma - 1) \omega_{\text{orb}}}. \quad (13.25)$$

The direction of $\mathbf{a} \times \mathbf{v}$ is opposite to the orbital angular-velocity vector, so the comoving axes precess backward relative to the orbital motion.

Physical meaning

Thomas precession is kinematic rather than dynamical: no extra physical torque has been introduced. It is a direct consequence of the noncommutativity of Lorentz boosts. The famous factor of one half in the nonrelativistic limit is essential in the relativistic explanation of atomic spin–orbit coupling.

Checkpoint

This lecture contains three apparently unrelated phenomena, but all three reveal limits of a naive picture of electromagnetic propagation and motion. Cherenkov radiation depends on coherent propagation through a medium; radiation reaction reminds us that emitted fields feed back on their source; Thomas precession reminds us that changing inertial frames is geometrically nontrivial in relativity.

Lecture 14: Special Relativity and Spacetime

Aim of this lecture

Motivate special relativity from electrodynamics, derive and interpret the Lorentz transformation, understand simultaneity, time dilation, length contraction and causal structure, and introduce proper time and four-velocity as the natural language of relativistic motion.

Motivation for this lecture

Electrodynamics itself creates the need for special relativity. Maxwell's equations predict a distinguished wave speed c , whereas Galilean transformations would make that speed depend on the observer. Electric and magnetic effects also exchange roles when the observer changes frame. To make these facts consistent we must replace Galilean spacetime with Lorentzian spacetime. This lecture therefore develops the Lorentz transformation and its consequences—simultaneity, time dilation, length contraction, causal structure and proper time—as the kinematical framework in which the rest of electrodynamics must be formulated.

14.1 Why relativity enters electrodynamics

Maxwell's equations predict electromagnetic waves with speed

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}. \quad (14.1)$$

The right-hand side contains constants of electromagnetism, not the velocity of an observer. If Galilean transformations were correct,

$$x' = x - vt, \quad t' = t, \quad (14.2)$$

then a light ray with $x = ct$ would have speed $c - v$ in S' . Maxwell's equations and Galilean relativity would therefore be incompatible.

This tension is especially vivid in induction problems. A magnet moving past a conducting loop and a loop moving past a magnet produce the same observable current when the relative motion is the same, but a purely Galilean account describes the two cases using apparently different electric and magnetic mechanisms. Relativity resolves the asymmetry: electric and magnetic fields are frame-dependent parts of one electromagnetic field.

Einstein's two postulates are

- (1) the laws of physics have the same form in all inertial frames;
- (2) the speed of light in vacuum is the same for all inertial observers.

14.2 Deriving the Lorentz transformation

Let S' move with speed v along $+x$ relative to S , with the origins coincident at $t = t' = 0$. Homogeneity of space and time implies that the transformation between inertial coordinates must

be linear. Since the origin of S' follows $x = vt$, write

$$x' = A(x - vt). \quad (14.3)$$

The most general corresponding linear time transformation can be written

$$t' = B(t - \alpha x), \quad (14.4)$$

where A , B and α may depend on v but not on x or t .

Now impose the invariance of the speed of light. A light pulse travelling in the $+x$ direction satisfies

$$x = ct, \quad x' = ct'. \quad (14.5)$$

Substitution gives

$$A(c - v) = cB(1 - \alpha c). \quad (14.6)$$

A light pulse travelling in the opposite direction satisfies

$$x = -ct, \quad x' = -ct', \quad (14.7)$$

which gives

$$A(c + v) = cB(1 + \alpha c). \quad (14.8)$$

Adding these two equations yields

$$A = B, \quad (14.9)$$

while subtracting them gives

$$\alpha = \frac{v}{c^2}. \quad (14.10)$$

Thus

$$x' = A(x - vt), \quad t' = A\left(t - \frac{vx}{c^2}\right). \quad (14.11)$$

It remains to determine A . By the principle of relativity, the inverse transformation must have the same form with $v \rightarrow -v$:

$$x = A(x' + vt'). \quad (14.12)$$

Substitute the expressions for x' and t' :

$$x = A^2 \left[(x - vt) + v \left(t - \frac{vx}{c^2} \right) \right] \quad (14.13)$$

$$= A^2 x \left(1 - \frac{v^2}{c^2} \right). \quad (14.14)$$

Therefore

$$A^2 \left(1 - \frac{v^2}{c^2} \right) = 1, \quad (14.15)$$

so choosing the positive root continuously connected to $A = 1$ at $v = 0$ gives

$$\boxed{A = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}}. \quad (14.16)$$

We have therefore derived

$$\boxed{x' = \gamma(x - vt)}, \quad (14.17)$$

$$\boxed{t' = \gamma \left(t - \frac{vx}{c^2} \right)}. \quad (14.18)$$

The transverse coordinates are unchanged,

$$y' = y, \quad z' = z. \quad (14.19)$$

Equivalently, with $\beta = v/c$,

$$\begin{aligned} ct' &= \gamma(ct - \beta x), \\ x' &= \gamma(x - \beta ct), \\ y' &= y, \\ z' &= z. \end{aligned} \tag{14.20}$$

The inverse transformation is obtained simply by replacing v with $-v$.

14.3 Relativity of simultaneity

The time transformation contains position:

$$t' = \gamma \left(t - \frac{vx}{c^2} \right). \tag{14.21}$$

Two events that are simultaneous in S have $\Delta t = 0$, but

$$\Delta t' = -\gamma \frac{v\Delta x}{c^2}. \tag{14.22}$$

Unless the events occur at the same place, they are generally not simultaneous in S' .

Physical meaning

Time dilation and length contraction are not independent ad hoc effects. They both follow from the relativity of simultaneity. Different frames slice spacetime into “space at one instant” in different ways.

14.4 Time dilation from a light clock

Imagine a clock consisting of two mirrors separated by a distance h perpendicular to the direction of motion. In the clock’s rest frame one tick corresponds to a round-trip light path $2h$, so

$$\Delta\tau = \frac{2h}{c}. \tag{14.23}$$

To a laboratory observer the clock moves sideways while the light travels, so each half-path is the hypotenuse of a right triangle:

$$\left(c \frac{\Delta t}{2} \right)^2 = h^2 + \left(v \frac{\Delta t}{2} \right)^2. \tag{14.24}$$

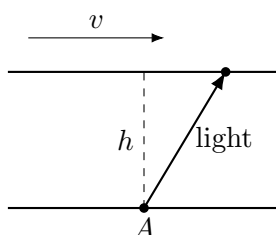
Using $h = c\Delta\tau/2$ gives

$$\Delta t^2(1 - \beta^2) = \Delta\tau^2, \tag{14.25}$$

and therefore

$$\Delta t = \gamma \Delta\tau. \tag{14.26}$$

The moving clock accumulates less time between the same two events on its worldline.



14.5 Length contraction

Let a rod have rest length L_0 in S' . To measure its length in S , the positions of its two ends must be recorded at the *same laboratory time*, so $\Delta t = 0$. The Lorentz transformation gives

$$\Delta x' = \gamma \Delta x. \quad (14.27)$$

Since $\Delta x' = L_0$ and $\Delta x = L$,

$$L = \frac{L_0}{\gamma}. \quad (14.28)$$

The requirement of simultaneity in the measuring frame is essential. Without specifying which endpoint events are compared, the phrase “the length of a moving object” is incomplete.

14.6 The invariant spacetime interval

For two events define

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \quad (14.29)$$

Substituting the Lorentz transformation shows directly that

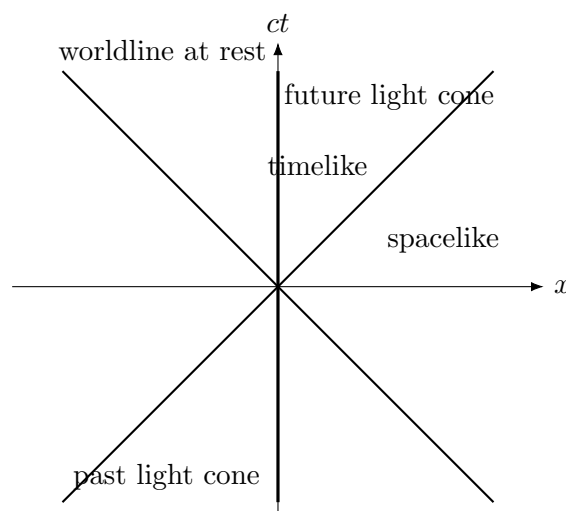
$$\Delta s'^2 = \Delta s^2. \quad (14.30)$$

This invariant classifies separations:

$$\Delta s^2 > 0 : \quad \text{timelike}, \quad (14.31)$$

$$\Delta s^2 = 0 : \quad \text{lightlike or null}, \quad (14.32)$$

$$\Delta s^2 < 0 : \quad \text{spacelike}. \quad (14.33)$$



For timelike-separated events, every inertial observer agrees on their time order. A causal influence travelling at or below c can connect them. For spacelike-separated events, different inertial observers can disagree on which event occurred first, but no signal travelling at or below c can connect them.

Physical meaning

Causality survives the relativity of simultaneity because causal ordering is invariant for timelike and lightlike separations. Only spacelike-separated events can have their order reversed by a Lorentz transformation, and those events cannot influence one another without superluminal propagation.

14.7 Proper time

Along the worldline of a massive particle,

$$c^2 d\tau^2 = c^2 dt^2 - d\mathbf{r}^2. \quad (14.34)$$

Since $d\mathbf{r}^2 = u^2 dt^2$,

$$\boxed{d\tau = dt \sqrt{1 - \frac{u^2}{c^2}} = \frac{dt}{\gamma_u}}. \quad (14.35)$$

The proper time is exactly the time read by a clock moving with the particle. Because it is built from the invariant interval, every inertial observer agrees on the proper time accumulated between two events on the same worldline.

14.8 Velocity addition

Differentiate the Lorentz transformation:

$$dx' = \gamma(dx - vdt), \quad (14.36)$$

$$dt' = \gamma \left(dt - \frac{vdx}{c^2} \right). \quad (14.37)$$

Therefore

$$\boxed{u'_x = \frac{u_x - v}{1 - u_x v / c^2}}. \quad (14.38)$$

For the transverse components,

$$\boxed{u'_y = \frac{u_y}{\gamma(1 - u_x v / c^2)}}, \quad (14.39)$$

$$\boxed{u'_z = \frac{u_z}{\gamma(1 - u_x v / c^2)}}. \quad (14.40)$$

If $u_x = c$, then

$$u'_x = \frac{c - v}{1 - v/c} = c, \quad (14.41)$$

as required.

14.9 Four-position and four-velocity

The Lorentz transformation becomes much simpler when space and time are combined into a four-vector,

$$\boxed{x^\mu = (ct, \mathbf{r})}. \quad (14.42)$$

The natural derivative parameter is proper time, because $d\tau$ is invariant. Define the four-velocity

$$\boxed{U^\mu = \frac{dx^\mu}{d\tau} = \gamma_u(c, \mathbf{u})}. \quad (14.43)$$

Its Minkowski norm is

$$U^\mu U_\mu = \gamma_u^2(c^2 - u^2) \quad (14.44)$$

$$= \gamma_u^2 c^2 (1 - u^2/c^2) \quad (14.45)$$

$$= \boxed{c^2}. \quad (14.46)$$

Thus every massive particle has a four-velocity of the same invariant magnitude c , even though its ordinary three-velocity can vary.

Checkpoint

The central conceptual move is to stop asking which quantities remain numerically unchanged and ask which *spacetime objects* transform simply. Proper time is invariant, x^μ and U^μ are four-vectors, and the invariant interval replaces the Newtonian separation of absolute space and absolute time.

Lecture 15: Relativistic Energy, Momentum and Dynamics

Aim of this lecture

Construct relativistic energy and momentum from four-velocity, use four-momentum conservation to solve decay and scattering problems efficiently, understand how force and acceleration are related relativistically, and introduce the four-force needed for covariant electrodynamics.

Motivation for this lecture

Relativistic kinematics tells us how spacetime coordinates and velocities transform, but a physical theory also needs dynamics: energy, momentum, force and conservation laws. The Newtonian definitions cannot simply be carried over because they would not assemble into Lorentz-covariant quantities. Four-velocity leads naturally to four-momentum, whose invariant norm gives the relativistic energy-momentum relation. Once this structure is in place, decay and scattering problems become much simpler and the Lorentz force can be prepared for a fully covariant formulation.

15.1 Four-momentum

For a particle of rest mass m , multiply the four-velocity by m :

$$p^\mu = mU^\mu = (\gamma mc, \gamma m\mathbf{u}). \quad (15.1)$$

We identify

$$\mathbf{p} = \gamma m\mathbf{u} \quad (15.2)$$

as the relativistic three-momentum and

$$E = \gamma mc^2 \quad (15.3)$$

as the relativistic total energy. Thus

$$p^\mu = \left(\frac{E}{c}, \mathbf{p} \right). \quad (15.4)$$

The invariant norm follows immediately from $U^\mu U_\mu = c^2$:

$$p^\mu p_\mu = m^2 c^2 \quad (15.5)$$

$$= \frac{E^2}{c^2} - p^2. \quad (15.6)$$

Hence

$$E^2 = p^2 c^2 + m^2 c^4. \quad (15.7)$$

For a massless particle,

$$E = pc. \quad (15.8)$$

15.2 Rest energy and kinetic energy

Even at rest, a particle has energy

$$\boxed{E_{\text{rest}} = mc^2.} \quad (15.9)$$

The kinetic energy is the excess above the rest energy,

$$\boxed{K = (\gamma - 1)mc^2.} \quad (15.10)$$

For $u \ll c$, use

$$\gamma = (1 - u^2/c^2)^{-1/2} = 1 + \frac{1}{2} \frac{u^2}{c^2} + \frac{3}{8} \frac{u^4}{c^4} + \cdots, \quad (15.11)$$

so

$$\boxed{K = \frac{1}{2}mu^2 + \frac{3}{8} \frac{mu^4}{c^2} + \cdots.} \quad (15.12)$$

Newtonian kinetic energy is simply the leading low-speed approximation.

15.3 Conservation of four-momentum

For an isolated system,

$$\boxed{\sum_{\text{in}} p^\mu = \sum_{\text{out}} p^\mu.} \quad (15.13)$$

The time component is conservation of total energy; the spatial components are conservation of three-momentum. Writing them as one four-vector equation has two advantages: covariance is manifest, and invariant squares often eliminate unknown momenta immediately.

Rest mass of the individual particles need not sum to the same value before and after a reaction. For example, rest energy can become kinetic energy. What is invariant for the *complete isolated system* is the norm of its total four-momentum,

$$M_{\text{sys}}^2 c^2 = P_{\text{tot}}^\mu P_{\text{tot},\mu}. \quad (15.14)$$

15.4 Two-body decay at rest: a line-by-line derivation

Let a parent particle of mass M decay at rest into particles of masses m_1 and m_2 :

$$P^\mu = (Mc, \mathbf{0}) = p_1^\mu + p_2^\mu. \quad (15.15)$$

Therefore

$$p_2^\mu = P^\mu - p_1^\mu. \quad (15.16)$$

Square both sides using the Minkowski inner product:

$$p_2^2 = (P - p_1)^2 \quad (15.17)$$

$$= P^2 + p_1^2 - 2P \cdot p_1. \quad (15.18)$$

Now

$$p_2^2 = m_2^2 c^2, \quad P^2 = M^2 c^2, \quad p_1^2 = m_1^2 c^2. \quad (15.19)$$

Because the parent is at rest,

$$P \cdot p_1 = Mc \frac{E_1}{c} = ME_1. \quad (15.20)$$

Hence

$$m_2^2 c^2 = M^2 c^2 + m_1^2 c^2 - 2ME_1, \quad (15.21)$$

and therefore

$$E_1 = \frac{M^2 + m_1^2 - m_2^2}{2M} c^2. \quad (15.22)$$

Similarly,

$$E_2 = \frac{M^2 + m_2^2 - m_1^2}{2M} c^2. \quad (15.23)$$

Momentum conservation then gives $|\mathbf{p}_1| = |\mathbf{p}_2|$. This invariant method is usually much shorter than solving energy and momentum components separately.

For a pion at rest decaying into a muon and an approximately massless neutrino,

$$E_\mu \simeq \frac{m_\pi^2 + m_\mu^2}{2m_\pi} c^2, \quad (15.24)$$

which is the result appearing in the earlier handwritten notes.

15.5 Compton scattering

A photon of initial four-momentum k_0^μ scatters from an electron initially at rest with four-momentum p^μ . After scattering the photon has four-momentum k^μ and the electron p'^μ :

$$k_0 + p = k + p'. \quad (15.25)$$

Rearrange,

$$p' = p + k_0 - k, \quad (15.26)$$

and square both sides. Since

$$p'^2 = p^2 = m^2 c^2, \quad k_0^2 = k^2 = 0, \quad (15.27)$$

we find

$$0 = 2p \cdot (k_0 - k) - 2k_0 \cdot k. \quad (15.28)$$

The initial electron is at rest, so

$$p^\mu = (mc, \mathbf{0}), \quad (15.29)$$

using $k^\mu = (E/c, \mathbf{k})$ with $|\mathbf{k}| = E/c$,

$$p \cdot k_0 = mE_0, \quad p \cdot k = mE. \quad (15.30)$$

For a photon scattering angle θ ,

$$k_0 \cdot k = \frac{E_0 E}{c^2} (1 - \cos \theta). \quad (15.31)$$

Thus

$$m(E_0 - E) = \frac{E_0 E}{c^2} (1 - \cos \theta). \quad (15.32)$$

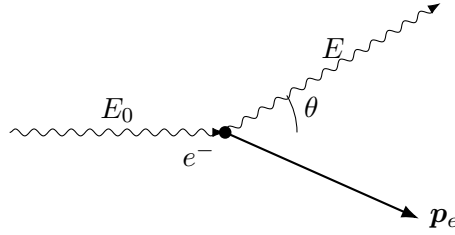
Divide by $E_0 E$:

$$\frac{1}{E} - \frac{1}{E_0} = \frac{1 - \cos \theta}{mc^2}. \quad (15.33)$$

Using $E = hc/\lambda$ gives

$$\lambda - \lambda_0 = \frac{h}{mc} (1 - \cos \theta). \quad (15.34)$$

The scale $h/(mc)$ is the Compton wavelength of the electron.



15.6 Relativistic dynamics

Newton's second law retains the form

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}, \quad (15.35)$$

but $\mathbf{p} = \gamma m \mathbf{u}$. Differentiate:

$$\mathbf{F} = m \frac{d}{dt}(\gamma \mathbf{u}) \quad (15.36)$$

$$= m\gamma \mathbf{a} + m\mathbf{u} \frac{d\gamma}{dt}. \quad (15.37)$$

Since

$$\frac{d\gamma}{dt} = \gamma^3 \frac{\mathbf{u} \cdot \mathbf{a}}{c^2}, \quad (15.38)$$

we have

$$\mathbf{F} = m\gamma \mathbf{a} + m\gamma^3 \frac{\mathbf{u} \cdot \mathbf{a}}{c^2} \mathbf{u}. \quad (15.39)$$

For acceleration parallel to the velocity,

$$F_{\parallel} = \gamma^3 m a_{\parallel}, \quad (15.40)$$

whereas for transverse acceleration,

$$F_{\perp} = \gamma m a_{\perp}. \quad (15.41)$$

These relations explain, for example, the factor γ in the magnetic bending equation used in Lecture 12.

15.7 Motion under a constant longitudinal force

If a constant force F acts along the motion and the particle starts from rest,

$$\frac{dp}{dt} = F \implies p = Ft. \quad (15.42)$$

But

$$p = \frac{mu}{\sqrt{1 - u^2/c^2}}. \quad (15.43)$$

Solving for u gives

$$u(t) = \frac{Ft/m}{\sqrt{1 + (Ft/mc)^2}}. \quad (15.44)$$

For short times this reduces to $u \simeq Ft/m$. For long times,

$$\lim_{t \rightarrow \infty} u(t) = c. \quad (15.45)$$

The momentum and energy continue to grow, but the speed approaches c asymptotically rather than exceeding it.

15.8 Power delivered by a force

Starting from

$$E^2 = p^2 c^2 + m^2 c^4 \quad (15.46)$$

and differentiating,

$$2E \frac{dE}{dt} = 2c^2 \mathbf{p} \cdot \frac{d\mathbf{p}}{dt}. \quad (15.47)$$

Using $\mathbf{p} = \gamma m \mathbf{u}$, $E = \gamma m c^2$ and $d\mathbf{p}/dt = \mathbf{F}$ gives

$$\boxed{\frac{dE}{dt} = \mathbf{F} \cdot \mathbf{u}.} \quad (15.48)$$

Thus the familiar work–energy relation survives unchanged in form.

15.9 Four-force

Define

$$\boxed{K^\mu = \frac{dp^\mu}{d\tau}.} \quad (15.49)$$

Since $dt/d\tau = \gamma_u$,

$$K^\mu = \gamma_u \left(\frac{1}{c} \frac{dE}{dt}, \frac{d\mathbf{p}}{dt} \right). \quad (15.50)$$

Therefore

$$\boxed{K^\mu = \gamma_u \left(\frac{\mathbf{F} \cdot \mathbf{u}}{c}, \mathbf{F} \right).} \quad (15.51)$$

Because $p^\mu p_\mu = m^2 c^2$ is constant for fixed rest mass,

$$0 = \frac{d}{d\tau} (p^\mu p_\mu) \quad (15.52)$$

$$= 2p_\mu K^\mu, \quad (15.53)$$

so

$$\boxed{K^\mu p_\mu = 0.} \quad (15.54)$$

The four-force is Minkowski-orthogonal to the four-momentum.

Physical meaning

In Newtonian mechanics, force determines acceleration directly through one scalar mass. Relativistically the relation depends on whether the force changes the magnitude or the direction of the velocity. Four-momentum is the cleaner object: its rate of change transforms covariantly, while the three-dimensional force–acceleration relation is frame dependent.

Checkpoint

Energy and momentum are not separate conservation laws accidentally packaged together. They are the components of one conserved four-vector. This is why invariant four-momentum methods make decay and scattering calculations so efficient, and why they prepare us naturally for the covariant Lorentz force in Lecture 17.

Worked example: threshold energy from an invariant

A good illustration of four-vector methods is a threshold calculation. Suppose a high-energy photon of energy E collides head-on with a low-energy photon of energy ϵ . What is the minimum E required to produce an electron-positron pair?

$$\gamma + \gamma \rightarrow e^- + e^+. \quad (15.55)$$

The invariant quantity is

$$s = (p_1 + p_2)^2. \quad (15.56)$$

For photons $p_1^2 = p_2^2 = 0$, so

$$s = 2p_1 \cdot p_2. \quad (15.57)$$

For a head-on collision, the spatial momenta are antiparallel, and

$$p_1 \cdot p_2 = \frac{E\epsilon}{c^2} - \mathbf{p}_1 \cdot \mathbf{p}_2 = \frac{E\epsilon}{c^2} + \frac{E\epsilon}{c^2} = \frac{2E\epsilon}{c^2}. \quad (15.58)$$

Thus

$$s = \frac{4E\epsilon}{c^2}. \quad (15.59)$$

At threshold the final electron and positron have no kinetic energy in the centre-of-momentum frame, so

$$s_{\text{th}} = (2m_e c)^2 = 4m_e^2 c^2. \quad (15.60)$$

Equating the two expressions gives

$$\boxed{E_{\text{th}} = \frac{m_e^2 c^4}{\epsilon}}. \quad (15.61)$$

The calculation is short because we never transform explicitly to the centre-of-momentum frame; the invariant s carries that information for us.

Lecture 16: Electromagnetic Fields under Lorentz Transformations

Aim of this lecture

Show explicitly how electric and magnetic fields mix between inertial frames, derive the transformation laws from physical examples, use the field invariants to classify electromagnetic configurations, and understand hidden momentum as a relativistic consistency requirement.

Motivation for this lecture

If special relativity is correct, the electric and magnetic fields measured by one observer cannot be absolute objects: a source that is purely electric in one frame may carry a current and therefore a magnetic field in another. We now determine explicitly how $\mathbf{E}$ and $\mathbf{B}$ mix under Lorentz transformations and identify combinations of the fields that remain invariant. The same relativistic consistency also explains apparently paradoxical systems such as a stationary magnetic dipole in an electric field, where hidden mechanical momentum is required to balance electromagnetic field momentum.

16.1 Why electric and magnetic fields must mix

Suppose a set of charges is at rest in one frame. It produces a charge density and an electric field. An observer moving relative to those charges sees them moving, so the same source now also constitutes a current and therefore produces a magnetic field. The distinction between “electric source” and “magnetic source” is therefore frame dependent.

This observation already tells us that $\mathbf{E}$ and $\mathbf{B}$ cannot transform independently. They must mix in exactly the way required to make the Lorentz force describe the same physical particle trajectory in every inertial frame.

For a boost with velocity $\mathbf{v}$ from S to S' , split each field into parts parallel and perpendicular to $\mathbf{v}$:

$$\mathbf{E} = \mathbf{E}_{\parallel} + \mathbf{E}_{\perp}, \quad \mathbf{B} = \mathbf{B}_{\parallel} + \mathbf{B}_{\perp}. \quad (16.1)$$

The transformation laws are

$$\boxed{\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}}, \quad (16.2)$$

$$\boxed{\mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}}, \quad (16.3)$$

$$\boxed{\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B})}, \quad (16.4)$$

$$\boxed{\mathbf{B}'_{\perp} = \gamma\left(\mathbf{B}_{\perp} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}\right)}. \quad (16.5)$$

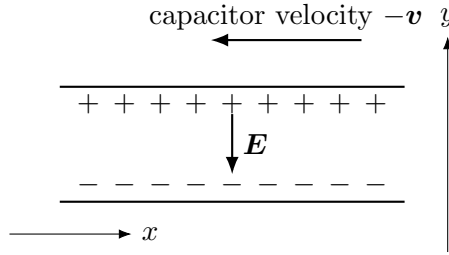
We will motivate these relations physically before recognising them in Lecture 17 as one tensor transformation law.

16.2 A capacitor at rest

Consider a parallel-plate capacitor at rest in frame S_0 . Let its plates lie in the xz plane, with surface charge densities $+\sigma_0$ and $-\sigma_0$. Between the plates,

$$\mathbf{E}_0 = E_0 \hat{\mathbf{y}}, \quad E_0 = \frac{\sigma_0}{\epsilon_0}, \quad \mathbf{B}_0 = 0. \quad (16.6)$$

Now observe the capacitor from a frame S moving in the $+x$ direction with speed v relative to S_0 . In S the capacitor moves in the $-x$ direction.



The dimension of each plate parallel to the motion is Lorentz contracted. The total charge on a plate is invariant, so the surface charge density increases:

$$\boxed{\sigma = \gamma \sigma_0}. \quad (16.7)$$

Hence the electric field perpendicular to the boost becomes

$$\boxed{E_y = \gamma E_{0y}}. \quad (16.8)$$

This is the special case $\mathbf{E}_\perp = \gamma \mathbf{E}_{0\perp}$ when the original frame has $\mathbf{B}_0 = 0$.

16.3 The same moving plates generate a magnetic field

In frame S , the moving surface charge is a surface current. The upper positive plate and lower negative plate carry oppositely directed surface currents, and their magnetic fields add between the plates. The result is

$$\boxed{B_z = -\frac{v}{c^2} E_y}. \quad (16.9)$$

Using $E_y = \gamma E_{0y}$,

$$B_z = -\gamma \frac{v}{c^2} E_{0y}. \quad (16.10)$$

This is exactly the transformation

$$\mathbf{B}_\perp = \gamma \left(\mathbf{B}_{0\perp} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E}_0 \right) \quad (16.11)$$

with $\mathbf{B}_0 = 0$ and $\mathbf{v} = v \hat{\mathbf{x}}$.

If instead the capacitor is oriented so that its electric field is parallel to the boost, the plate area is not Lorentz contracted and one obtains

$$\boxed{\mathbf{E}_\parallel = \mathbf{E}_{0\parallel}}. \quad (16.12)$$

Rotating these examples supplies the remaining component relations.

16.4 Component form for a boost along $\mathbf{x}$

For $\mathbf{v} = v\hat{\mathbf{x}}$,

$$E'_x = E_x, \quad (16.13)$$

$$E'_y = \gamma(E_y - vB_z), \quad (16.14)$$

$$E'_z = \gamma(E_z + vB_y), \quad (16.15)$$

$$B'_x = B_x, \quad (16.16)$$

$$B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right), \quad (16.17)$$

$$B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right). \quad (16.18)$$

These six equations are often the most convenient form for calculations.

The equivalent compact vector expressions are

$$\boxed{\mathbf{E}' = \gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1} \frac{\mathbf{v}}{c^2} (\mathbf{v} \cdot \mathbf{E})}, \quad (16.19)$$

$$\boxed{\mathbf{B}' = \gamma\left(\mathbf{B} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}\right) - \frac{\gamma^2}{\gamma + 1} \frac{\mathbf{v}}{c^2} (\mathbf{v} \cdot \mathbf{B})}. \quad (16.20)$$

The apparently more complicated final terms simply ensure that the parallel components remain unchanged.

16.5 A uniformly moving point charge revisited

In the rest frame of a point charge,

$$\mathbf{B}_0 = 0, \quad \mathbf{E}_0 = \frac{q}{4\pi\epsilon_0 R_0^2} \hat{\mathbf{R}}_0. \quad (16.21)$$

Lorentz-transforming this field gives, in a frame where the charge moves uniformly,

$$\boxed{\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{1 - \beta^2}{(1 - \beta^2 \sin^2 \theta)^{3/2}} \frac{\hat{\mathbf{R}}}{R^2}}, \quad (16.22)$$

where $\mathbf{R}$ is the vector from the instantaneous position of the charge to the field point. The magnetic field is

$$\boxed{\mathbf{B} = \frac{1}{c^2} \mathbf{v} \times \mathbf{E}}. \quad (16.23)$$

These are exactly the uniform-motion fields obtained from the Liénard–Wiechert potentials in Lecture 10. Thus two very different routes—retarded potentials and Lorentz transformation—give the same answer.

16.6 The electromagnetic field invariants

Although $\mathbf{E}$ and $\mathbf{B}$ separately change between frames, two combinations do not:

$$\boxed{B^2 - \frac{\mathbf{E}^2}{c^2} \text{ is invariant}}, \quad (16.24)$$

$$\boxed{\mathbf{E} \cdot \mathbf{B} \text{ is invariant.}} \quad (16.25)$$

We will see in Lecture 17 that these arise naturally by contracting the electromagnetic field tensor with itself and with its dual.

These invariants allow us to classify fields without finding the transformation explicitly.

- If $\mathbf{E} \cdot \mathbf{B} \neq 0$, no inertial frame can make either field vanish completely.
- If $\mathbf{E} \cdot \mathbf{B} = 0$ and $E^2 > c^2 B^2$, the field is *electric-like*: there exists a frame in which $\mathbf{B}' = 0$.
- If $\mathbf{E} \cdot \mathbf{B} = 0$ and $c^2 B^2 > E^2$, the field is *magnetic-like*: there exists a frame in which $\mathbf{E}' = 0$.
- If both invariants vanish, the field is *null*. A plane electromagnetic wave is the canonical example; no inertial frame can transform away its field.

For a plane wave,

$$\mathbf{E} \perp \mathbf{B}, \quad E = cB, \quad (16.26)$$

so indeed

$$\mathbf{E} \cdot \mathbf{B} = 0, \quad B^2 - \frac{E^2}{c^2} = 0. \quad (16.27)$$

Physical meaning

The invariants answer a qualitative question before any algebra is done: is a magnetic field merely the relativistic appearance of a purely electric field, or is there irreducible magnetic content? The answer is encoded in $\mathbf{E} \cdot \mathbf{B}$ and $E^2 - c^2 B^2$.

16.7 Lorentz force and work

The three-force on charge q is

$$\boxed{\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}).} \quad (16.28)$$

Dotting with the particle velocity gives

$$\mathbf{F} \cdot \mathbf{u} = q\mathbf{u} \cdot \mathbf{E} + q\mathbf{u} \cdot (\mathbf{u} \times \mathbf{B}) \quad (16.29)$$

$$= q\mathbf{u} \cdot \mathbf{E}. \quad (16.30)$$

Therefore

$$\boxed{\frac{dE}{dt} = q\mathbf{u} \cdot \mathbf{E}.} \quad (16.31)$$

The magnetic field changes the direction of momentum but does no instantaneous work.

16.8 Hidden momentum: the puzzle

Consider a rectangular current loop in a uniform external electric field. The loop is stationary as a whole and has magnetic moment $\mathbf{m}$. Classically one might expect its total mechanical momentum to be zero. Relativity changes this conclusion because charges moving through different parts of the loop have slightly different energies and therefore different momenta.

Take a loop of horizontal length ℓ and vertical width w . Let the moving charge carriers have charge Q and rest mass M . Suppose there are N_+ carriers in the upper horizontal segment moving with

speed u_+ and N_- in the lower segment moving oppositely with speed u_- . The same current flows everywhere, so

$$I = \frac{QN_+u_+}{\ell} = \frac{QN_-u_-}{\ell}. \quad (16.32)$$

Thus

$$N_+u_+ = N_-u_- = \frac{I\ell}{Q}. \quad (16.33)$$

The relativistic momentum in the two horizontal segments does not cancel if the Lorentz factors differ:

$$p_{\text{mech}} = M(\gamma_+N_+u_+ - \gamma_-N_-u_-) \quad (16.34)$$

$$= M(\gamma_+ - \gamma_-)\frac{I\ell}{Q}. \quad (16.35)$$

As a carrier moves through the vertical side of the loop, the electric field does work QEw . Hence

$$(\gamma_+ - \gamma_-)Mc^2 = QEw. \quad (16.36)$$

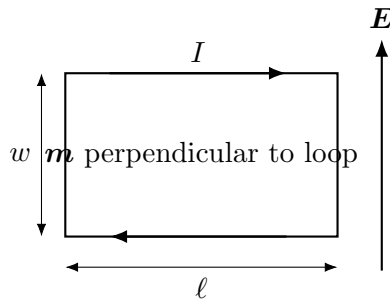
Substituting this into the momentum gives

$$p_{\text{mech}} = \frac{I\ell w E}{c^2}. \quad (16.37)$$

Since $A = \ell w$ and $m = IA$,

$$\boxed{p_{\text{hidden}} = \frac{\mathbf{m} \times \mathbf{E}}{c^2}}. \quad (16.38)$$

The vector direction follows from the orientation of the current loop and the electric field.



The electromagnetic field itself carries an equal and opposite momentum in the static closed system, so the total momentum can still vanish.

Physical meaning

“The object is at rest” does not imply “all microscopic momenta add to zero.” Relativistic energy flow through a stationary current distribution can carry mechanical momentum. Hidden momentum is required so that mechanical and electromagnetic momentum transform consistently.

Checkpoint

Electric and magnetic fields are frame-dependent pieces of one relativistic electromagnetic field. The transformation laws explain the magnetic field of moving charge, the invariants tell us what cannot be transformed away, and hidden momentum shows that the same relativistic consistency extends to the momentum carried by matter and fields.

Worked example: finding a frame with no magnetic field

Suppose at some point in spacetime the electromagnetic field satisfies

$$\mathbf{E} \cdot \mathbf{B} = 0, \quad E^2 > c^2 B^2. \quad (16.39)$$

The field is electric-like. There should be an inertial frame in which the magnetic field vanishes. Choose a boost velocity perpendicular to both fields,

$$\mathbf{v} = \frac{c^2 \mathbf{E} \times \mathbf{B}}{E^2}. \quad (16.40)$$

Because $E^2 > c^2 B^2$, this speed is less than c :

$$\frac{v^2}{c^2} = \frac{c^2 B^2}{E^2} < 1. \quad (16.41)$$

Using the transformation law for the magnetic field perpendicular to the boost,

$$\mathbf{B}' = \gamma \left(\mathbf{B} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \right), \quad (16.42)$$

and substituting $\mathbf{v} = c^2 \mathbf{E} \times \mathbf{B} / E^2$, we find

$$\mathbf{v} \times \mathbf{E} = \frac{c^2}{E^2} (\mathbf{E} \times \mathbf{B}) \times \mathbf{E} = c^2 \mathbf{B}, \quad (16.43)$$

where $\mathbf{E} \cdot \mathbf{B} = 0$ has been used. Hence

$$\boxed{\mathbf{B}' = 0}. \quad (16.44)$$

This example gives a concrete meaning to the invariant classification. If $E^2 - c^2 B^2 > 0$ and $\mathbf{E} \cdot \mathbf{B} = 0$, the field can be made purely electric by going to the right inertial frame.

Lecture 17: Covariant Electrodynamics

Aim of this lecture

Combine electric and magnetic fields into the electromagnetic field tensor, write Maxwell's equations and the Lorentz force covariantly, describe gauge transformations and charge conservation, and connect the field equations with the particle Lagrangian.

Motivation for this lecture

The component-by-component field transformations of Lecture 16 work, but they hide the underlying simplicity. Relativity suggests that $\mathbf{E}$ and $\mathbf{B}$ should be different components of a single spacetime object, just as energy and momentum are components of four-momentum. Introducing the electromagnetic field tensor makes Lorentz covariance manifest and compresses Maxwell's equations and the Lorentz force into compact tensor equations. Gauge transformations, the four-current and covariant conservation laws then appear as parts of the same structure rather than as separate facts.

17.1 Four-vectors and tensors

With metric

$$g_{\mu\nu} = \text{diag}(1, -1, -1, -1),$$

we write

$$x^\mu = (ct, x, y, z), \quad \partial_\mu = \left(\frac{1}{c} \partial_t, \nabla \right), \quad \partial^\mu = \left(\frac{1}{c} \partial_t, -\nabla \right).$$

A rank-two tensor transforms as

$$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta T^{\alpha\beta}. \quad (17.1)$$

Symmetry or antisymmetry is preserved under Lorentz transformations.

17.2 Four-potential and field tensor

Define the four-potential

$$A^\mu = \left(\frac{\phi}{c}, \mathbf{A} \right). \quad (17.2)$$

The electromagnetic field tensor is

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu. \quad (17.3)$$

With our conventions,

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}. \quad (17.4)$$

A different placement of signs is possible if a different metric or index convention is chosen; physical predictions are unchanged provided the convention is used consistently.

Physical meaning

Electric and magnetic fields are not separate relativistic objects. They are six components of one antisymmetric tensor $F^{\mu\nu}$. The field-transformation laws of Lecture 16 are simply the tensor transformation law written component by component.

17.3 Dual tensor and field invariants

Define

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}. \quad (17.5)$$

The two independent Lorentz scalars constructed quadratically from the fields are proportional to

$$F^{\mu\nu}F_{\mu\nu} = 2\left(B^2 - \frac{E^2}{c^2}\right), \quad (17.6)$$

$$F^{\mu\nu}\tilde{F}_{\mu\nu} \propto -\frac{1}{c}\mathbf{E} \cdot \mathbf{B}. \quad (17.7)$$

These reproduce the invariants identified in Lecture 16.

17.4 Four-current and charge conservation

The charge and current densities form the four-current

$$J^\mu = (c\rho, \mathbf{J}). \quad (17.8)$$

The ordinary continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

becomes

$$\partial_\mu J^\mu = 0. \quad (17.9)$$

This is local charge conservation written as a Lorentz scalar equation.

17.5 Covariant Maxwell equations

The inhomogeneous Maxwell equations are

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu. \quad (17.10)$$

For $\nu = 0$ this is Gauss's law; for the spatial components it is the Ampère–Maxwell law.

The homogeneous equations are

$$\partial_\mu \tilde{F}^{\mu\nu} = 0, \quad (17.11)$$

or equivalently

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0. \quad (17.12)$$

These contain $\nabla \cdot \mathbf{B} = 0$ and Faraday's law.

17.6 Gauge transformations

The potentials are not unique. Under

$$\boxed{A^\mu \rightarrow A'^\mu = A^\mu + \partial^\mu \chi,} \quad (17.13)$$

the field tensor is unchanged because mixed partial derivatives commute:

$$F'^{\mu\nu} = F^{\mu\nu}.$$

The Lorenz gauge is

$$\boxed{\partial_\mu A^\mu = 0.} \quad (17.14)$$

In this gauge Maxwell's equations reduce to

$$\boxed{\partial_\mu \partial^\mu A^\nu = \mu_0 J^\nu.} \quad (17.15)$$

This is the covariant form of the wave equations from Lecture 2.

17.7 Covariant Lorentz force

The Lorentz force law becomes

$$\boxed{\frac{dp^\mu}{d\tau} = q F^\mu{}_\nu U^\nu.} \quad (17.16)$$

Its spatial components reproduce $d\mathbf{p}/dt = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$, while the time component reproduces the work-energy equation $dE/dt = q\mathbf{u} \cdot \mathbf{E}$.

17.8 Electromagnetic energy–momentum conservation

The field carries energy and momentum. A covariant description uses the electromagnetic stress-energy tensor

$$\boxed{T^{\mu\nu} = \frac{1}{\mu_0} \left(-F^{\mu\alpha} F^\nu{}_\alpha + \frac{1}{4} g^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right),} \quad (17.17)$$

with our metric/sign convention. Its components contain the familiar energy density, Poynting vector and Maxwell stress tensor. Local conservation reads

$$\partial_\mu T^{\mu\nu} = -F^{\nu\lambda} J_\lambda. \quad (17.18)$$

The right-hand side is the four-force density exerted by the field on matter. When matter and field contributions are added, the total stress-energy tensor is conserved.

17.9 Particle Lagrangian and minimal coupling

For completeness, recall first the nonrelativistic Lagrangian of N interacting point particles,

$$L = \sum_i \frac{1}{2} m_i \dot{\mathbf{r}}_i^2 - V(\mathbf{r}_1, \dots, \mathbf{r}_N). \quad (17.19)$$

The Euler–Lagrange equations reproduce Newton's equations.

For a charge q moving in prescribed electromagnetic potentials, the relativistic Lagrangian is

$$L = -mc^2 \sqrt{1 - \frac{u^2}{c^2}} + q\mathbf{u} \cdot \mathbf{A} - q\phi. \quad (17.20)$$

The canonical momentum is

$$\mathbf{P} = \frac{\partial L}{\partial \mathbf{u}} = \gamma m \mathbf{u} + q\mathbf{A}, \quad (17.21)$$

which differs from the mechanical momentum by $q\mathbf{A}$. Applying the Euler–Lagrange equation gives

$$\frac{d}{dt}(\gamma m \mathbf{u}) = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}). \quad (17.22)$$

Thus the Lorentz force follows directly from minimal coupling to the potentials.

Checkpoint

The course has now come full circle. We began by introducing ϕ and $\mathbf{A}$ as convenient potentials for solving Maxwell's equations. Covariantly they combine into A^μ ; their derivatives form $F^{\mu\nu}$; gauge freedom expresses their non-uniqueness; and the same potentials enter the particle Lagrangian. Potentials are therefore not merely a computational trick: they are the natural variables of relativistic electrodynamics.

Lecture 18: The Action Principle, Symmetry and Noether's Theorem

Aim of this lecture

Understand why the action principle is a useful reformulation of dynamics, derive the Euler–Lagrange equations from Hamilton's principle, connect continuous symmetries with conserved quantities through Noether's theorem, and formulate relativistic particle dynamics and Maxwell's equations from covariant actions.

Motivation for this lecture

The covariant equations of Lecture 17 give a complete local description of particles and fields, but there is an even more structural formulation. The action principle encodes the equations of motion in a single scalar functional, making coordinate freedom and Lorentz covariance especially transparent. Its deepest payoff is the link between symmetry and conservation: Noether's theorem explains why translation, rotation and gauge symmetries are tied to conserved quantities. This final lecture therefore unifies particle dynamics, electromagnetism and conservation laws within one variational framework.

18.1 Why introduce an action principle?

Up to this point we have usually formulated dynamics locally. Newton's law tells us the acceleration at a given instant, Maxwell's equations tell us the local derivatives of the electromagnetic fields, and the Lorentz force tells us how a charged particle responds to those fields. The action principle packages the same dynamics in a rather different way: instead of asking what happens at one instant, it assigns a number to an entire possible history and identifies the physical history by a stationarity condition.

At first this may seem like an unnecessarily indirect way of stating the laws of motion. Its power becomes clear for three reasons.

1. It works naturally in arbitrary coordinates, including constrained systems where Cartesian components are inconvenient.
2. Relativistic covariance is often much easier to see at the level of the action than in component equations of motion.
3. Most importantly for this lecture, symmetries of the action lead systematically to conservation laws. This is the content of Noether's theorem.

Physical meaning

The action principle does not introduce new physics on top of Newtonian or electromagnetic dynamics. It is a more structural way of encoding the same physics. What it exposes particularly clearly is the link

$$\boxed{\text{continuous symmetry} \iff \text{conserved quantity.}}$$

Momentum, angular momentum and energy are therefore not three unrelated pieces of good fortune: they arise because the laws of physics do not depend on where we place the origin, how we orient the axes, or when we start the clock.

18.2 Generalised coordinates, the Lagrangian and the action

Consider a mechanical system described by generalised coordinates

$$q_a(t), \quad a = 1, \dots, N.$$

The coordinates need not be Cartesian positions. They can be angles, distances along constrained surfaces, or any variables that uniquely specify the configuration of the system.

The Lagrangian is a function

$$L = L(q_a, \dot{q}_a, t). \quad (18.1)$$

For the familiar class of conservative nonrelativistic systems,

$$\boxed{L = T - V,} \quad (18.2)$$

where T is the kinetic energy and V the potential energy.

To every path $q_a(t)$ joining a specified initial configuration at t_1 to a specified final configuration at t_2 , we associate the *action*

$$\boxed{S[q] = \int_{t_1}^{t_2} L(q_a, \dot{q}_a, t) dt.} \quad (18.3)$$

The action has dimensions of energy multiplied by time, the same dimensions as Planck's constant. Classically, however, no value of $\hbar$ is involved: the physical path is selected by the condition that S be stationary.

18.3 Hamilton's principle: stationary, not necessarily minimum

Suppose $q_a(t)$ is the physical path. Consider a nearby family of paths

$$q_a(t) \longrightarrow q_a(t) + \epsilon \eta_a(t), \quad (18.4)$$

where ϵ is an arbitrarily small parameter and the functions $\eta_a(t)$ vanish at the endpoints,

$$\eta_a(t_1) = \eta_a(t_2) = 0. \quad (18.5)$$

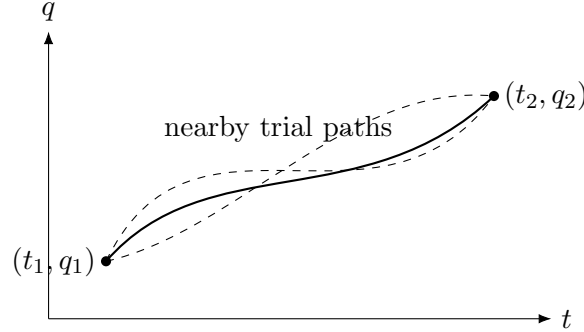
The endpoint condition means that all the trial histories begin and end at the same configurations; only the path between them is varied.

Hamilton's principle states

$$\boxed{\delta S = 0} \quad (18.6)$$

for the actual motion.

The word *stationary* matters. The physical action need not be the absolute minimum among all conceivable paths. Just as a function can have a maximum, minimum or saddle point where its first derivative vanishes, the action is required only to have zero first variation.



18.4 Derivation of the Euler–Lagrange equations

We now carry out the variation explicitly. Under $q_a \rightarrow q_a + \epsilon \eta_a$,

$$\delta q_a = \epsilon \eta_a, \quad \delta \dot{q}_a = \epsilon \dot{\eta}_a. \quad (18.7)$$

The corresponding first-order change in the action is

$$\delta S = \int_{t_1}^{t_2} dt \sum_a \left(\frac{\partial L}{\partial q_a} \delta q_a + \frac{\partial L}{\partial \dot{q}_a} \delta \dot{q}_a \right) \quad (18.8)$$

$$= \epsilon \int_{t_1}^{t_2} dt \sum_a \left(\frac{\partial L}{\partial q_a} \eta_a + \frac{\partial L}{\partial \dot{q}_a} \dot{\eta}_a \right). \quad (18.9)$$

The second term contains $\dot{\eta}_a$, whereas we want the arbitrary variation itself. We therefore integrate by parts:

$$\int_{t_1}^{t_2} dt \frac{\partial L}{\partial \dot{q}_a} \dot{\eta}_a = \left[\frac{\partial L}{\partial \dot{q}_a} \eta_a \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} dt \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_a} \right) \eta_a. \quad (18.10)$$

The boundary term vanishes because $\eta_a(t_1) = \eta_a(t_2) = 0$. Hence

$$\delta S = \epsilon \int_{t_1}^{t_2} dt \sum_a \left[\frac{\partial L}{\partial q_a} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_a} \right) \right] \eta_a(t). \quad (18.11)$$

The functions $\eta_a(t)$ are otherwise arbitrary. The integral can therefore vanish for every allowed variation only if each coefficient vanishes:

$$\boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_a} \right) - \frac{\partial L}{\partial q_a} = 0.} \quad (18.12)$$

These are the Euler–Lagrange equations.

Physical meaning

A differential equation of motion has emerged from a global statement about an entire path. The reason the final result is nevertheless local is that the variation $\eta_a(t)$ can be chosen independently in an arbitrarily small neighbourhood of any time.

Worked example: the harmonic oscillator from stationary action

For a mass m attached to an ideal spring,

$$L(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2. \quad (18.13)$$

Then

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \quad \frac{\partial L}{\partial x} = -kx. \quad (18.14)$$

The Euler–Lagrange equation gives

$$\frac{d}{dt}(m\dot{x}) + kx = 0, \quad (18.15)$$

so

$$\boxed{m\ddot{x} = -kx.} \quad (18.16)$$

Thus Hamilton’s principle reproduces the familiar Newton equation. The advantage is not that this elementary problem becomes shorter, but that the same procedure continues to work when the coordinates, constraints and symmetries become much less obvious.

18.5 A closed system of nonrelativistic point particles

For N particles with positions $\mathbf{r}_i$ and a potential that depends only on their positions,

$$L = \sum_{i=1}^N \frac{1}{2}m_i\dot{\mathbf{r}}_i^2 - V(\mathbf{r}_1, \dots, \mathbf{r}_N). \quad (18.17)$$

Treating every Cartesian component as a generalised coordinate gives

$$\frac{d}{dt}(m_i\dot{\mathbf{r}}_i) = -\nabla_i V, \quad (18.18)$$

which is simply Newton’s second law for each particle.

For a closed system the interaction potential commonly depends only on relative positions, for example

$$V = V(\mathbf{r}_1 - \mathbf{r}_2, \mathbf{r}_1 - \mathbf{r}_3, \dots). \quad (18.19)$$

This apparently innocent statement already contains the seeds of momentum and angular-momentum conservation: a common translation of all particles leaves every relative separation unchanged, and if V depends only on distances, a common rotation also leaves V unchanged.

18.6 Canonical momentum and cyclic coordinates

The momentum conjugate to q_a is defined by

$$\boxed{p_a \equiv \frac{\partial L}{\partial \dot{q}_a}.} \quad (18.20)$$

The Euler–Lagrange equation may therefore be written

$$\dot{p}_a = \frac{\partial L}{\partial q_a}. \quad (18.21)$$

If L does not depend explicitly on one coordinate q_a , then

$$\frac{\partial L}{\partial q_a} = 0 \quad \implies \quad \boxed{p_a = \text{constant.}} \quad (18.22)$$

Such a coordinate is called *cyclic* or *ignorable*. This is the simplest example of a symmetry producing a conservation law.

18.7 Time-translation invariance and energy conservation

Define

$$H \equiv \sum_a p_a \dot{q}_a - L. \quad (18.23)$$

At this stage H is simply a useful combination of the Lagrangian variables. Differentiate it:

$$\frac{dH}{dt} = \sum_a (\dot{p}_a \dot{q}_a + p_a \ddot{q}_a) - \sum_a \left(\frac{\partial L}{\partial q_a} \dot{q}_a + \frac{\partial L}{\partial \dot{q}_a} \ddot{q}_a \right) - \frac{\partial L}{\partial t}. \quad (18.24)$$

Using $p_a = \partial L / \partial \dot{q}_a$ and $\dot{p}_a = \partial L / \partial q_a$, all terms cancel except the last:

$$\boxed{\frac{dH}{dt} = -\frac{\partial L}{\partial t}.} \quad (18.25)$$

Hence, if the Lagrangian has no explicit time dependence,

$$\boxed{H = \text{constant.}} \quad (18.26)$$

For $L = T - V$ with the usual velocity-independent potential and a quadratic kinetic energy,

$$H = T + V = E, \quad (18.27)$$

so this constant is the ordinary mechanical energy.

Physical meaning

The statement “energy is conserved” is therefore tied to a symmetry: the laws describing a closed system are unchanged if the experiment is carried out tomorrow rather than today. Time-translation invariance is the origin of energy conservation.

18.8 Noether’s theorem in mechanics

We now make the symmetry–conservation connection systematic. Consider an infinitesimal continuous transformation

$$q_a \longrightarrow q_a + \epsilon X_a(q, t), \quad (18.28)$$

where ϵ is small. Suppose the Lagrangian changes at most by a total time derivative,

$$\delta L = \epsilon \frac{dF}{dt}. \quad (18.29)$$

A total derivative changes the action only by endpoint terms, so it does not change the equations of motion when the endpoints are fixed.

On a trajectory that satisfies the Euler–Lagrange equations,

$$\delta L = \epsilon \sum_a \left[\frac{\partial L}{\partial q_a} X_a + \frac{\partial L}{\partial \dot{q}_a} \frac{dX_a}{dt} \right] \quad (18.30)$$

$$= \epsilon \sum_a \left[\dot{p}_a X_a + p_a \dot{X}_a \right] \quad (18.31)$$

$$= \epsilon \frac{d}{dt} \left(\sum_a p_a X_a \right). \quad (18.32)$$

Comparing this with $\delta L = \epsilon dF/dt$ gives

$$Q = \sum_a p_a X_a - F = \text{constant}. \quad (18.33)$$

This is a useful form of Noether’s theorem for transformations that do not explicitly change the time coordinate.

Checkpoint

Noether’s theorem says more than “if you notice a cyclic coordinate, something is conserved.” It provides an algorithm: identify an infinitesimal continuous symmetry, calculate how the coordinates transform, and contract that transformation with the canonical momenta to obtain the conserved quantity.

18.9 Translations give momentum conservation

Translate every particle by the same infinitesimal vector $\epsilon \mathbf{a}$:

$$\delta \mathbf{r}_i = \epsilon \mathbf{a}. \quad (18.34)$$

If the system is isolated and its Lagrangian depends only on relative positions, then $\delta L = 0$. Here $F = 0$, and the Noether charge is

$$Q = \sum_i \mathbf{p}_i \cdot \mathbf{a} = \mathbf{a} \cdot \sum_i \mathbf{p}_i. \quad (18.35)$$

Because $\mathbf{a}$ is arbitrary,

$$\mathbf{P} \equiv \sum_i \mathbf{p}_i = \text{constant}. \quad (18.36)$$

Thus translational invariance produces conservation of total momentum.

18.10 Rotations give angular-momentum conservation

For an infinitesimal rotation by vector angle $\epsilon \boldsymbol{\omega}$,

$$\delta \mathbf{r}_i = \epsilon \boldsymbol{\omega} \times \mathbf{r}_i. \quad (18.37)$$

If the Lagrangian is rotationally invariant, then $\delta L = 0$. The Noether charge becomes

$$Q = \sum_i \mathbf{p}_i \cdot (\boldsymbol{\omega} \times \mathbf{r}_i) \quad (18.38)$$

$$= \boldsymbol{\omega} \cdot \sum_i (\mathbf{r}_i \times \mathbf{p}_i). \quad (18.39)$$

Since ω is arbitrary,

$$\boxed{\mathbf{L} \equiv \sum_i \mathbf{r}_i \times \mathbf{p}_i = \text{constant.}} \quad (18.40)$$

Rotational invariance is therefore the origin of angular-momentum conservation.

Worked example: a particle in a central potential

Consider

$$L = \frac{1}{2} m \dot{\mathbf{r}}^2 - V(r), \quad r = |\mathbf{r}|. \quad (18.41)$$

Because r is unchanged by any spatial rotation, L is rotationally invariant. Noether's theorem therefore gives

$$\boxed{\mathbf{L} = \mathbf{r} \times m \dot{\mathbf{r}} = \text{constant.}} \quad (18.42)$$

The trajectory must lie in a fixed plane perpendicular to $\mathbf{L}$. Notice that we have learned this geometrical property without solving the radial equation of motion.

In spherical polar coordinates,

$$L = \frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2 \right) - V(r). \quad (18.43)$$

The coordinate ϕ is cyclic, so

$$p_\phi = m r^2 \sin^2 \theta \dot{\phi} \quad (18.44)$$

is conserved. This is the component of angular momentum along the chosen z axis. The coordinate statement and the rotational Noether theorem are two views of the same symmetry.

18.11 The basic symmetry–conservation dictionary

For ordinary closed mechanical systems the most important examples can be summarised as

Continuous symmetry	What is unchanged?	Conserved quantity
Time translations	choice of time origin	energy
Spatial translations	choice of spatial origin	linear momentum
Spatial rotations	orientation of axes	angular momentum
Shift of a cyclic coordinate	$q_a \rightarrow q_a + \text{constant}$	conjugate momentum p_a

These are not merely mnemonic correspondences. Each row is a special case of the same variational argument.

18.12 The relativistic free-particle action

The nonrelativistic action $\int (T - V) dt$ singles out the coordinate time t . For a free relativistic particle we would prefer an expression that is manifestly Lorentz invariant. The invariant interval along a timelike worldline is

$$ds = c d\tau = c \sqrt{1 - \frac{u^2}{c^2}} dt. \quad (18.45)$$

The simplest scalar action that can be constructed from the worldline alone is therefore proportional to its invariant proper length:

$$\boxed{S_0 = -mc \int ds = -mc^2 \int d\tau.} \quad (18.46)$$

The minus sign is conventional and ensures the correct nonrelativistic limit.

Writing the same action using coordinate time gives

$$S_0 = \int L_0 dt, \quad \boxed{L_0 = -mc^2 \sqrt{1 - \frac{u^2}{c^2}}.} \quad (18.47)$$

Its canonical momentum is

$$\mathbf{p} = \frac{\partial L_0}{\partial \mathbf{u}} = \frac{m\mathbf{u}}{\sqrt{1 - u^2/c^2}} = \gamma m\mathbf{u}, \quad (18.48)$$

exactly the relativistic mechanical momentum introduced in Lecture 15. The corresponding conserved energy is

$$H = \mathbf{p} \cdot \mathbf{u} - L_0 \quad (18.49)$$

$$= \gamma mu^2 + \frac{mc^2}{\gamma} = \gamma mc^2. \quad (18.50)$$

Thus the familiar relativistic expressions follow directly from Lorentz invariance of the action.

18.13 Minimal electromagnetic coupling

For a particle of charge q , the relativistically invariant interaction with the four-potential is the line integral

$$\boxed{S_{\text{int}} = -q \int A_\mu dx^\mu.} \quad (18.51)$$

With the conventions of Lecture 17,

$$A_\mu = \left(\frac{\phi}{c}, -\mathbf{A} \right), \quad dx^\mu = (c dt, d\mathbf{r}), \quad (18.52)$$

so

$$A_\mu dx^\mu = \phi dt - \mathbf{A} \cdot d\mathbf{r}. \quad (18.53)$$

Consequently

$$S = S_0 + S_{\text{int}} = \int L dt \quad (18.54)$$

with

$$\boxed{L = -mc^2 \sqrt{1 - \frac{u^2}{c^2}} + q\mathbf{u} \cdot \mathbf{A} - q\phi,} \quad (18.55)$$

which is precisely the particle Lagrangian quoted at the end of Lecture 17.

18.14 Deriving the Lorentz force from the action

It is worth doing the algebra rather than merely quoting the result. The canonical momentum is

$$\frac{\partial L}{\partial \mathbf{u}} = \gamma m\mathbf{u} + q\mathbf{A}. \quad (18.56)$$

Therefore

$$\frac{d}{dt} \frac{\partial L}{\partial \mathbf{u}} = \frac{d}{dt}(\gamma m\mathbf{u}) + q \frac{\partial \mathbf{A}}{\partial t} + q(\mathbf{u} \cdot \nabla)\mathbf{A}. \quad (18.57)$$

When taking the spatial gradient of L , the velocity is held fixed:

$$\frac{\partial L}{\partial \mathbf{r}} = q \nabla(\mathbf{u} \cdot \mathbf{A}) - q \nabla \phi. \quad (18.58)$$

The Euler–Lagrange equation gives

$$\frac{d}{dt}(\gamma m \mathbf{u}) = q \left[-\nabla \phi - \frac{\partial \mathbf{A}}{\partial t} + \nabla(\mathbf{u} \cdot \mathbf{A}) - (\mathbf{u} \cdot \nabla) \mathbf{A} \right]. \quad (18.59)$$

Since $\mathbf{u}$ is independent of position during the gradient operation,

$$\nabla(\mathbf{u} \cdot \mathbf{A}) - (\mathbf{u} \cdot \nabla) \mathbf{A} = \mathbf{u} \times (\nabla \times \mathbf{A}). \quad (18.60)$$

Using

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}, \quad (18.61)$$

we obtain

$$\boxed{\frac{d}{dt}(\gamma m \mathbf{u}) = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})}. \quad (18.62)$$

The Lorentz force is therefore not an additional rule that must be appended to the action formalism; it follows from the same stationary-action principle.

18.15 Gauge invariance of the particle action

Under the gauge transformation

$$A_\mu \longrightarrow A_\mu + \partial_\mu \chi, \quad (18.63)$$

the interaction action changes by

$$\Delta S_{\text{int}} = -q \int \partial_\mu \chi dx^\mu \quad (18.64)$$

$$= -q \int d\chi \quad (18.65)$$

$$= -q [\chi(x_2) - \chi(x_1)]. \quad (18.66)$$

This is purely an endpoint term. If the endpoints are fixed, its variation is zero, so the equations of motion are unchanged.

Physical meaning

This gives a particularly clean interpretation of gauge freedom. Two electromagnetic potentials related by a gauge transformation need not give exactly the same numerical action; they may differ by a boundary term. What matters physically is that they give the same stationary paths and therefore the same fields and equations of motion.

18.16 A manifestly covariant derivation of the Lorentz force

We can make the covariance even more explicit by parametrising the worldline by an arbitrary parameter λ and writing

$$S = -mc \int d\lambda \sqrt{\dot{x}^\mu \dot{x}_\mu} - q \int d\lambda A_\mu \dot{x}^\mu, \quad (18.67)$$

where now the dot means $d/d\lambda$. Varying $x^\mu(\lambda)$ and only afterwards choosing $\lambda = \tau$ gives

$$m \frac{dU^\mu}{d\tau} = q(\partial_\mu A_\nu - \partial_\nu A_\mu)U^\nu. \quad (18.68)$$

Since

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (18.69)$$

we obtain

$$\boxed{m \frac{dU^\mu}{d\tau} = q F^\mu{}_\nu U^\nu.} \quad (18.70)$$

This is exactly the covariant Lorentz-force equation from Lecture 17, now derived from a Lorentz-scalar action.

18.17 The electromagnetic field also has an action

The action principle is not restricted to particle coordinates. A field has infinitely many degrees of freedom, one for every point in space. The field analogue of the Lagrangian is a *Lagrangian density* $\mathcal{L}$, and

$$S = \frac{1}{c} \int d^4x \mathcal{L}, \quad d^4x = d(ct) d^3r. \quad (18.71)$$

For the electromagnetic field coupled to a prescribed four-current,

$$\boxed{\mathcal{L}_{\text{EM}} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - J_\mu A^\mu.} \quad (18.72)$$

The first term describes the field itself; the second describes its coupling to the source.

The Euler–Lagrange equation for a field $A_\nu(x)$ is

$$\boxed{\frac{\partial \mathcal{L}}{\partial A_\nu} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) = 0.} \quad (18.73)$$

Rather than use this formula blindly, it is instructive to vary the action directly. Since

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu, \quad (18.74)$$

and $F^{\mu\nu}$ is antisymmetric,

$$\delta S = \frac{1}{c} \int d^4x \left[-\frac{1}{\mu_0} F^{\mu\nu} \partial_\mu \delta A_\nu - J^\nu \delta A_\nu \right]. \quad (18.75)$$

Integrating the first term by parts and taking the variation to vanish on the spacetime boundary,

$$\delta S = \frac{1}{c} \int d^4x \left[\frac{1}{\mu_0} \partial_\mu F^{\mu\nu} - J^\nu \right] \delta A_\nu. \quad (18.76)$$

Because δA_ν is arbitrary,

$$\boxed{\partial_\mu F^{\mu\nu} = \mu_0 J^\nu.} \quad (18.77)$$

Thus the inhomogeneous Maxwell equations follow from stationary action.

What about the other two Maxwell equations? They are already contained in the definition

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (18.78)$$

because this automatically implies the Bianchi identity

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0. \quad (18.79)$$

Hence the full set of Maxwell equations is encoded by the four-potential and the action.

18.18 Gauge invariance and charge conservation

The field-strength term is exactly gauge invariant because $F_{\mu\nu}$ is gauge invariant. The source term changes according to

$$-J_\mu A^\mu \longrightarrow -J_\mu A^\mu - J_\mu \partial^\mu \chi \quad (18.80)$$

$$= -J_\mu A^\mu - \partial^\mu (J_\mu \chi) + \chi \partial^\mu J_\mu. \quad (18.81)$$

The first extra contribution is a spacetime boundary term. Therefore, provided

$$\boxed{\partial_\mu J^\mu = 0}, \quad (18.82)$$

the action is gauge invariant up to a boundary term.

Conversely, if the coupling to an external current is to be gauge invariant for arbitrary χ , the current must satisfy the continuity equation. Gauge invariance and local charge conservation are therefore deeply linked.

Notation and convention

There is a more general field-theoretic statement in which Noether's theorem is applied to local gauge symmetries. That subject leads into modern gauge theory and lies beyond the present course. For our purposes the important result is the concrete one above: the electromagnetic source coupling is compatible with gauge freedom precisely because electric charge is locally conserved.

18.19 Noether's theorem for fields

For completeness, the mechanical Noether theorem has a covariant field version. Let fields $\varphi_A(x)$ be described by

$$S = \frac{1}{c} \int d^4x \mathcal{L}(\varphi_A, \partial_\mu \varphi_A). \quad (18.83)$$

Suppose an infinitesimal continuous transformation

$$\delta \varphi_A = \epsilon \Delta \varphi_A \quad (18.84)$$

changes the Lagrangian density only by a four-divergence,

$$\delta \mathcal{L} = \epsilon \partial_\mu K^\mu. \quad (18.85)$$

On fields satisfying the Euler–Lagrange equations, the Noether current

$$j^\mu = \sum_A \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_A)} \Delta \varphi_A - K^\mu \quad (18.86)$$

obeys

$$\boxed{\partial_\mu j^\mu = 0}. \quad (18.87)$$

A conserved current is the relativistic local version of a conserved quantity. Integrating its time component over space gives a conserved charge, provided no flux escapes through the boundary.

Spacetime translation symmetry of a closed field theory produces energy–momentum conservation. The electromagnetic stress-energy tensor introduced in Lecture 17 is the gauge-invariant, symmetrised form of the corresponding energy–momentum current. Thus the conservation equation

$$\partial_\mu T_{\text{total}}^{\mu\nu} = 0 \quad (18.88)$$

is another manifestation of the same symmetry principle.

18.20 What has been unified?

We can now read much of the second half of the course backwards from one compact structure:

1. Lorentz invariance suggests the invariant worldline action $-mc \int ds$.
2. Coupling a charge to the four-potential through $-q \int A_\mu dx^\mu$ produces the Lorentz force.
3. The field action built from $F_{\mu\nu}F^{\mu\nu}$ produces Maxwell's equations.
4. Gauge transformations change the particle action only by endpoint terms and the field-source action only by a spacetime boundary term when charge is conserved.
5. Continuous spacetime symmetries lead, through Noether's theorem, to conservation of energy, momentum and angular momentum.

Physical meaning

The action formulation is a natural endpoint for the course because it explains why several structures encountered earlier fit together. The four-potential is simultaneously the variable from which $F^{\mu\nu}$ is built, the quantity that couples directly to a charged particle, and the field whose gauge freedom leaves the action physically unchanged. Lorentz symmetry, gauge symmetry and conservation laws are therefore not separate add-ons to electrodynamics; they are structural properties of the same theory.

Checkpoint

The essential ideas of Topic B5 are now in place:

- the physical motion makes the action stationary;
- varying the action gives the Euler–Lagrange equations;
- invariance under continuous transformations produces conserved quantities through Noether's theorem;
- the free-particle, charged-particle and electromagnetic-field actions can all be written covariantly;
- the Lorentz force and Maxwell equations follow by variation rather than being imposed independently.

Summary of Part B

The logical chain of the course is:

1. Green's functions solve linear field equations by building the response to point sources.
2. The wave Green's function introduces retardation and finite propagation speed.
3. Retarded potentials generate radiation fields, whose $1/r$ decay allows energy to escape to infinity.
4. Small sources are organised by multipoles: electric dipole, magnetic dipole, electric quadrupole, and so on.
5. A moving point charge produces the Liénard–Wiechert potentials; acceleration generates the radiative $1/R$ field.
6. Relativistic motion produces beaming, synchrotron radiation and the full Liénard power formula; media allow Cherenkov radiation; energy loss motivates radiation reaction.
7. Special relativity unifies electric and magnetic fields as components of $F^{\mu\nu}$, while Maxwell's equations, charge conservation, the Lorentz force and gauge freedom all acquire compact covariant forms.
8. The action principle provides a unified variational formulation: the Euler–Lagrange equations generate the dynamics, continuous symmetries generate conservation laws through Noether's theorem, and covariant particle and field actions reproduce the Lorentz force and Maxwell equations.